pybind11——[illegible]C+11[illegible]Python

pybind11[illegible]C++[illegible]Python[illegible]Python[illegible]C++[illegible]C++[illegible]Python[illegible]David Abrahams[illegible]Boost.Python[illegible]

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1.1 [illegible]

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- Internal references with correct reference counting[illegible]

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1.2 □□□□□

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- □□Python2.7, 3.5+, PyPy/PyPy3 7.3□□□□□□□□□
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- □□Python□buffer□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□C++□□□□□□Eigen□□NumPy□□□□□□□□□□□□□□□□□□□
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- □□□□□□□□□□□□Python□□□□□□□□□□□□□□
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- □□`constexpr`□□□□□□□□□□□□□□□□□□□□□□□□□
- □□□□□□C++□□□□Python pickle□unpickle□□□

1.3 □□□□□□

1. Clang/LLVM 3.3□□ (Apple Xcode’s clang□□5.0.0□□□□)
2. GCC 4.8□□
3. Microsoft Visual Studio 2015 Update 3□□
4. Intel classic C++ compiler 18 or newer (ICC 20.2 tested in CI)
5. Cygwin/GCC (previously tested on 2.5.1)
6. NVCC (CUDA 11.0 tested in CI)
7. NVIDIA PGI (20.9 tested in CI)

1.4 □□

This project was created by Wenzel Jakob. Significant features and/or improvements to the code were contributed by Jonas Adler, Lori A. Burns, Sylvain Corlay, Eric Cousineau, Aaron Gokaslan, Ralf Grosse-Kunstleve, Trent Houliston, Axel Huebl, @hulucc, Yannick Jadoul, Sergey Lyskov Johan Mabille, Tomasz Miąsko, Dean Moldovan, Ben Pritchard, Jason Rhinelander, Boris Schäling, Pim Schellart, Henry Schreiner, Ivan Smirnov, Boris Staletic, and Patrick Stewart.

We thank Google for a generous financial contribution to the continuous integration infrastructure used by this project.

1.5 □□

See the contributing guide for information on building and contributing to pybind11.

1.6 License

pybind11 is provided under a BSD-style license that can be found in the LICENSE file. By using, distributing, or contributing to this project, you agree to the terms and conditions of this license.

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3. □□□□

□□□□□pybind/pybind11 on GitHub□□□pybind11□□□□□□pybind11□□□□□□□□□□□□□□□□
□□□□□pybind11□

3.1 □□□□□□□□

□□□□□□□Git□□□□□□□□pybind11□□□□□□□□□□□□□□□□□□□git□□□□□□□□□□□□□pybind11□

```
git submodule add -b stable ../../pybind/pybind11 extern/pybind11
git submodule update --init
```

□□□□□□□□□□□□□□`extern`□□□□□□□□GitHub□□□□□□□□GitHub□□□□□□□□https□ssh URL □□□□□□□□URL`../../pybind/pybind11`□□□□□□□□□□`.git`□□□GitHub□□□□

□□□□□□□□□□□include `extern/pybind11/include`□□□□□□□□□□□□□□□□□□□□□Build System□□)□□□pybind11□

3.2 □□PyPI□□□

□□□□□pip□□PyPI□□□Pybind11□Python□□□□□□□□□□□CMake□□□□□□□

```
pip install pybind11
```

□□pybind11□□□□□Python□□□□□□□□□□□□root□□□□□□□pybind11□□□□□□□

```
pip install "pybind11[global]"
```

□□□□□□□□□□Python□□□□□□□□□root□□□□□□□□□□`/usr/local/include/pybind11` □ `/usr/local/share/cmake/pybind11`□□□□□□□□□□□□□□□□□□□□□□□□□□ `pyproject.toml`□□□□

3.3 □□conda-forge□□

You can use pybind11 with conda packaging via conda-forge:

```
conda install -c conda-forge pybind11
```

3.4 □□vcpkg□□

□□□□□Microsoft vcpkg□□□□□□□□□□□□pybind11□

```
git clone https://github.com/Microsoft/vcpkg.git
cd vcpkg
./bootstrap-vcpkg.sh
./vcpkg integrate install
vcpkg install pybind11
```

3.5 □□brew□□□□

brew□□□□Homebrew on macOS, or Linuxbrew on Linux□□pybind11□□□□□□□

```
brew install pybind11
```

3.6 □□□□

Other locations you can find pybind11 are listed here; these are maintained by various packagers and the community.

pybind11□□□□

1. □□□□

1.1 □□□□□

□□□python3-dev□□□□pybind11□□□□□□□□□□□include pybind11□□□□□□python3□□□□□□□□cmake□□□□□

```cmake
set(PYTHON_TARGET_VER 3.6)
find_package(PythonInterp ${PYTHON_TARGET_VER} EXACT)
find_package(PythonLibs ${PYTHON_TARGET_VER} EXACT REQUIRED)

include_directories(pybind11_include_path)
include_directories(${PYTHON_INCLUDE_DIRS})
```

1.2 □□□□

```cpp
#include <pybind11/pybind11.h>

int add(int i, int j) {
    return i + j;
}

PYBIND11_MODULE(example, m) {
    m.doc() = "pybind11 example plugin"; // optional module docstring
    m.def("add", &add, "A function which adds two numbers");
}
```

□ `PYBIND11_MODULE` □□□□□□□□□□□□Python□ `import` □□□□□□□□□□□□□□□□□□□ `py::module_` □□□□m□□□□□□□□□□□□ `module_::def()` □□□□□□□□□□□□

1.2.1 □□□□□

□□ `py::arg` □□□□□□□□□□Python□□□□□□□□□□□□□□□□□□□□□□□□

```
m.def("add", &add, "A function which adds two numbers",
      py::arg("i"), py::arg("j"));
```

□□□□□□□

```
// regular notation
m.def("add1", &add, py::arg("i"), py::arg("j"));
// shorthand
using namespace pybind11::literals;
m.def("add2", &add, "i"_a, "j"_a);
```

Python□□□□□

```python
import example
example.add(i=1, j=2)  #3L
```

1.2.2 □□□□□

pybind11□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□ `arg` □□□□□□□□□□□

```
m.def("add", &add, "A function which adds two numbers",
      py::arg("i") = 1, py::arg("j") = 2);
```

□□□□□□□□□

```
// regular notation
m.def("add1", &add, py::arg("i") = 1, py::arg("j") = 2);
// shorthand
m.def("add2", &add, "i"_a=1, "j"_a=2);
```

1.2.3 □□□□

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□□Python□□□□□□□□□□□□□□Python□□□□□□□□□□□□□□□□□□□

```
m.def("add", static_cast<int(*)(int, int)>(&add), "A function which adds
two int numbers");
m.def("add", static_cast<double(*)(double, double)>(&add), "A function
which adds two double numbers");
```

□□□□□□□□□C++14□□□□□□□□□□□□□□□□□□□□

```
m.def("add", py::overload_cast<int, int>(&add), "A function which adds two
int numbers");
m.def("add", py::overload_cast<double, double>(&add), "A function which
adds two double numbers");
```

□□□`py::overload_cast`□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□(`void (Pet::*)`)□□□□□□const□□□□□□□□`py::const`□□□

1.3 □□□□

□□□□□□`attr`□□□□□□□□□□Python□□□□C++□□□□□□□□□□□□□□□□□attriutes□□□□□□□□□□□`py::cast`□□□□□□

```
PYBIND11_MODULE(example, m) {
    m.attr("the_answer") = 42;
    py::object world = py::cast("World");
    m.attr("what") = world;
}
``

Python□□□□□□
```pyhton
>>> import example
>>> example.the_answer
42
>>> example.what
'World'
```

1.4 □□□□□□□

□□□□□□□□□C++□□□□□□□`Pet`□□□□□□

```
struct Pet {
    Pet(const std::string &name) : name(name) { }
    void setName(const std::string &name_) { name = name_; }
    const std::string &getName() const { return name; }

    std::string name;
};
```

□□□□□□□□□
```

```cpp
#include <pybind11/pybind11.h>
namespace py = pybind11;

PYBIND11_MODULE(example, m) {
    py::class_<Pet>(m, "Pet")
        .def(py::init<const std::string &>())
        .def("setName", &Pet::setName)
        .def("getName", &Pet::getName)
        .def("__repr__",
            [](const Pet &a) {
                return "<example.Pet named '" + a.name + "'>";
            });
}
```

`class_`□□□C++ class□ struct□□□□`init()`□□□□□□□□□□□□□□□□□□□□□□□□□
□□□□□□□□□□

□□`print(p)`□□□□□□□□□□□□□□□□□□□□□□□□□□`__repr__`□□□□□□□□□□□
□□□□Pet□□□□□□□□□□□□□□□□□□□□□□□

Python□□□□□

```python
>>> import example
>>> p = example.Pet("Molly")
>>> print(p)
<example.Pet named 'Molly'>
>>> p.getName()
u'Molly'
>>> p.setName("Charly")
>>> p.getName()
u'Charly'
```

□□□□□□□□`class_::def_static`□□□

1.4.1 □□□□

□□`class_::def_readwrite`□□□□□□□□□□□`class_::def_readonly`□□□□□□□□□

```
py::class_<Pet>(m, "Pet")
    .def(py::init<const std::string &>())
    .def_readwrite("name", &Pet::name)
    // ... remainder ...
```

Python□□□□□□□□

```python
>>> p = example.Pet("Molly")
>>> p.name
u'Molly'
>>> p.name = "Charly"
>>> p.name
u'Charly'
```

□□`Pet::name`□□□□□□□□□□□□□□setter□getters□□□

```
class Pet {
public:
    Pet(const std::string &name) : name(name) { }
    void setName(const std::string &name_) { name = name_; }
    const std::string &getName() const { return name; }
private:
    std::string name;
};
```

□□□□`class_::def_property()`(□□□□□□`class_::def_property_readonly()`)□□□□□□□□□□□□□□□setter□geter□□□

```
py::class_<Pet>(m, "Pet")
    .def(py::init<const std::string &>())
    .def_property("name", &Pet::getName, &Pet::setName)
    // ... remainder ...
```

□□□□□□□read□□□□□nullptr□□□□

□□□□□`class_::def_readwrite_static()`, `class_::def_readonly_static()` `class_::def_property_static()`, `class_::def_property_readonly_static()`□□□□□

□□□□□□□

1.4.2 □□□□

□□□Pyhton□□□□□□□□□□□□

```python
>>> class Pet:
...     name = "Molly"
...
>>> p = Pet()
>>> p.name = "Charly"  # overwrite existing
>>> p.age = 2  # dynamically add a new attribute
```

□□□□□□□C++□□□□□□□□□□□□□□□□□□□□□□`class_::def_readwrite`□`class_::def_property`□□□□□□□□□□□□□□□□□□

```python
>>> p = example.Pet()
>>> p.name = "Charly"  # OK, attribute defined in C++
>>> p.age = 2  # fail
AttributeError: 'Pet' object has no attribute 'age'
```

□□C++□□□□□□□□□□□□□□`py::class_`□□□□□□□`py::dynamic_attr`□□□

```cpp
py::class_<Pet>(m, "Pet", py::dynamic_attr())
    .def(py::init<>())
    .def_readwrite("name", &Pet::name);
```

□□□□□□□□□□□□□□□□□□□

```
>>> p = example.Pet()
>>> p.name = "Charly"  # OK, overwrite value in C++
>>> p.age = 2  # OK, dynamically add a new attribute
>>> p.__dict__  # just like a native Python class
{'age': 2}
```

□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□`__dict__`□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□Python□□□□□□□□□□□□□pybind11□□□□□□Python□□□□□□□□□□□□□□□□□□□□□□□

1.4.3 □□□□

□□□□□□□□□□□□□□□□□□□□□□□□□

```cpp
struct Pet {
    Pet(const std::string &name, int age) : name(name), age(age) { }

    void set(int age_) { age = age_; }
    void set(const std::string &name_) { name = name_; }

    std::string name;
    int age;
};

// method 1
py::class_<Pet>(m, "Pet")
   .def(py::init<const std::string &, int>())
   .def("set", static_cast<void (Pet::*)(int)>(&Pet::set), "Set the pet's age")
   .def("set", static_cast<void (Pet::*)(const std::string &)>(&Pet::set), "Set the pet's name");

// method 2
py::class_<Pet>(m, "Pet")
    .def("set", py::overload_cast<int>(&Pet::set), "Set the pet's age")
    .def("set", py::overload_cast<const std::string &>(&Pet::set), "Set the pet's name");
```

1.5 □□□□□□

□□C□□□□□□□□□□□□□□□

```cpp
enum Flags {
    Read = 4,
    Write = 2,
    Execute = 1
};

py::enum_<Flags>(m, "Flags", py::arithmetic())
    .value("Read", Flags::Read)
    .value("Write", Flags::Write)
    .value("Execute", Flags::Execute)
    .export_values();
```

`enum_::export_values()`□□□□□□□□□□□□□□C++11□□□□□□□□□□□□□

□□□□□□□□□□□□□□`__members__`□□□□`name`□□□□□□□□□□□□□unicode□□□□`str(enum)`□□□□□□□□□□□□□□□□□

1.6 □□*args□**kwargs□□

Python□□□□□□□□□□□□□□□□□□□□□

```python
def generic(*args, **kwargs):
    ...  # do something with args and kwargs
```

□□□□□□□pybind11□□□□□□□□□

```
void generic(py::args args, const py::kwargs& kwargs) {
    /// .. do something with args
    if (kwargs)
        /// .. do something with kwargs
}

/// Binding code
m.def("generic", &generic);
```

`py::args`□□□`py::tuple`□`py::kwargs`□□□`py::dict`□

2. □□□□□

2.1 □□□□

Python□C++□□□□□□□□□□□□□□□□□□□□□□□□□□□no-trivial□□□□□□□□□□□□□□□□□□□□□□Python□□□□□□□□□□□□□□□□□□□□C++□□□□□□□pybind11□□□□□□□□□□□□□□□□□□□□□□□□`model::def()`□`class_def()`□□□□□□□`return_value_policy::automatic`□

□□

```
/* Function declaration */
Data *get_data() { return _data; /* (pointer to a static data structure)
*/ }
...

/* Binding code */
m.def("get_data", &get_data); // <-- KABOOM, will cause crash when called
from Python
```

□Python□□□`get_data()`□□□□□□□□C++□□□□□□□□□□□□Python□□□□□□□□□□□□□□□□□□`return_value_policy::automatic`□□pybind11□□□□□□`_data`□□□□□

□Python□□□□□□□□ `_data` □Python□□□□pybind11□□□□C++□□□□□operator delete()□□
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□□□□□□□□□□□□□□□□□□ `return_value_policy::reference` □□□□□□□□□□□□□□□□□
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```
m.def("get_data", &get_data, py::return_value_policy::reference);
```

□□□pybind11□□□□□□□□□□
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`return_value_policy::take_ownership`	□□□□□□□□□□□□□□□□□□□□□□ □□Pyhton□□□□□□□□delete□□□□
`return_value_policy::copy`	□□□□□□□Python□□□□□□□□□□□ □□□□□□□□□□□□□□□□
`return_value_policy::move`	□□ `std::move` □□□□□□□□□□□□□ □Python□□□□□□□□□□□□□□□□□ □□□
`return_value_policy::reference`	□□□□□□□□□□□□□C++□□□□□□ □□□□□□□□□□□□□□□□□□□□Pytho □□□□□C++□□□□□□□□□□□□□
`return_value_policy::reference_internal`	□□□□□□□□□□□□□□□□□□□□□□ `this` □ `self` □□□□□□□□reference □□ `keep_alive<0, 1>` □□□□□□□ Python□□□□□□□□□□□□□□□□□□ `def_property` □ `def_readwrite` getter□□□□□□□□□□□
`return_value_policy::automatic`	□□□□□□□□□□□□□ `return_value_policy::take_ow` □□□□□□□□□□□ `return_value_policy::copy` □□ □□□□□□□□□□□□□□□□□□□ `py::cl` □□□□□□

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`return_value_policy::automatic_reference`	□□□□□□□□□□□□□□□□□□□□ `return_value_policy::referer` □C++□□□□□□Python□□□□□ `pybind11/stl.h` □□casters□□□□ □□□□□□□□□□□□□□

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```
class_<MyClass>(m, "MyClass")
    .def_property("data", &MyClass::getData, &MyClass::setData,
                  py::return_value_policy::copy);
```

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```
class_<MyClass>(m, "MyClass")
    .def_property("data"
        py::cpp_function(&MyClass::getData,
py::return_value_policy::copy),
        py::cpp_function(&MyClass::setData)
    );
```

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2.2 □□□□□□□

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□□□keep alive□

□□□C++□□□□□□□□□C++□□□□□□□□□□□□□□`keep_alive<Nurse, Patient>`□□□□□□□Nurse□□□□□□□Patient□□□□□□0□□□□□□1□□□□□□□□□□1□□□□□□□this□□□□□□□□□□□□2□□□□Nurse□□□□□□□□□□□None□□□□□□□□□□□□□

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```
py::class_<List>(m, "List").def("append", &List::append, py::keep_alive<1,
2>());
```

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```
py::class_<Nurse>(m, "Nurse").def(py::init<Patient &>(), py::keep_alive<1,
2>());
```

> Note: `keep_alive` □Boost.Python□□ `with_custodian_and_ward` □ `with_custodian_and_ward_postcall` □□□

Call guard

`call_guard<T>`□□□□□□T□□□scope guard□□□□□□□□□□□□□□□

```
m.def("foo", foo, py::call_guard<T>());
```

□□□□□□□□□

```
m.def("foo", [](args...) {
    T scope_guard;
    return foo(args...); // forwarded arguments
});
```

□□□□□□□T□□□□□□□ `gil_scoped_release` □□□□□□□□□□□□

`call_guard` □□□□□□□□□□□□□ `call_guard<T1, T2, T3 ...>` □□□□□□□□□□□□□□□□□□□□

> See also: `test/test_call_policies.cpp` □□□□□□□□□□□ `keep_alive` □ `call_guard` □□□□

2.3 Keyword-only□□

Python3□□□keyword-only□□□□□□□□□□□ `*` □□□□□□□□

```python
def f(a, *, b):  # a can be positional or via keyword; b must be via keyword
    pass

f(a=1, b=2)  # good
f(b=2, a=1)  # good
f(1, b=2)  # good
f(1, 2)  # TypeError: f() takes 1 positional argument but 2 were given
```

pybind11□□□ `py::kw_only` □□□□□□□□□□□

```
m.def("f", [](int a, int b) { /* ... */ },
      py::arg("a"), py::kw_only(), py::arg("b"));
```

□□□□□□□□□`py::args`□□□□□

2.4 Positional-only□□

python3.8□□□Positional-only□□□□□pybind11□□`py::pos_only()`□□□□□□□□□

```
m.def("f", [](int a, int b) { /* ... */ },
      py::arg("a"), py::pos_only(), py::arg("b"));
```

□□□□□□□□□□□□□□`a`□□□□□□□□□keyword-only□□□□□□□

2.5 Non-converting□□

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- □□`py::implicitly_convertible<A,B>()`□□□□□□
- □□□□□□□□□□□□□□□□□
- □□□□□□□□float□□□□□□`std::complex<float>`□□□□□
- Calling a function taking an Eigen matrix reference with a numpy array of the wrong type or of an incompatible data layout.

□□`py::arg`□□□`.noconvert()`□□□□□□□□□□□

```
m.def("floats_only", [](double f) { return 0.5 * f; },
py::arg("f").noconvert());
m.def("floats_preferred", [](double f) { return 0.5 * f; }, py::arg("f"));
```

□□□□□□□□□□□`TypeError`□□□

```
>>> floats_preferred(4)
2.0
>>> floats_only(4)
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
TypeError: floats_only(): incompatible function arguments. The following
argument types are supported:
    1. (f: float) -> float

Invoked with: 4
```

□□□□□□□□□□`_a`□□□□□□□□□□□□□`py::arg().noconvert()`□

3. □□□□□

3.1 □□□□□

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```cpp
struct Pet {
    Pet(const std::string &name) : name(name) { }
    std::string name;
};

struct Dog : Pet {
    Dog(const std::string &name) : Pet(name) { }
    std::string bark() const { return "woof!"; }
};
```

pybind11□□□□□□□□□□□□□□□1□□C++□□□□□□□`class_`□□□□□□2□□□□□□□`class_`□□□□□
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```cpp
py::class_<Pet>(m, "Pet")
   .def(py::init<const std::string &>())
   .def_readwrite("name", &Pet::name);

// Method 1: template parameter:
py::class_<Dog, Pet /* <- specify C++ parent type */>(m, "Dog")
    .def(py::init<const std::string &>())
    .def("bark", &Dog::bark);

// Method 2: pass parent class_ object:
py::class_<Dog>(m, "Dog", pet /* <- specify Python parent type */)
    .def(py::init<const std::string &>())
    .def("bark", &Dog::bark);
```

[illegible]

```python
>>> p = example.Dog("Molly")
>>> p.name
u'Molly'
>>> p.bark()
u'woof!'
```

[illegible]Python[illegible]

```cpp
// [illegible]
m.def("pet_store", []() { return std::unique_ptr<Pet>(new Dog("Molly"));
});
```

```python
>>> p = example.pet_store()
>>> type(p)  # `Dog` instance behind `Pet` pointer
Pet          # no pointer downcasting for regular non-polymorphic types
>>> p.bark()
AttributeError: 'Pet' object has no attribute 'bark'
```

`pet_store` [illegible]Dog[illegible]Python[illegible]Pet[illegible]C++[illegible]pybind11[illegible]

```cpp
struct PolymorphicPet {
    virtual ~PolymorphicPet() = default;
};

struct PolymorphicDog : PolymorphicPet {
    std::string bark() const { return "woof!"; }
};

// Same binding code
py::class_<PolymorphicPet>(m, "PolymorphicPet");
py::class_<PolymorphicDog, PolymorphicPet>(m, "PolymorphicDog")
    .def(py::init<>())
    .def("bark", &PolymorphicDog::bark);

// Again, return a base pointer to a derived instance
m.def("pet_store2", []() { return std::unique_ptr<PolymorphicPet>(new
PolymorphicDog); });
```

```python
>>> p = example.pet_store2()
>>> type(p)
PolymorphicDog  # automatically downcast
>>> p.bark()
u'woof!'
```

pybind11□□□□□□□□□□□□□□□□□□□□□□□□□□C++□□□

3.2 Python□□C++□

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```cpp
class Animal {
public:
    virtual ~Animal() { }
    virtual std::string go(int n_times) = 0;
};

class Dog : public Animal {
public:
    std::string go(int n_times) override {
        std::string result;
        for (int i=0; i<n_times; ++i)
            result += "woof! ";
        return result;
    }
};

std::string call_go(Animal *animal) {
    return animal->go(3);
}

PYBIND11_MODULE(example, m) {
    py::class_<Animal>(m, "Animal")
        .def("go", &Animal::go);

    py::class_<Dog, Animal>(m, "Dog")
        .def(py::init<>());

    m.def("call_go", &call_go);
}
```

□□□□□□□□□□Python□□□□□Animal□□□□□“No constructor defined!“□

□□□□□□□□□□□□□Animal□□□□□□□□

```
class PyAnimal : public Animal {
public:
    /* Inherit the constructors */
    using Animal::Animal;

    /* Trampoline (need one for each virtual function) */
    std::string go(int n_times) override {
        PYBIND11_OVERRIDE_PURE(
            std::string, /* Return type */
            Animal,      /* Parent class */
            go,          /* Name of function in C++ (must match Python
name) */
            n_times      /* Argument(s) */
        );
    }
};
```

□□□□□□□□□□□□`PYBIND11_OVERRIDE_PURE`□□□□□□□□□□□□□□□□`PYBIND11_OVERRIDE`□`PYBIND11_OVERRIDE_PURE_NAME`□`PYBIND11_OVERRIDE_NAME`□□□□□□□□□□□□C□□□□□Python□□□□□□□□□□□□`__str__`□□□

```
std::string toString() override {
  PYBIND11_OVERRIDE_NAME(
      std::string, // Return type (ret_type)
      Animal,      // Parent class (cname)
      "__str__",   // Name of method in Python (name)
      toString,    // Name of function in C++ (fn)
  );
}
```

Animal□□□□□□□□□□□□□□□

```
PYBIND11_MODULE(example, m) {
    py::class_<Animal, PyAnimal /* <--- trampoline*/>(m, "Animal")
        .def(py::init<>())
        .def("go", &Animal::go);

    py::class_<Dog, Animal>(m, "Dog")
        .def(py::init<>());

    m.def("call_go", &call_go);
}
```

pybind11□□□ `class_` □□□□□□□□□PyAnimal□□□□□□□Python□□□Animal□□

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```
py::class_<Animal, PyAnimal /* <--- trampoline*/>(m, "Animal");
    .def(py::init<>())
    .def("go", &PyAnimal::go); /* <--- THIS IS WRONG, use &Animal::go */
```

□□□Python□□□□□□□□□□□□□□ `Animal::go` □□□□□□□□□□□□□□□

```python
from example import *
d = Dog()
call_go(d)     # u'woof! woof! woof! '
class Cat(Animal):
    def go(self, n_times):
        return "meow! " * n_times

c = Cat()
call_go(c)   # u'meow! meow! meow! '
```

□□□□□□Python□□□□□□□□□□□□□□□□□□□□□□□□□□C++□□□□(□□ `__init__`)□□□□□□□□□□□□□□□□□□□□□□C++□□□□□□□□□□□□□□□□□□□□□□□□□pybind11 2.6□□□□□□□□□□□□□ `TypeError` □□□

```python
class Dachshund(Dog):
    def __init__(self, name):
        Dog.__init__(self)  # Without this, a TypeError is raised.
        self.name = name

    def bark(self):
        return "yap!"
```

□□□□□□□□□`__init__`□□□□□□□`supper()`□□□□□□□□□□□□□`supper()`□□□□□□□□□□□□□□Python□C++□□□□□□□□□□Python MRO□C++□□□□□□□□□□□

3.3 □□□□□□□

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```cpp
class Animal {
public:
    virtual std::string go(int n_times) = 0;
    virtual std::string name() { return "unknown"; }
};
class Dog : public Animal {
public:
    std::string go(int n_times) override {
        std::string result;
        for (int i=0; i<n_times; ++i)
            result += bark() + " ";
        return result;
    }
    virtual std::string bark() { return "woof!"; }
};
```

□□□□□□Animal□□□□□□□□□□□□□Python□□□□□□`Dog`□□□□□□□□`Dog`□□□□□□□□□□□□`bark()`□□□□Animal□`go()`□`name()`□□□□□□□□Dog□□□□□□name□□□□

```
class PyAnimal : public Animal {
public:
    using Animal::Animal; // Inherit constructors
    std::string go(int n_times) override {
PYBIND11_OVERRIDE_PURE(std::string, Animal, go, n_times); }
    std::string name() override { PYBIND11_OVERRIDE(std::string, Animal,
name, ); }
};
class PyDog : public Dog {
public:
    using Dog::Dog; // Inherit constructors
    std::string go(int n_times) override { PYBIND11_OVERRIDE(std::string,
Dog, go, n_times); }
    std::string name() override { PYBIND11_OVERRIDE(std::string, Dog,
name, ); }
    std::string bark() override { PYBIND11_OVERRIDE(std::string, Dog,
bark, ); }
};
```

> □□□ `name()` □ `bark()` □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

□□□□□□□pybind11□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

```
class Husky : public Dog {};
class PyHusky : public Husky {
public:
    using Husky::Husky; // Inherit constructors
    std::string go(int n_times) override {
PYBIND11_OVERRIDE_PURE(std::string, Husky, go, n_times); }
    std::string name() override { PYBIND11_OVERRIDE(std::string, Husky,
name, ); }
    std::string bark() override { PYBIND11_OVERRIDE(std::string, Husky,
bark, ); }
};
```

□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

```
template <class AnimalBase = Animal> class PyAnimal : public AnimalBase {
public:
    using AnimalBase::AnimalBase; // Inherit constructors
    std::string go(int n_times) override {
PYBIND11_OVERRIDE_PURE(std::string, AnimalBase, go, n_times); }
    std::string name() override { PYBIND11_OVERRIDE(std::string,
AnimalBase, name, ); }
};
template <class DogBase = Dog> class PyDog : public PyAnimal<DogBase> {
public:
    using PyAnimal<DogBase>::PyAnimal; // Inherit constructors
    // Override PyAnimal's pure virtual go() with a non-pure one:
    std::string go(int n_times) override { PYBIND11_OVERRIDE(std::string,
DogBase, go, n_times); }
    std::string bark() override { PYBIND11_OVERRIDE(std::string, DogBase,
bark, ); }
};
```

□□

□□□□pybind11□□□□□

```
py::class_<Animal, PyAnimal<>> animal(m, "Animal");
py::class_<Dog, Animal, PyDog<>> dog(m, "Dog");
py::class_<Husky, Dog, PyDog<Husky>> husky(m, "Husky");
// ... add animal, dog, husky definitions
```

□□□Husky□□□□□□□□□□□□□□□□□□□□□□□□□□

Python□□□□□

```
class ShihTzu(Dog):
    def bark(self):
        return "yip!"
```

3.4 □□□□□□□

□□□□□□□□□□□□□□□□□□□□□pybind11□□□□□□□□□□□□□□`std::unique_ptr`□□□□□□□□□□□□□□□□□□□□□□□□□□□Pybind11□□□□□`py::nodelete`□□□□□□□□□□□□□□C++□□□□□□□□□□□□□□□

```cpp
/* ... definition ... */

class MyClass {
private:
    ~MyClass() { }
};

/* ... binding code ... */

py::class_<MyClass, std::unique_ptr<MyClass, py::nodelete>>(m, "MyClass")
    .def(py::init<>())
```

3.5 □□□□

□□□□□A□B□□□□□A□□□□□□□B□

```cpp
py::class_<A>(m, "A")
    /// ... members ...

py::class_<B>(m, "B")
    .def(py::init<A>())
    /// ... members ...

m.def("func",
    [](const B &) { /* .... */ }
);
```

□□□func□□□□A□□□□□a□Pyhton□□□□□□`func(B(a))`□□C++□□□□□□□`func(a)`□□□□A□□□□□B□□□

□□□□□□B□□□□□A□□□□□□□□□□□□□□□□□□□□□□□□□Python□□□□□□□□□□□□□

```
py::implicitly_convertible<A, B>();
```

3.6 □□□□□

□□□□□□□□ `Vector2` □□□□□□□□□□□□□□□□□□□□□□

```cpp
class Vector2 {
public:
    Vector2(float x, float y) : x(x), y(y) { }

    Vector2 operator+(const Vector2 &v) const { return Vector2(x + v.x, y + v.y); }
    Vector2 operator*(float value) const { return Vector2(x * value, y * value); }
    Vector2& operator+=(const Vector2 &v) { x += v.x; y += v.y; return *this; }
    Vector2& operator*=(float v) { x *= v; y *= v; return *this; }

    friend Vector2 operator*(float f, const Vector2 &v) {
        return Vector2(f * v.x, f * v.y);
    }

    std::string toString() const {
        return "[" + std::to_string(x) + ", " + std::to_string(y) + "]";
    }
private:
    float x, y;
};
```

□□□□□□□□□□

```
#include <pybind11/operators.h>

PYBIND11_MODULE(example, m) {
    py::class_<Vector2>(m, "Vector2")
        .def(py::init<float, float>())
        .def(py::self + py::self)
        .def(py::self += py::self)
        .def(py::self *= float())
        .def(float() * py::self)
        .def(py::self * float())
        .def(-py::self)
        .def("__repr__", &Vector2::toString);
}
```

`.def(py::self * float())`□□□□□□□□□□

```
.def("__mul__", [](const Vector2 &a, float b) {
    return a * b;
}, py::is_operator())
```

3.7 □□□□□

Python□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□`copy`□□□□□□□□□□□

□Python3□□□pickle□□□□□□□□□□□□□□□□`__copy__`□`__deepcopy__`□□□□□□□□□□□Python2.7□□□□pybind11□□□cPickle□□□□□□□□□□□□□□□□□□□

□□□□□□□□□□□□□□□□□□□□□□□□□□□

```
py::class_<Copyable>(m, "Copyable")
    .def("__copy__",  [](const Copyable &self) {
        return Copyable(self);
    })
    .def("__deepcopy__", [](const Copyable &self, py::dict) {
        return Copyable(self);
    }, "memo"_a);
```

> Note: □□□□□□□□□□□□

3.8 □□□□

pybind11□□□□□□□□□□□□□□□□□□`class_`□□□□□□□

```
py::class_<MyType, BaseType1, BaseType2, BaseType3>(m, "MyType")
   ...
```

□□□□□□□□□□□□□□□□□□holder□□□pybind11□□□□□□□□□□□□□□□□□□□□□□□□□
□□

□□Python□□□□□□□□C++□□□□□□C++□□Python□□

□□□□□□□□□□□□□□□□□□□□□□□□□□□□pybind11□□□□□□□□□□□□□□□□□□□□□□
□□□□□□□□□□□□□□□`multiple_inheritance`□□□

```
py::class_<MyType, BaseType2>(m, "MyType", py::multiple_inheritance());
```

□□□□□□□□□□□□□□□□□□□

3.9 □□protected□□□□

□□□□□Python□□protected □□□□

```
class A {
protected:
    int foo() const { return 42; }
};

py::class_<A>(m, "A")
    .def("foo", &A::foo); // error: 'foo' is a protected member of 'A'
```

□□□□□□□□□□□□□□□□□□□□□□□□Python□□□□□protected □□□□□□□□□□□□□□□□

```
class A {
protected:
    int foo() const { return 42; }
};

class Publicist : public A { // helper type for exposing protected
functions
public:
    using A::foo; // inherited with different access modifier
};

py::class_<A>(m, "A") // bind the primary class
    .def("foo", &Publicist::foo); // expose protected methods via the
publicist
```

□□ `&Publicist::foo` □ `&A::foo` □□□□□□□□□□□□□□□□□□□□□□□□□□□□ `Publicist` □□□□□□□□□□□□□□□ `public` □

□□□□□□□Python□□□□ `protected` □□□□□□publicist pattern□□□□□trampoline□□□□

```cpp
class A {
public:
    virtual ~A() = default;

protected:
    virtual int foo() const { return 42; }
};

class Trampoline : public A {
public:
    int foo() const override { PYBIND11_OVERRIDE(int, A, foo, ); }
};

class Publicist : public A {
public:
    using A::foo;
};

py::class_<A, Trampoline>(m, "A") // <-- `Trampoline` here
    .def("foo", &Publicist::foo); // <-- `Publicist` here, not
`Trampoline`!
```

3.10 □□final□

□C++11□□□□□□□□□ `findal` □□□□□□□□□□□□□□□ `py::is_final` □□□□□□□□□□□□□□Python□□□□□□□□□C++□□□□□□□□□final□

```cpp
class IsFinal final {};

py::class_<IsFinal>(m, "IsFinal", py::is_final());
```

□Python□□□□□□□□□□□□□□□□□

```python
class PyFinalChild(IsFinal):
    pass

TypeError: type 'IsFinal' is not an acceptable base type
```

4. □□□□

4.1 C++□□□□□Python□□□□□

□Python□□pybind11□□C++□□□□pybind11□□□C++□□□□□□□□□□□Python□□□□□□□□Python□□□□□□□□□□

pybind11□□□`std::exception`□□□□□□□□□□□□□Python□□□□□□□□□□□□□□Python□□□□□□□□Python C API□□□□□□□□□□□C++□□□□□□□□□□□□□□pybind11□□□□□□□Python□□□

Exception thrown by C++	Translated to Python exception type
`std::exception`	`RuntimeError`
`std::bad_alloc`	`MemoryError`
`std::domain_error`	`ValueError`
`std::invalid_argument`	`ValueError`
`std::length_error`	`ValueError`
`std::out_of_range`	`IndexError`
`std::range_error`	`ValueError`
`std::overflow_error`	`OverflowError`
`pybind11::stop_iteration`	`StopIteration` (used to implement custom iterators)
`pybind11::index_error`	`IndexError` (used to indicate out of bounds access in `__getitem__`, `__setitem__`, etc.)
`pybind11::key_error`	`KeyError` (used to indicate out of bounds access in `__getitem__`, `__setitem__` in dict-like objects, etc.)
`pybind11::value_error`	`ValueError` (used to indicate wrong value passed in `container.remove(...)`)
`pybind11::type_error`	`TypeError`

Exception thrown by C++	Translated to Python exception type
`pybind11::buffer_error`	`BufferError`
`pybind11::import_error`	`ImportError`
`pybind11::attribute_error`	`AttributeError`
Any other exception	`RuntimeError`

□□□□□□□□□□□□□□□□□□□□□Python□□□□Python□□□□□□□□
`pybind11::error_already_set`□

□□□□□□□□□□□□□□□□□□Python□□□□`handle::call()`□□□`cast_error`□□□

4.2 □□□□□□□□

□□□□□□□□□□□□□□□□pybind11□□□□□□□□□□□□□□□□□□□□pybind11 class□□□□□□□□□□□□ □□□global□□□□□□□□C++□□□`what()`□□□C++□Python□□□□□□□□□□□□□□□□

```
py::register_exception<CppExp>(module, "PyExp");
```

□□□□□□□□□□□□□□□□□PyExp□Python□□□□□□□CppExp□□□□□□□□PyExp□□□

□□□□□□□□□□□□□□□□□□

```
py::register_local_exception<CppExp>(module, "PyExp");
```

□□□□□□□□handle□□□□□□□□□□

```
py::register_exception<CppExp>(module, "PyExp", PyExc_RuntimeError);
py::register_local_exception<CppExp>(module, "PyExp", PyExc_RuntimeError);
```

□□□PyExp□□□□□□PyExp□RuntimeError□

Python□□□□□□□□□□□Python□□Standard Exceptions□□□□□□□`PyExc_Exception`□

`py::register_exception_translator(translator)` □
`py::register_local_exception_translator(translator)` □□□□□□□□□□□□□□□□□□□□□
□□□□□□□□□□□□□□□□□□□□□□`void(std::exception_ptr)`□

9.3 □C++□□□Python□□

□C++□□Python□□□□□□□□□□□□Python□□□□□Python□□□□□□pybind11□□Python□□□□□ `pybind11::error_already_set` □□□□□□□□□□□□C++□□□□□□□□□Python□□□ `error_already_set` □□□Python□□□□Python□□□□C++□□□□□

Exception raised in Python	Thrown as C++ exception type
Any Python `Exception`	`pybind11::error_already_set`

□□□□□

```
try {
    // open("missing.txt", "r")
    auto file = py::module_::import("io").attr("open")("missing.txt",
"r");
    auto text = file.attr("read")();
    file.attr("close")();
} catch (py::error_already_set &e) {
    if (e.matches(PyExc_FileNotFoundError)) {
        py::print("missing.txt not found");
    } else if (e.matches(PyExc_PermissionError)) {
        py::print("missing.txt found but not accessible");
    } else {
        throw;
    }
}
```

□□□□□□□□C++□Python□□□□Python□□□□□□□□□□□□□`error_already_set`.

```cpp
try {
    py::eval("raise ValueError('The Ring')");
} catch (py::value_error &boromir) {
    // Boromir never gets the ring
    assert(false);
} catch (py::error_already_set &frodo) {
    // Frodo gets the ring
    py::print("I will take the ring");
}

try {
    // py::value_error is a request for pybind11 to raise a Python
exception
    throw py::value_error("The ball");
} catch (py::error_already_set &cat) {
    // cat won't catch the ball since
    // py::value_error is not a Python exception
    assert(false);
} catch (py::value_error &dog) {
    // dog will catch the ball
    py::print("Run Spot run");
    throw;  // Throw it again (pybind11 will raise ValueError)
}
```

9.4 □□Python C API□□□

□□□□□□pybind11 wrappers□□□□□□Python C API□□□□□□□□□□□□Python C API□□□□□□□□□□□□□□□□□□□□□pybind11□□□□□□□□

□□□Python C API□□□□Python□□□□□□□□□□`throw py::error_already_set();`□□□□pybind11□□□□□□□□□Python□□□□□□□□□□□□□□□□□□□□`PyErr_SetString`□

```cpp
PyErr_SetString(PyExc_TypeError, "C API type error demo");
throw py::error_already_set();

// But it would be easier to simply...
throw py::type_error("pybind11 wrapper type error");
```

□□□□□`PyErr_Clear`□□□□□

□□Python□□□□□□□□□□□□□Python/pybind11□□□□□□□□□

9.5 □□unraiseable□□

□□Python□□□C++□□□□□□□□□□`noexcept(true)`□□c++□□□□□std::terminate()□□□□□□□□C++□□□□□□□Python□□□□□□□□□□□□□□□□□□`error_already_set`□□□□□□□□□`error_already_set::discard_as_unraisable()`□□□Python□□□

□□□□□□`__del__`□□□□□Python□□□□□□□□□□Python□□unraisable□□□□□□□□Python 3.8+□□□□□system hook□□□□auditing event□□□

□□noexcept□□□□□□try-catch□□□□□□`error_already_set`□□□□□□□□□□□□□pybind11□□□Python□□□□□□□Python□□□□□pybind11□□□□□□C++□□□noexcept□□□□□□□□□□□□□□□□□□□□□Python□□□□□□□`discard_as_unraisable`□□□□□□

```
void nonthrowing_func() noexcept(true) {
    try {
        // ...
    } catch (py::error_already_set &eas) {
        // Discard the Python error using Python APIs, using the C++ magic
        // variable __func__. Python already knows the type and value and
of the
        // exception object.
        eas.discard_as_unraisable(__func__);
    } catch (const std::exception &e) {
        // Log and discard C++ exceptions.
        third_party::log(e);
    }
}
```

5. □□□□

6. python C++□□

7. □□

7.1 □□□□□□□□

pybind11□□□□□□□□`PYBIND11_DECLARE_HOLDER_TYPE()`□`PYBIND11_OVERRIDE_*`□□□□□□□□□□□□□□(□□□□□□□□□□□)□□□□□□□□□□□□□□□□

```
PYBIND11_OVERRIDE(MyReturnType<T1, T2>, Class<T3, T4>, func)
```

□□□□□□□□□5□□□□□□□□□□□3□□□□□□□□□□□□□□□□□□□□□□□□`PYBIND11_TYPE`□□□□□

```
// Version 1: using a type alias
using ReturnType = MyReturnType<T1, T2>;
using ClassType = Class<T3, T4>;
PYBIND11_OVERRIDE(ReturnType, ClassType, func);

// Version 2: using the PYBIND11_TYPE macro:
PYBIND11_OVERRIDE(PYBIND11_TYPE(MyReturnType<T1, T2>),
                  PYBIND11_TYPE(Class<T3, T4>), func)
```

`PYBIND11_MAKE_OPAQUE`□□□□□□□□□□

7.2 □□□□□□□GIL□

□Python□□□C++□□□□□□□□GIL□`gil_scoped_release`□`gil_scoped_acquire`□□□□□□□□□□□□□GIL□□□□□□□□C++□□□□□Python□□□□□□□□□□□□

```
class PyAnimal : public Animal {
public:
    /* Inherit the constructors */
    using Animal::Animal;

    /* Trampoline (need one for each virtual function) */
    std::string go(int n_times) {
        /* Acquire GIL before calling Python code */
        py::gil_scoped_acquire acquire;

        PYBIND11_OVERRIDE_PURE(
            std::string, /* Return type */
            Animal,      /* Parent class */
            go,          /* Name of function */
            n_times      /* Argument(s) */
        );
    }
};

PYBIND11_MODULE(example, m) {
    py::class_<Animal, PyAnimal> animal(m, "Animal");
    animal
        .def(py::init<>())
        .def("go", &Animal::go);

    py::class_<Dog>(m, "Dog", animal)
        .def(py::init<>());

    m.def("call_go", [](Animal *animal) -> std::string {
        /* Release GIL before calling into (potentially long-running) C++
code */
        py::gil_scoped_release release;
        return call_go(animal);
    });
}
```

□□□□□□□□`call_guard`□□□□□□□`call_go`□□□□□

```
m.def("call_go", &call_go, py::call_guard<py::gil_scoped_release>());
```

mdbook-mermaid

[illegible]Mermaid[illegible]

[illegible] `Mermaid` [illegible],[illegible],[illegible],[illegible],[illegible],[illegible] `markdown` [illegible].

[illegible]:

- [illegible]`Mermaid`[illegible];
- [illegible]`Mermaid`[illegible];
- [illegible] `Gitbook` [illegible].

[illegible]Mermaid[illegible]

[illegible]

```
- [illegible]
- [illegible]
- [illegible]
```

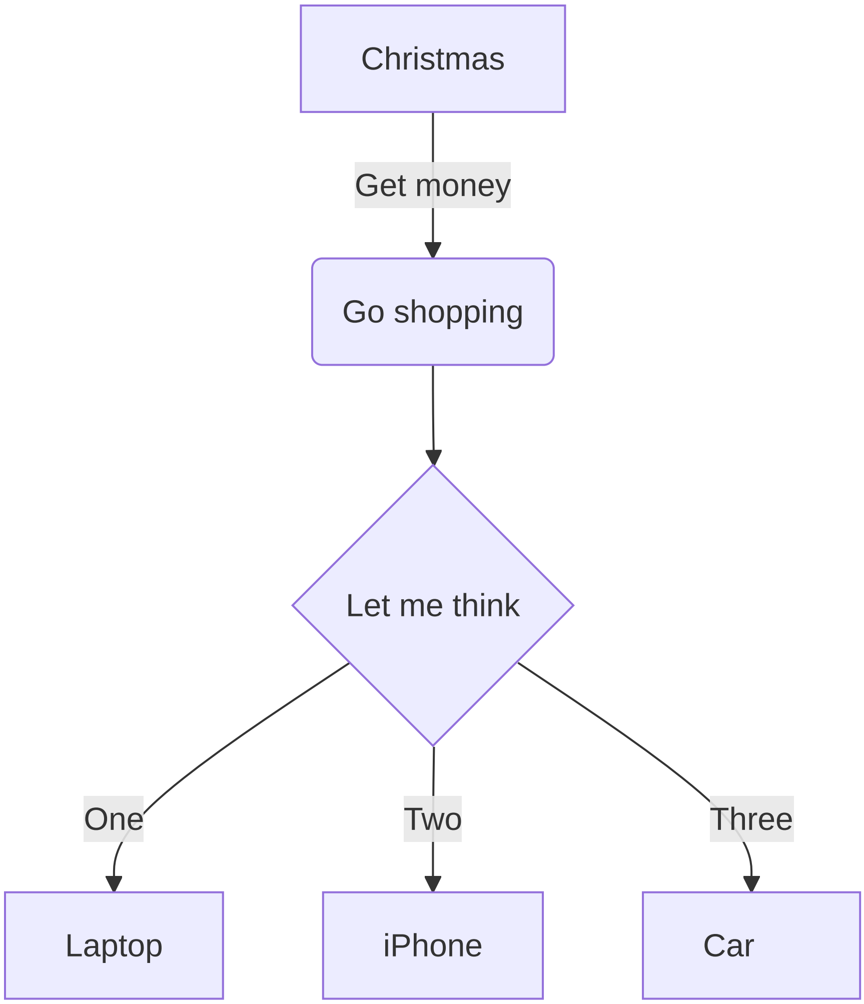
[illegible],[illegible].

`Mermaid`[illegible] `Javascript` [illegible].[illegible] `markdown` [illegible],[illegible].

[illegible]

```
graph TD
  A[Christmas] -->|Get money| B(Go shopping)
  B --> C{Let me think}
  C -->|One| D[Laptop]
  C -->|Two| E[iPhone]
  C -->|Three| F[fa:fa-car Car]
```

□□



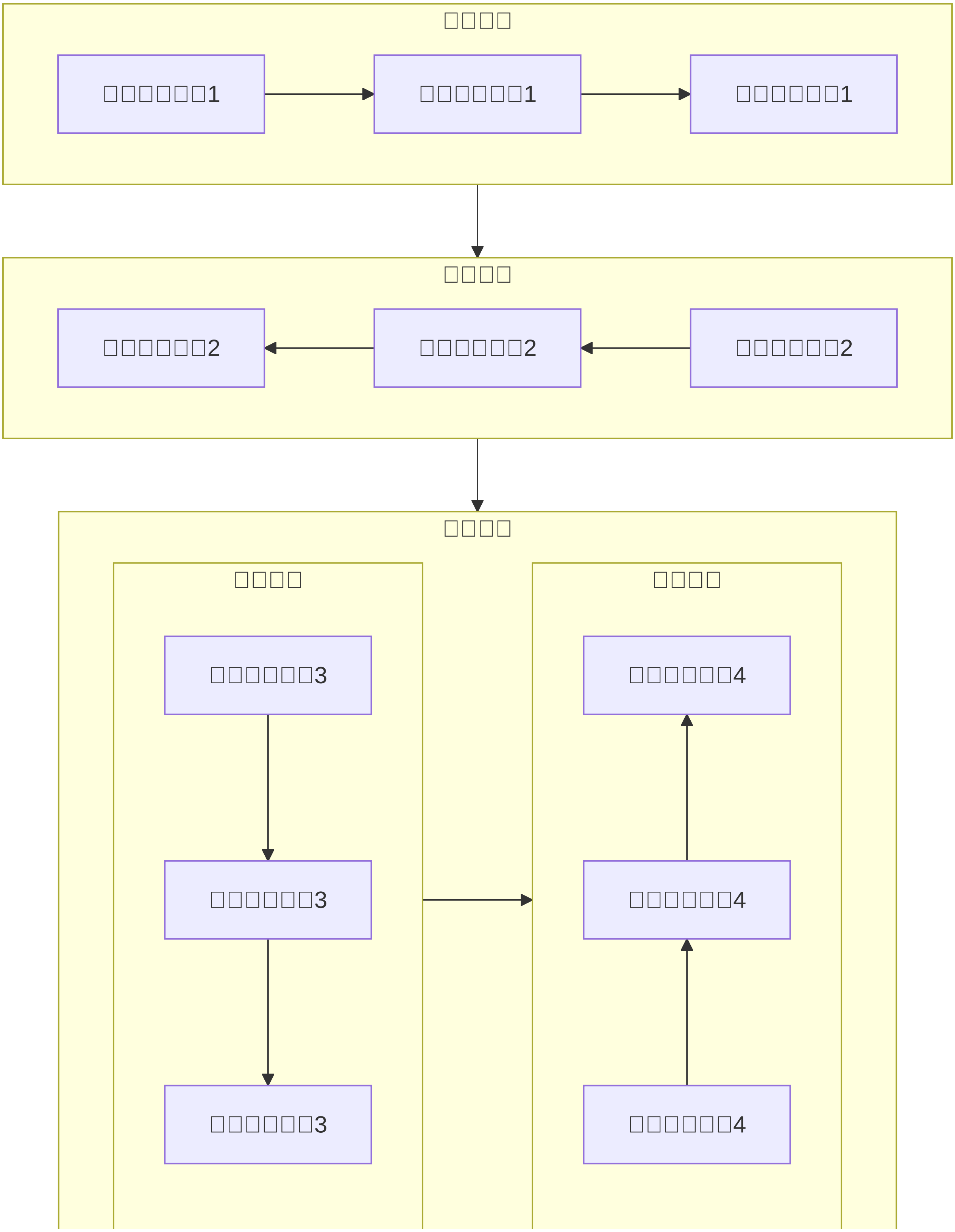
- □□□□: https://github.com/mermaid-js/mermaid
- □□□□: https://mermaidjs.github.io/mermaid-live-editor/
- □□□□: https://mermaid-js.github.io/mermaid/#/flowchart
- □□□□: https://mermaid.nodejs.cn/syntax/flowchart.html

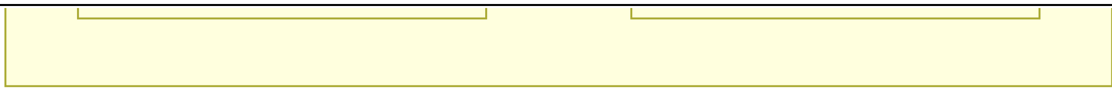
Mermaid□□□□□□□

□□□□

□□□

```
+ TB
+ BT
+ LR
+ RL
```





□□□□□□□,□□□□□□□□□,□□□□□□□: `top`(□), `bottom`(□), `left`(□)□ `right`(□).□□□□□: `TB` (□□□□), `BT` (□□□□), `LR` (□□□□)□ `RL` (□□□□)□□.

> □□: □□□□□□□□□□□□□,□□□□□□□□□□□□.

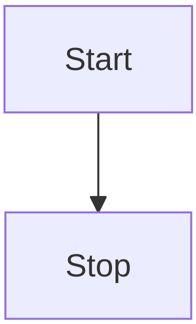
- TB

> □□□□: from **T**op to **B**ottom

□□

```
graph TB
    Start --> Stop
```

□□



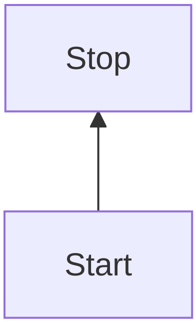
- BT

> □□□□: from **B**ottom to **T**op

□□

```
graph BT
    Start --> Stop
```

□□



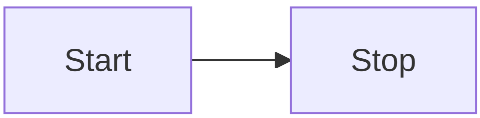
- LR

> □□□□: from **L**eft to **R**ight

□□

```
graph LR
    Start --> Stop
```

□□



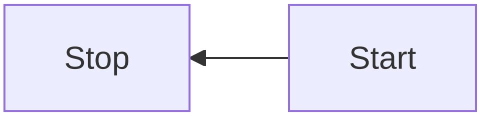
- RL

> □□□□: from **R**ight to **L**eft

□□

```
graph RL
    Start --> Stop
```

□□



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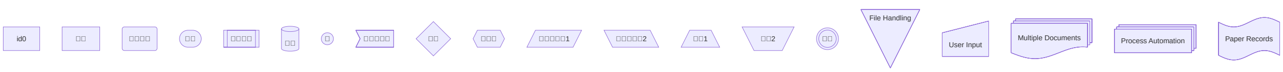
□□□

```
- □□□□
    + [□□]
        - [[□□□□]]
        - [(□□)]
        - [{□□□□}]
        - [/□□□□□/]
        - [\□□□□□\]
        - [/□□\]
        - [\□□/]
    + (□□□□)
        - ((□□))
        - ([□□□])
        - ({□□□□})
    + {□□}
        - {{□□□}}
        - {[□□□□]}
        - {(□□□□)}
    + >□□□□□]
```



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> □□: □□□□□□□□,□□□□□□.

□□□□□

□□□□□,□□□□□□□□□,□□□□□□□□□ `id` □□,□□□□□□□□□□□□□□□.

> `id` □□□□□□□□□□□□□□□□□□□□□□□.

□□

```
graph TD
    id
```

□□

□□□□□

□□□□□,□□□□□□,□□□□□□□□ `id` □□□□ `<node shape>` □□,□□□□□□□□□□□□□□□□□.

> `id` □□□□□□□□□,□□□□□□□□□□ `<node shape>` □□□□,□□ `id` □□□□□□□□□□.

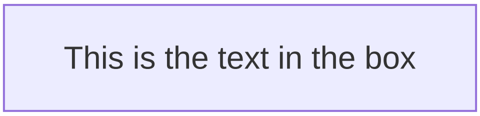
- □□

> □□□□: `[node description]` , `[]` □□□□□□□□□□□□□, `node description` □□□□□□□□.

□□

```
graph LR
    id1[This is the text in the box]
```

□□



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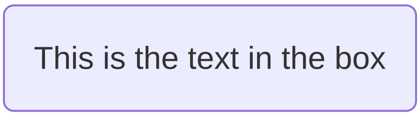
> □□□□: `(node description)` , `()` □□□□□□□□□□□□□□□, `node description` □□□□□□□□.

□□

```
graph LR
    id1(This is the text in the box)
```

□□



- □□□

> □□□□: `([node description])` , `()` □□□□□ `[]` □□□□□□□□□□□□□□□□□□□□,□□□□□□□

□, `node description` □□□□□□□□.

□□

```
graph LR
    id1([This is the text in the box])
```

□□

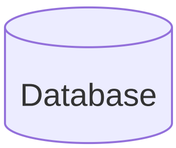
- □□

□□□□: `[(node description)]` , `[]` □□□□□ `()` □□□□□□□□□□□□, `node description` □□□□□□□□.

□□

```
graph LR
    id1[(Database)]
```

□□



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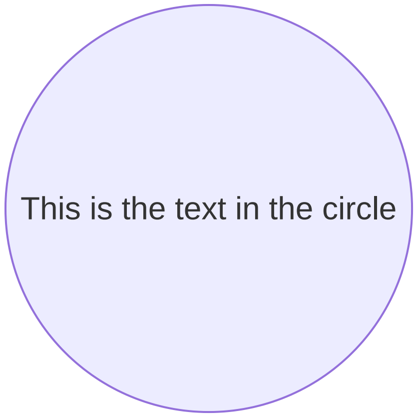
□□□□: `((node description))` , `()` □□□□□ `()` □□□□□□□□□□□□, `node description` □□□□□□□□.

□□

```
graph LR
    id1((This is the text in the circle))
```

□□



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> □□□□: `>node description]` ,□□□□□□□ `>` ,□□□□□□□ `]` □□□□□□□□□, `node description` □□□□□□□□.

□□

```
graph LR
    id1>This is the text in the box]
```

□□

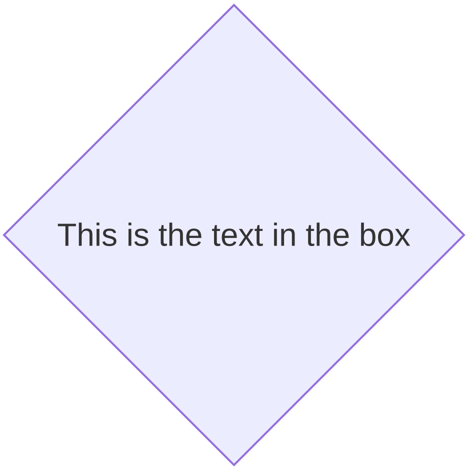
- □□

> □□□□: `{node description}` , `{}` □□□□□□□□□, `node description` □□□□□□□□.

□□

```
graph LR
    id1{This is the text in the box}
```

□□



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> □□□□: `{{node description}}` , `{}` □□□□□ `{}` □□□□□□□□□□, `node description` □□□□□□□□.

□□

```
graph LR
    id1\{\{This is the text in the box\}\}
```

> `Gitbook` □□□□□□□ `{}` □□□□□□,□□□□□□□□□□,□□□□□□□ `\` □□□□.

□□

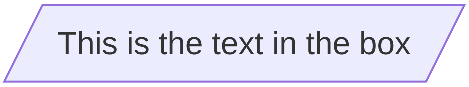
- □□□□□

> □□□□: `[/node description/]` , `[]` □□□□□ `//` □□□□□□□□□□□□□□□□, `node description` □□□□□□□□□.

□□

```
graph TD
    id1[/This is the text in the box/]
```

□□



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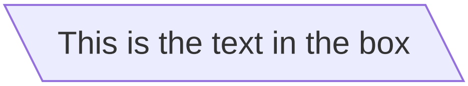
> □□□□: `[\node description\]` , `[]` □□□□□ `\\` □□□□□□□□□□□□□□□□, `node description` □□□□□□□□□.

□□

```
graph TD
    id1[\This is the text in the box\]
```

□□



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> □□□□: `[/node description\]` , `[]` □□□□□ `/\` □□□□□□□□□□□□□□□, `node description` □□□□□□□□.

□□

```
graph TD
    A[/Christmas\]
```

□□

- □□□□□

> □□□□: `[\node description/]` , `[]` □□□□□ `\/` □□□□□□□□□□□□□□□, `node description` □□□□□□□□.

□□

```
graph TD
    B[\Go shopping/]
```

□□

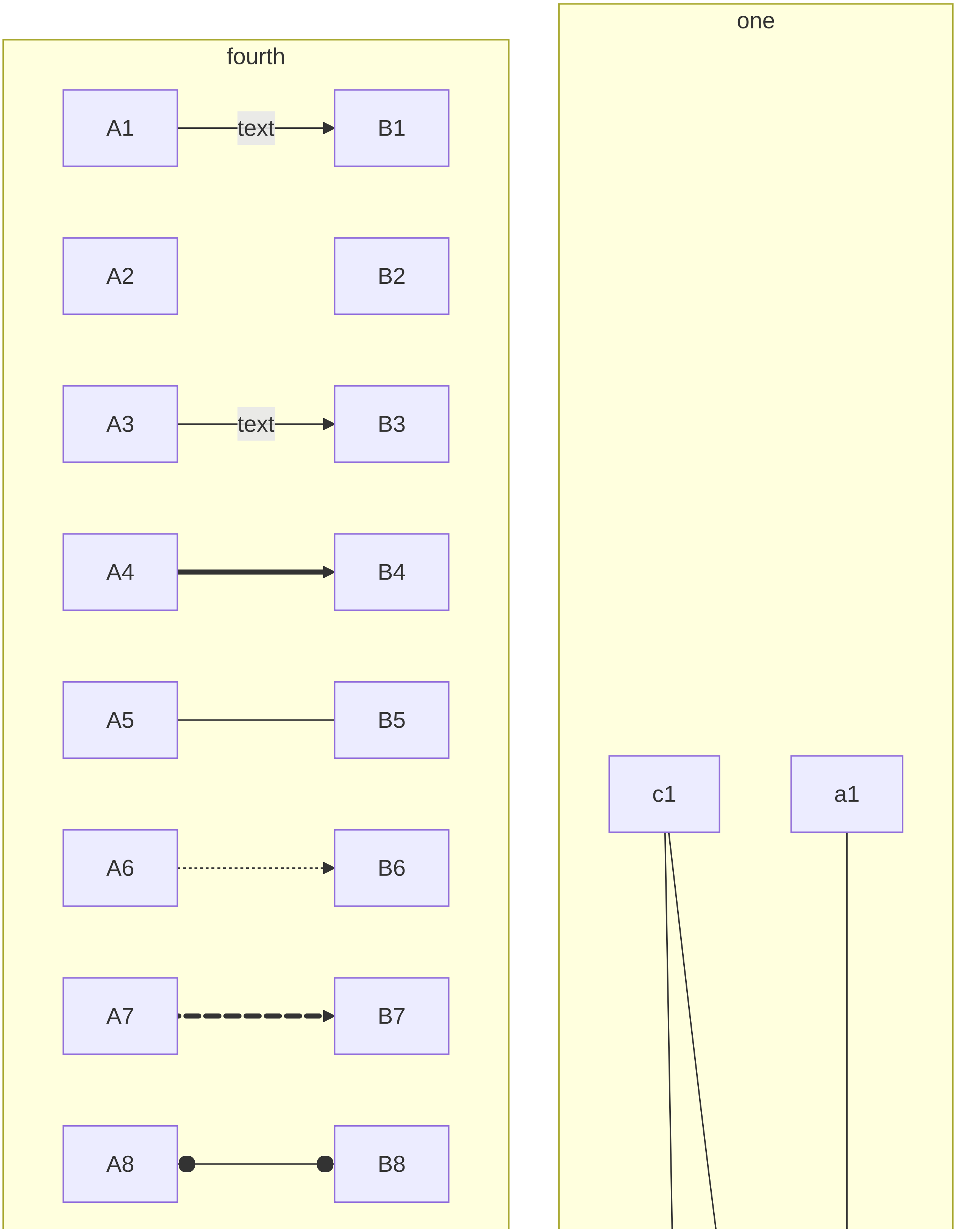

Go shopping

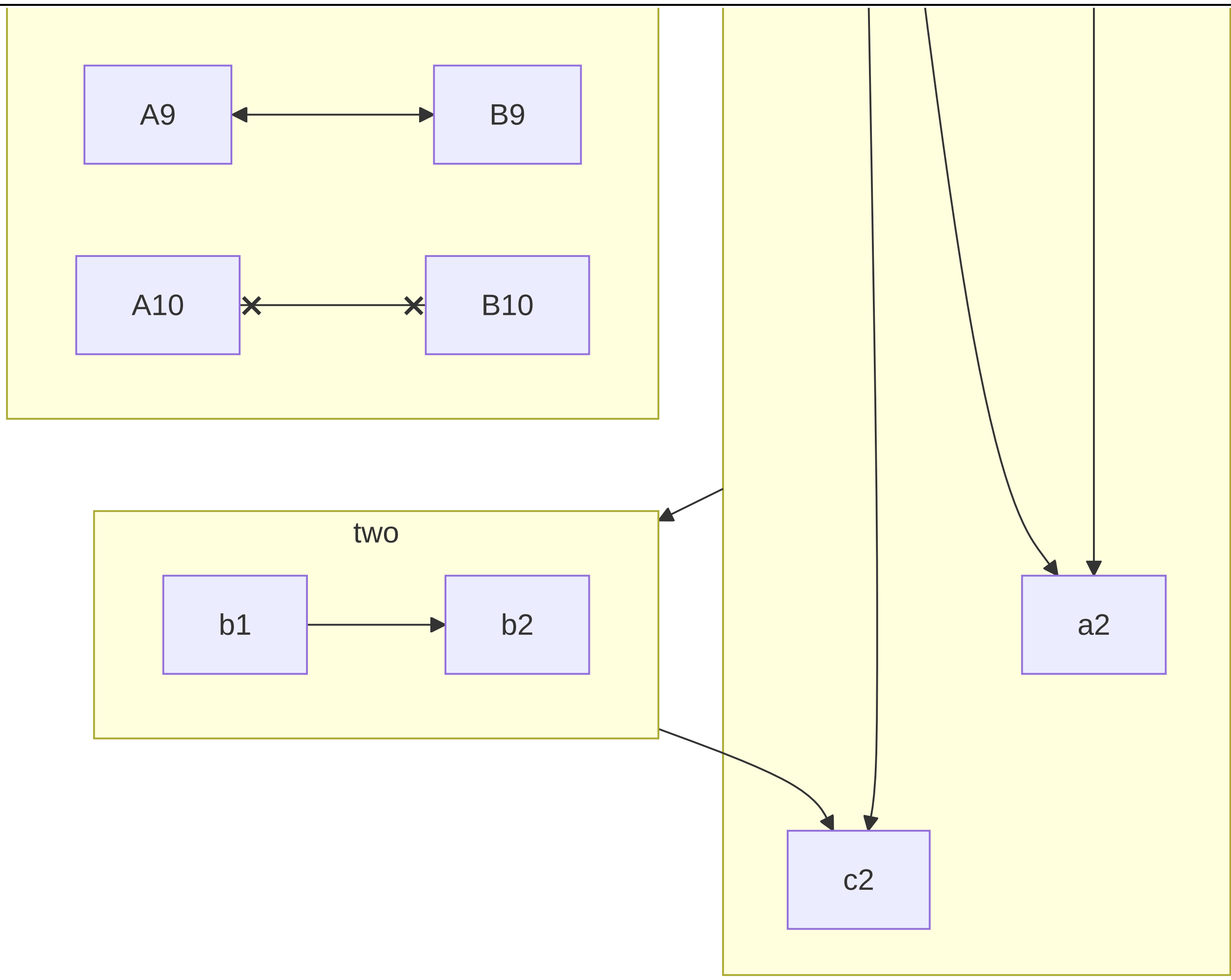
□□□

□□□

```
```

```
+ □□/□□
    - --
    - -.
+ □□□/□□□
    - >
    - -
+ □□□/□□□
    - □□
        + --□□□□
        + |□□□□|
    - □□
        + -.□□□□
        + |□□□□|
+ □□
    - ==
+ □□□□
    - -->
    - ---
    - -.->
    - -.-
    - □□□□□□□□
        + --□□□□-->
        + -->|□□□□|
    - □□□□□□□□
        + --□□□□---
        + ---|□□□□|
    - □□□□□□□□
        + -.□□□□-.->
        + -.->|□□□□|
    - □□□□□□□□
        + -.□□□□-.-
        + -.-|□□□□|
    - ==>
    - ===
    - □□□□□□□□□□(2)
        + ==□□□□==>
        + ==>|□□□□|
    - □□□□□□□□□□(2)
        + ==□□□□===
        + ===|□□□□|
```


fourth
A1
text
B1
A2
B2
A3
text
B3
A4
B4
A5
B5
A6
B6
A7
B7
A8
B8
one
c1
a1


A9
B9
A10
B10
two
b1
b2
a2
c2

□□□□□□□□,□□□□□□□□□□□□□□□□□□□□,□□□□□□□□□□□□□□□□,□□ `--` □□□□,□□□□□□□ `-.-` □□□□,□□□□□□□□□ `>` ,□□□□□□□□□□ `-`.

> □□: □□□□□□,□□□□□□□□,□□□□□□□□□□□□□□□□□□□□,□□□□□□□□□□□□□□.

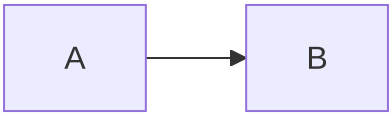
- □□□□□□□□

> □□□□: `-->` ,□□ `--` □□□□, `>` □□□□□.

□□

```
graph LR
    A-->B
```

□□



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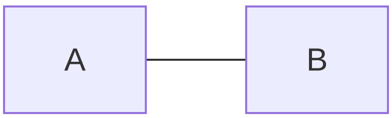
□□

```
graph LR
    A --- B
```

□□



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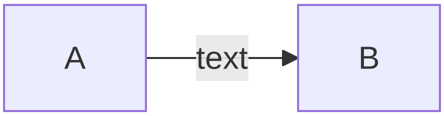
> □□□□: `--connection line description-->` ,□□□□□ `--` □□□□□□□□□,□□□ `-->` □□ □□□□□□.

□□

```
graph LR
    A-- text -->B
```

□□



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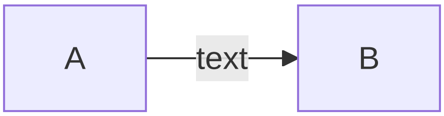
□□

```
graph LR
    A-->|text|B
```

□□



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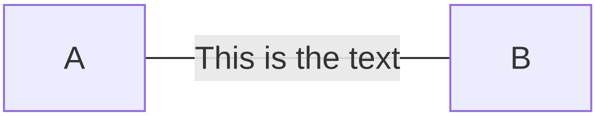
> □□□□: `--connection line description` ,□□□□□ `--` □□□□□□□□□,□□□ `---` □□□□□ □□□□.

□□

```
graph LR
    A-- This is the text ---B
```

□□

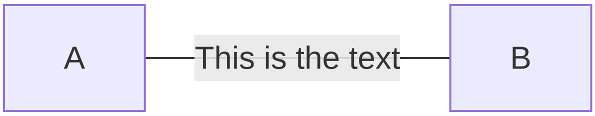


> □□□□: `|connection line description|` ,□□ `||` □□□□□□□□□□.

□□

```
graph LR
    A---|This is the text|B
```

□□



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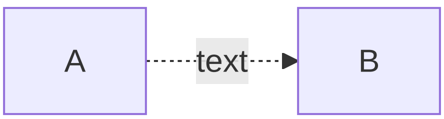
> □□□□: `-.connection line description.->` ,□□□□□ `-.` □□□□□□□□□,□□□ `.->` □□
> □□□□□□.

□□

```
graph LR
   A-. text .-> B
```

□□



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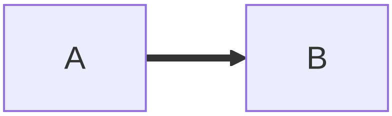
> □□□□: `==>` ,□□□□□□.

□□

```
graph LR
   A ==> B
```

□□



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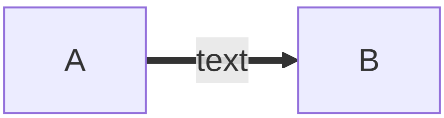
□□

```
graph LR
   A == text ==> B
```

□□



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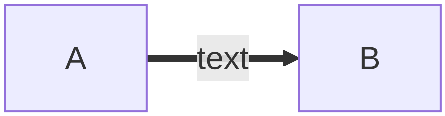
> □□□□: `|connection line description|` ,□□ `||` □□□□□□□□□□.

□□

```
graph LR
   A ==>|text| B
```

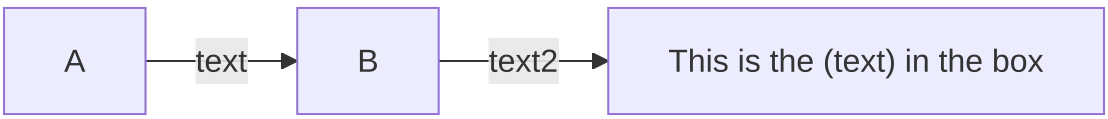
□□



□□□□

□□□

```
+ -->-->
+ &
+ ""
+ %%
+ subgraph
```



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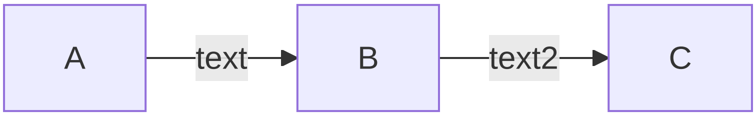
□□

□□□□□□□□, `A-->B-->C` □□□ `A-->B` □ `B-->C` □□.

```
graph LR
   A -- text --> B -- text2 --> C
```

□□



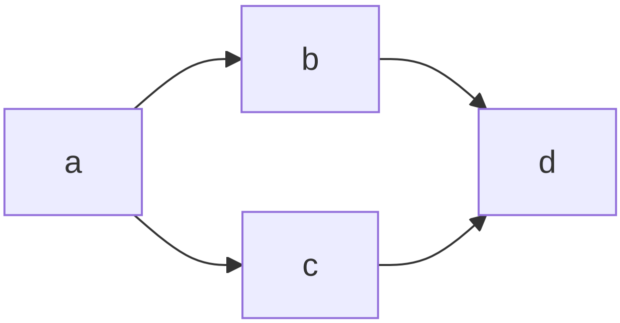
- □□□□□□□

□□□□□□□□,`A-->B & C` □□□ `A-->B` □ `A-->C` □□.

□□

```
graph LR
   a --> b & c--> d
```

□□



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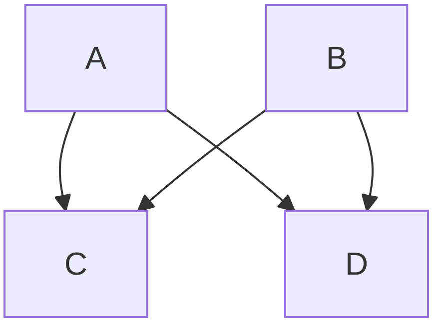
□□□□□□□□□□□□,`A & B --> C & D` □□□ `A-->C`,`A-->D`,`B-->C` □ `B-->D` □□□□□□.

□□

```
graph TB
    A & B--> C & D
```

□□



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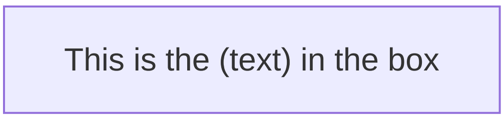
□□□□□□□□□□□□□□□□□□ `""` □□□□,□□□ `[]` □ `()` □□ `{}` □□□□□□□.

□□

```
graph LR
    id1["This is the (text) in the box"]
```

□□



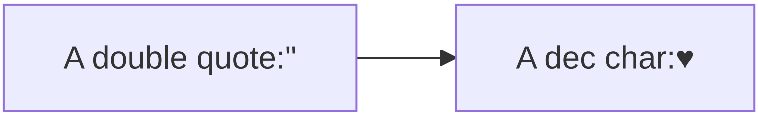
- □□□□□□□□□

□□ `Html` □□□□

□□

```
graph LR
    A["A double quote:#quot;"] -->B["A dec char:#9829;"]
```

□□



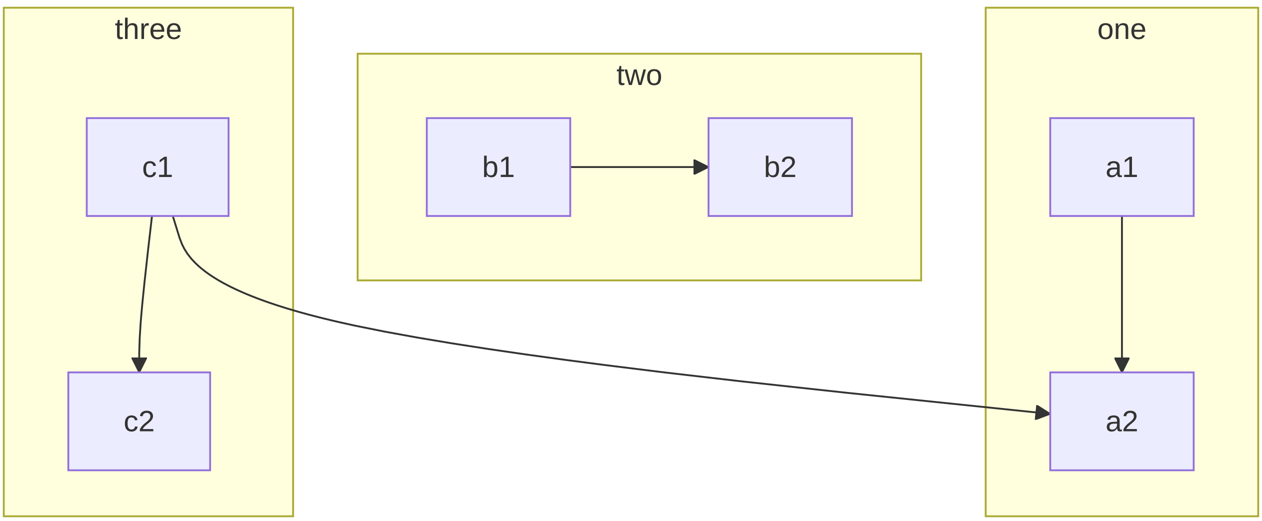
- □□□□□□

□□

```
subgraph title
    graph definition
end
```

□□

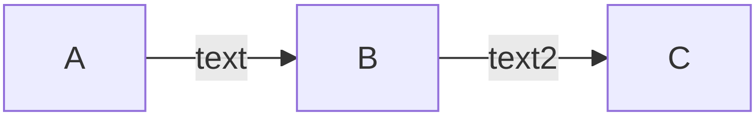
```
graph TB
    c1-->a2
    subgraph one
    a1-->a2
    end
    subgraph two
    b1-->b2
    end
    subgraph three
    c1-->c2
    end
```



- □□□□

□□□ `%%` □□□□□□□□.

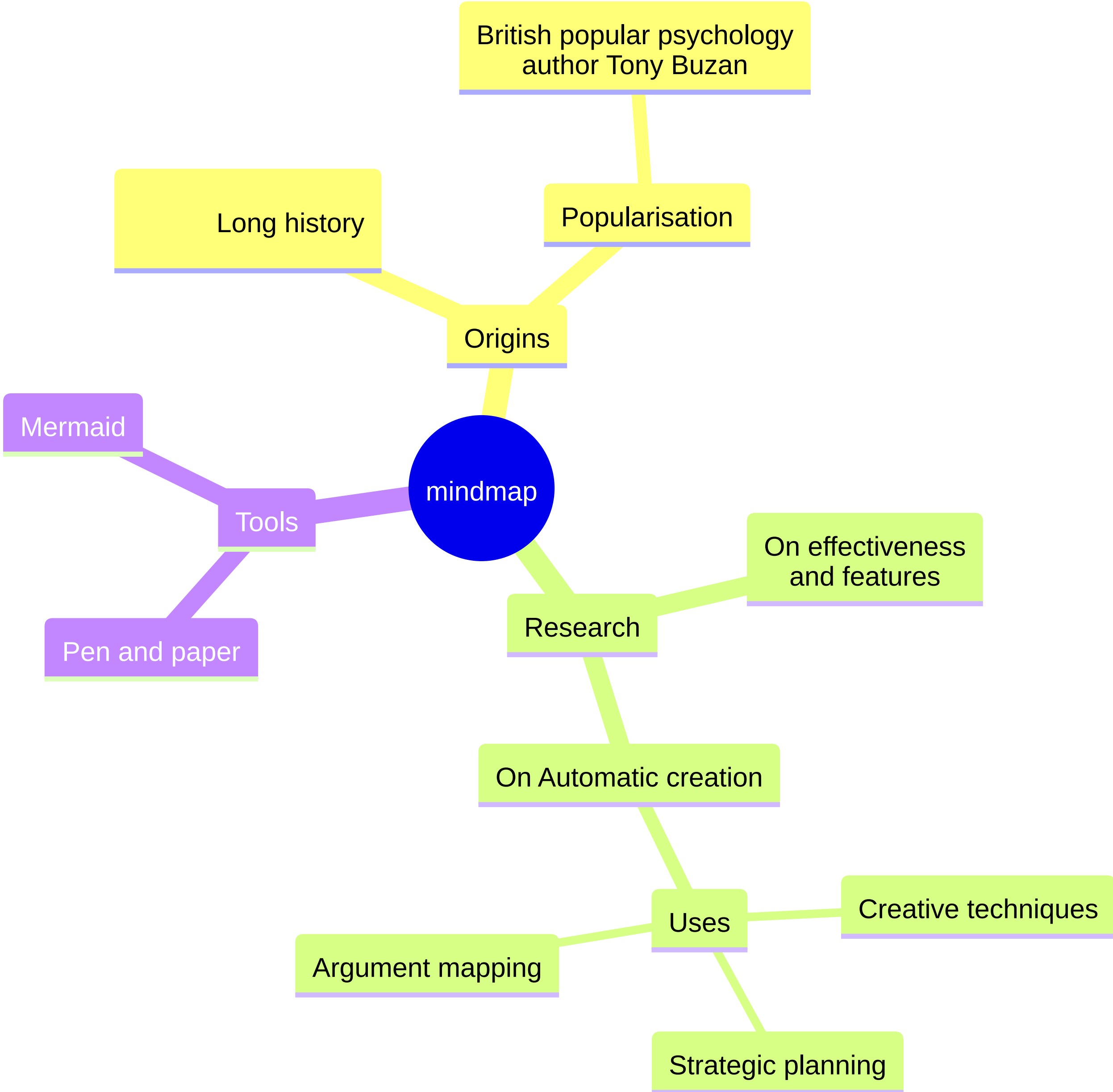
```
graph LR
%% this is a comment A -- text --> B{node}
   A -- text --> B -- text2 --> C
```



□□□□□□□□□□□

□□□

```
- □□□□□□
- □□□□□
- □□□□
```



`Mermaid` □□□□□□□□□□,□□□ `Markdown` □□□□□□□,□□□□□□□□□□□,□□□□□□□□□□□□□□□□□.

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□□	□□	□□
□□	graph	`graph` □□□□□
□□	subgraph	`subgraph` □□□□□□
□	top	`TB` □ `BT`,□□□□□□□□□□
□	bottom	`BT` □ `TB`,□□□□□□□□□□
□	left	`LR` □ `RL`,□□□□□□□□□□
□	right	`RL` □ `LR`,□□□□□□□□□□

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> □□□□□□□□□□□,□□□□□□□□□□□,□□□□□□□□□□□,□□□□□□□□□.

- □□□□

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`[]`	□□	□□□□	□□
`()`	□□□□	□□□□	□□
`{}`	□□	□□□□	□□
`<>`	□□	□□□□	□□□
`--`	□□	□□□□□	□□
`-.`	□□	□□□□□	□□
`==`	□□□□	□□□□□	□□
`=:`	□□□□	□□□□□	□□□

[illegible]	[illegible]	[illegible]	[illegible]
`>`	[illegible]	[illegible]	[illegible]
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`--`	[illegible]	[illegible]	[illegible]
`-.`	[illegible]	[illegible]	[illegible]
`==`	[illegible]	[illegible]	[illegible]
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- [illegible]

[illegible]	[illegible]	[illegible]	[illegible]
`[[]]`	[illegible]	[illegible]	[illegible]
`[()]`	[illegible]	[illegible]	[illegible]
`[{}]`	[illegible]	[illegible]	[illegible]
`(())`	[illegible]	[illegible]	[illegible]
`([])`	[illegible]	[illegible]	[illegible]
`({})`	[illegible]	[illegible]	[illegible]
[illegible]	[illegible]	[illegible]	[illegible]
`{[]}`	[illegible]	[illegible]	[illegible]
`{()}`	[illegible]	[illegible]	[illegible]
`-->`	[illegible]	[illegible]	[illegible]
`---`	[illegible]	[illegible]	[illegible]
`-.>`	[illegible]	[illegible]	[illegible]

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`-.->`	□□□□□	□□□□□	□□
`.->`	□□□□□	□□□□□	□□
`-.-`	□□□□□	□□□□□	□□
`.-`	□□□□□	□□□□□	□□
`==>`	□□□□□□□	□□□□□	□□
`===`	□□□□□□□	□□□□□	□□
`=:>`	□□□□□□□	□□□□□	□□□
`=:=>`	□□□□□□□	□□□□□	□□□
`=:=`	□□□□□□□	□□□□□	□□□
`:=`	□□□□□□□	□□□□□	□□□
`□□□`	□□□□□□□□□	□□□□□□□	□□
`--connection line description-->`	□□□□□□□□□□□□□□	□□□□□□□	□□
`-.connection line description-.->`	□□□□□□□□□□□□□□	□□□□□□□	□□
`--connection line description---`	□□□□□□□□□□□□□□	□□□□□□□	□□
`-.connection line description-.-`	□□□□□□□□□□□□□□	□□□□□□□	□□
`==connection line description==>`	□□□□□□□□□□□□□□□□	□□□□□□□	□□
`=:connection line description=:=>`	□□□□□□□□□□□□□□□□	□□□□□□□	□□□
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`=:connection line description=:=`	□□□□□□□□□□□□□□□□	□□□□□□□	□□□

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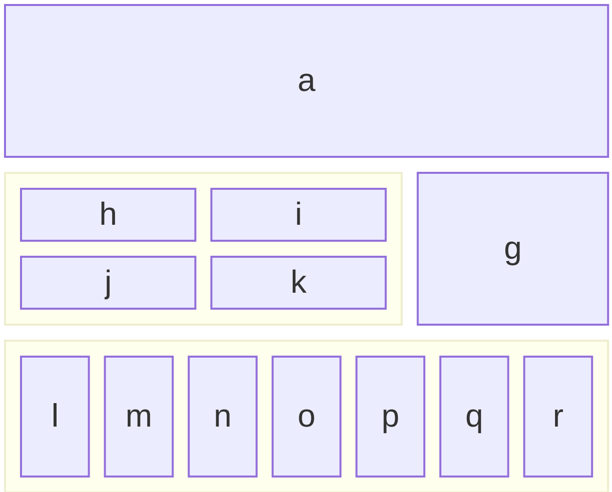
□□□□□□□□□□□□□□□□□,□□□□□ `JS` □ `CSS` □□□□□□□□□□□□□□□□,□□□□□□□□□.

> □□□□: https://mermaid-js.github.io/mermaid/#/flowchart?id=styling-and-classes

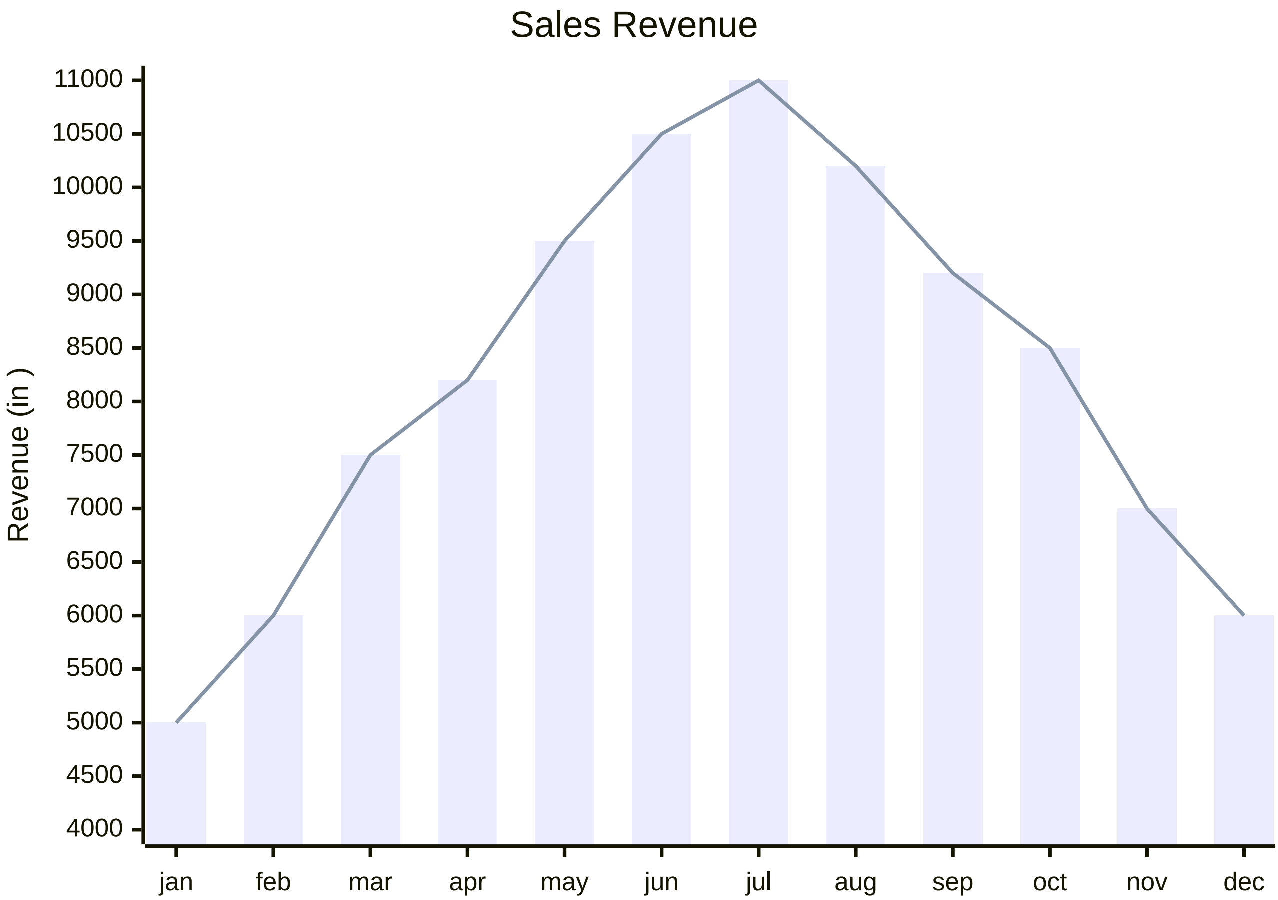
- □□□□ Interaction : https://mermaid-js.github.io/mermaid/#/flowchart?id=interaction
- □□□□ Styling and classes : https://mermaid-js.github.io/mermaid/#/flowchart?id=interaction
- □□□□ Basic support for fontawesome: https://mermaid-js.github.io/mermaid/#/flowchart?id=basic-support-for-fontawesome
- □□□□ https://mermaid-js.github.io/mermaid/#/flowchart?id=graph-declarations-with-spaces-between-vertices-and-link-and-without-semicolon

□□□

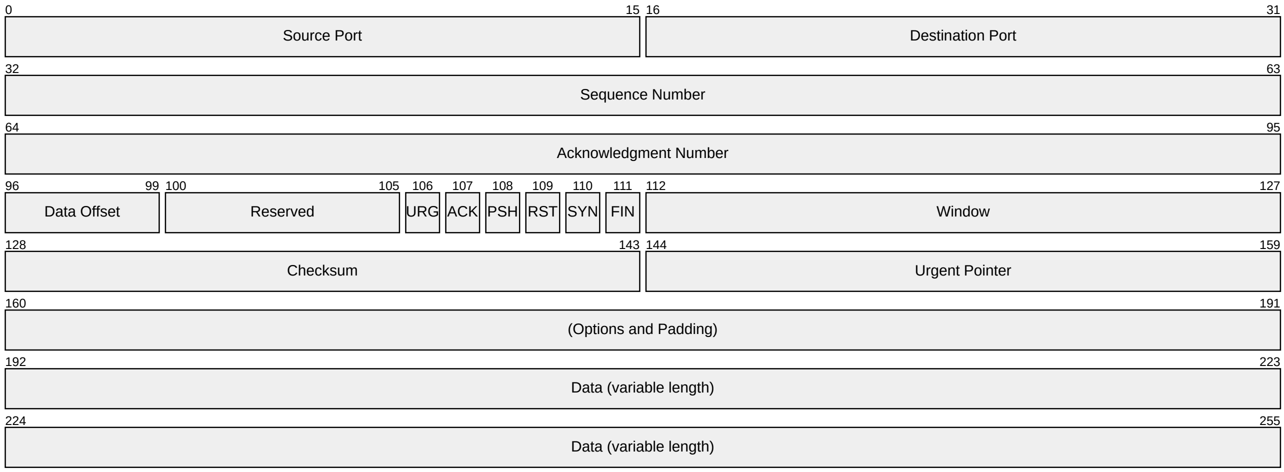
□□


a
h
i
g
j
k
l
m
n
o
p
q
r

XY图


Sales Revenue
Revenue (in)
11000
10500
10000
9500
9000
8500
8000
7500
7000
6500
6000
5500
5000
4500
4000
jan
feb
mar
apr
may
jun
jul
aug
sep
oct
nov
dec

□□□


0
15
16
31
Source Port
Destination Port
32
63
Sequence Number
64
95
Acknowledgment Number
96
99
100
105
106
107
108
109
110
111
112
127
Data Offset
Reserved
URG
ACK
PSH
RST
SYN
FIN
Window
128
143
144
159
Checksum
Urgent Pointer
160
191
(Options and Padding)
192
223
Data (variable length)
224
255
Data (variable length)

TCP Packet

mdbook-admonish

The following admonishments are implemented by the mdbook-admonish plugin and are automatically themed to match Catppuccin.

Directives

All supported directives are listed below.

`note`

Note

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

`abstract`, `summary`, `tldr`

Abstract

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

`info`, `todo`

Info

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

`tip`, `hint`, `important`

Tip

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

`success`, `check`, `done`

Success

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

`question`, `help`, `faq`

Question

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

`warning`, `caution`, `attention`

Warning

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

`failure`, `fail`, `missing`

Failure

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

`danger`, `error`

Danger

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

`bug`

Bug

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

`example`

Example

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

`quote`, `cite`

Quote

Rust is a multi-paradigm, general-purpose programming language designed for performance and safety, especially safe concurrency.

Bienvenue sur notre site de développement 3D !

Bienvenue sur notre site dédié au développement 3D. Ici, vous trouverez des ressources, des tutoriels et des informations utiles pour vous lancer dans le monde passionnant de la 3D.

Info

A beautifully styled message.

Un example

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla et euismod nulla. Curabitur feugiat, tortor non consequat finibus, justo purus auctor massa, nec semper lorem quam in massa.

Une note

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla et euismod nulla. Curabitur feugiat, tortor non consequat finibus, justo purus auctor massa, nec semper lorem quam in massa.

Un warning

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla et euismod nulla. Curabitur feugiat, tortor non consequat finibus, justo purus auctor massa, nec semper lorem quam in massa.

Collapsing note

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Le javascript c’est yolo préférez Typescript

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla et euismod nulla. Curabitur feugiat, tortor non consequat finibus, justo purus auctor massa, nec semper lorem quam in massa.

Referencing* and *dereferencing

The opposite of *referencing* by using `&` is *dereferencing*, which is accomplished with the dereference operator, `*`.

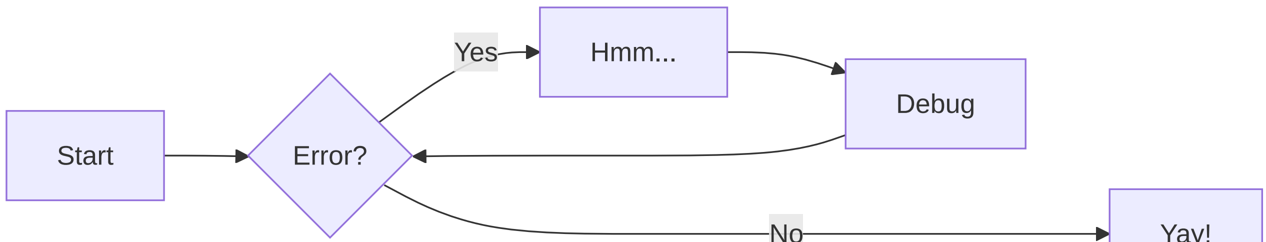
Bug

This syntax won't work in Python 3:

```python
print "Hello, world!"
```

À propos de nous

Nous sommes une équipe passionnée par la 3D et nous avons pour mission de partager nos connaissances avec la communauté. Vous trouverez ici des articles, des exemples de code et des démonstrations pour vous aider à démarrer votre voyage dans le développement 3D.



Pour commencer

Si vous êtes nouveau dans le domaine de la 3D, ne vous inquiétez pas ! Notre page “Getting Started” vous guidera à travers les étapes essentielles pour démarrer rapidement.

Restons en contact

N’hésitez pas à nous suivre sur les réseaux sociaux pour rester à jour avec nos dernières publications et annonces. Si vous avez des questions ou des commentaires, n’hésitez pas à nous contacter !

Mizux

mdbook_mathjax

Section 1

Section 2

Chapter 2

Section 1

Section 2

Chapter 3

Section 1

Section 2

Chapter 4

Section 1

Section 2

Chapter 5

Section 1

Section 2

Chapter 6

Section 1

> ⓘ Note
>
> Highlights information that users should take into account, even when skimming.

> Tip
>
> Optional information to help a user be more successful.

> Important
>
> Crucial information necessary for users to succeed.

> ⚠ Warning
>
> Critical content demanding immediate user attention due to potential risks.

> Caution
>
> Negative potential consequences of an action.

Section 2

Here is an inline example, $\pi(\theta)$,

an equation,

$$\nabla f(x) \in \mathbb{R}^n,$$

and a regular $ symbol.

Define $f(x)$:

$$f(x) = x^2$$
$$x \in \mathbb{R}$$

```
graph TD
  A[ Anyone ] -->|Can help | B( Go to github.com/yuzutech/kroki )
  B --> C{ How to contribute? }
  C --> D[ Reporting bugs ]
  C --> E[ Sharing ideas ]
  C --> F[ Advocating ]
```

mdbook-katex

HTML:

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$\Rightarrow x + iy = re^{i\theta}$$

Markdown (requires `mdbook-katex`):

$$\oint_C f(x, y)\,\mathrm{d}A$$

Inspect element and use `Sources` tab (under `Debugger` on Firefox) to check that all CSS and fonts are properly loaded from GitHub pages instead of external CDN.

▼ Proof that $e^{ix} = \cos x + i\sin x$

$$\begin{aligned}
e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} \implies e^{ix} = \sum_{n=0}^{\infty} \frac{(ix)^n}{n!} \\
\cos x &= \sum_{m=0}^{\infty} \frac{(-1)^m x^{2m}}{(2m)!} = \sum_{m=0}^{\infty} \frac{(ix)^{2m}}{(2m)!} \\
\sin x &= \sum_{s=0}^{\infty} \frac{(-1)^s x^{2s+1}}{(2s+1)!} = \sum_{s=0}^{\infty} \frac{(ix)^{2s+1}}{i(2s+1)!} \\
\cos x + i\sin x &= \sum_{l=0}^{\infty} \frac{(ix)^{2l}}{(2l)!} + \sum_{s=0}^{\infty} \frac{(ix)^{2s+1}}{(2s+1)!} = \sum_{n=0}^{\infty} \frac{(ix)^n}{n!} \\
&= e^{ix}
\end{aligned}$$

Fourier Transform:

$$f(t) = \int_{-\infty}^{\infty} F(\omega) i^{4t\omega} d\omega$$

$$F(\omega) = \int_{-\infty}^{\infty} f(t) i^{-4t\omega} dt$$

Pauli Matrices:

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

mdbook-whichlang

```c
#include <stdio.h>

int main(void) {
    printf("Hello World\n");
}
```

```cpp
#include <iostream>

int main()
{
   std::cout << "Hello World" << std::endl;
}
```

```js
console.log("Hello World");
```

```ts
console.log("Hello World");
```

```
Console.log("Hello World")
```

```css
```

```css
h1 {
  color: blue;
}
```

```vim
set syntax=ruby
```

init.lua

```lua
print("Hello World")
```

```python
print("Hello World")
```

```bash
echo "Hello World"
```

hello.zig

```zig
const std = @import("std");

pub fn main() !void {
    const stdout = std.io.getStdOut().writer();
    try stdout.print("Hello, {s}!\n", .{"world"});
}
```

```
package main

import "core:fmt"

main :: proc() {
	fmt.println("Hellope!")
}
```

hello.wat

```
(module
    (import "wasi_unstable" "fd_write"
        (func $fd_write (param i32 i32 i32 i32) (result i32))
    )

    (memory 1)
    (export "memory" (memory 0))

    (data (i32.const 0) "\08\00\00\00\0c\00\00\00Hello World\n")

    (func $main (export "_start")
        i32.const 1
        i32.const 0
        i32.const 1
        i32.const 20
        call $fd_write
        drop
    )
)
```

mdbook-langtabs

Some code

let’s try with a simple example:

```
LogChannel(n"DEBUG", "hello world");
```

```
print("Hello, World!")
```

```
fn main() { println!("hello world"); }
```

```
print("Hello World")
```

```
#include <iostream>

int main() {
    std::cout << "Hello World!";
    return 0;
}
```

```yaml
some:
  interesting:
    - property
```

```json
{
  "some": { "interesting": ["property"] }
}
```

```xml
<some>
  <interesting>
    <property />
  </interesting>
</some>
```

contrary to code blocks, `inline` and `fenced` are left untouched.

it should also work when deeply nested:

1. outer:

    1. inner:

        ```swift
        GameInstance
          .GetStatusEffectSystem(this.GetGame())
          .ApplyStatusEffect(
            this.GetEntityID(),
            t"BaseStatusEffect.SplinterAddicted",
            this.GetRecordID(),
            this.GetEntityID());
        ```

Hello World

Here’s a simple “Hello World” example in different languages:

```rust
fn main() {
    println!("Hello, World!");
}
```

```python
def main():
    print("Hello, World!")

if __name__ == "__main__":
    main()
```

```javascript
function main() {
    console.log("Hello, World!");
}

main();
```

```go
package main

import "fmt"

func main() {
    fmt.Println("Hello, World!")
}
```

```java
public class HelloWorld {
    public static void main(String[] args) {
        System.out.println("Hello, World!");
    }
}
```

Simple Function Example

Here’s how you might define a function to calculate factorial in different languages:

```rust
fn factorial(n: u64) -> u64 {
    match n {
        0 | 1 => 1,
        _ => n * factorial(n - 1)
    }
}

fn main() {
    println!("5! = {}", factorial(5)); // Outputs: 5! = 120
}
```

```python
def factorial(n):
    if n <= 1:
        return 1
    return n * factorial(n - 1)

print(f"5! = {factorial(5)}")  # Outputs: 5! = 120
```

```javascript
function factorial(n) {
    if (n <= 1) return 1;
    return n * factorial(n - 1);
}

console.log(`5! = ${factorial(5)}`);  // Outputs: 5! = 120
```

```go
package main

import "fmt"

func factorial(n uint64) uint64 {
    if n <= 1 {
        return 1
    }
    return n * factorial(n-1)
}

func main() {
    fmt.Printf("5! = %d\n", factorial(5))  // Outputs: 5! = 120
}
```

```java
public class FactorialExample {
    static long factorial(int n) {
        if (n <= 1) return 1;
        return n * factorial(n - 1);
    }

    public static void main(String[] args) {
        System.out.println("5! = " + factorial(5));  // Outputs: 5! = 120
    }
}
```

Data Structures Example

Here’s how you might implement a simple stack in different languages:

```rust
struct Stack<T> {
    items: Vec<T>,
}

impl<T> Stack<T> {
    fn new() -> Self {
        Stack { items: Vec::new() }
    }

    fn push(&mut self, item: T) {
        self.items.push(item);
    }

    fn pop(&mut self) -> Option<T> {
        self.items.pop()
    }

    fn is_empty(&self) -> bool {
        self.items.is_empty()
    }
}

fn main() {
    let mut stack = Stack::new();
    stack.push(1);
    stack.push(2);
    stack.push(3);

    while let Some(item) = stack.pop() {
        println!("Popped: {}", item);
    }
}
```

```python
class Stack:
    def __init__(self):
        self.items = []

    def push(self, item):
        self.items.append(item)

    def pop(self):
        if not self.is_empty():
            return self.items.pop()
        return None

    def is_empty(self):
        return len(self.items) == 0

# Usage
stack = Stack()
stack.push(1)
stack.push(2)
stack.push(3)

while not stack.is_empty():
    print(f"Popped: {stack.pop()}")
```

JS

```javascript
class Stack {
    constructor() {
        this.items = [];
    }

    push(item) {
        this.items.push(item);
    }

    pop() {
        if (!this.isEmpty()) {
            return this.items.pop();
        }
        return null;
    }

    isEmpty() {
        return this.items.length === 0;
    }
}

// Usage
const stack = new Stack();
stack.push(1);
stack.push(2);
stack.push(3);

while (!stack.isEmpty()) {
    console.log(`Popped: ${stack.pop()}`);
}
```

```go
package main

import "fmt"

type Stack struct {
    items []int
}

func (s *Stack) Push(item int) {
    s.items = append(s.items, item)
}

func (s *Stack) Pop() (int, bool) {
    if s.IsEmpty() {
        return 0, false
    }

    index := len(s.items) - 1
    item := s.items[index]
    s.items = s.items[:index]
    return item, true
}

func (s *Stack) IsEmpty() bool {
    return len(s.items) == 0
}

func main() {
    stack := Stack{}
    stack.Push(1)
    stack.Push(2)
    stack.Push(3)

    for !stack.IsEmpty() {
        item, _ := stack.Pop()
        fmt.Printf("Popped: %d\n", item)
    }
}
```

```java
import java.util.ArrayList;

public class StackExample {
    static class Stack<T> {
        private ArrayList<T> items = new ArrayList<>();

        public void push(T item) {
            items.add(item);
        }

        public T pop() {
            if (isEmpty()) {
                return null;
            }
            return items.remove(items.size() - 1);
        }

        public boolean isEmpty() {
            return items.isEmpty();
        }
    }

    public static void main(String[] args) {
        Stack<Integer> stack = new Stack<>();
        stack.push(1);
        stack.push(2);
        stack.push(3);

        while (!stack.isEmpty()) {
            System.out.println("Popped: " + stack.pop());
        }
    }
}
```

kroki

kroki-graphviz

```
digraph G {
  a -> b [dir=both color="red:blue"]
  c -> d [dir=none color="green:red;0.25:blue"]
}
```

kroki-plantuml

```
@startuml
clock   "Clock_0"   as C0 with period 50
clock   "Clock_1"   as C1 with period 50 pulse 15 offset 10
binary  "Binary"  as B
concise "Concise" as C
robust  "Robust"  as R
analog  "Analog"  as A

@0
C is Idle
R is Idle
A is 0

@100
B is high
C is Waiting
R is Processing
A is 3

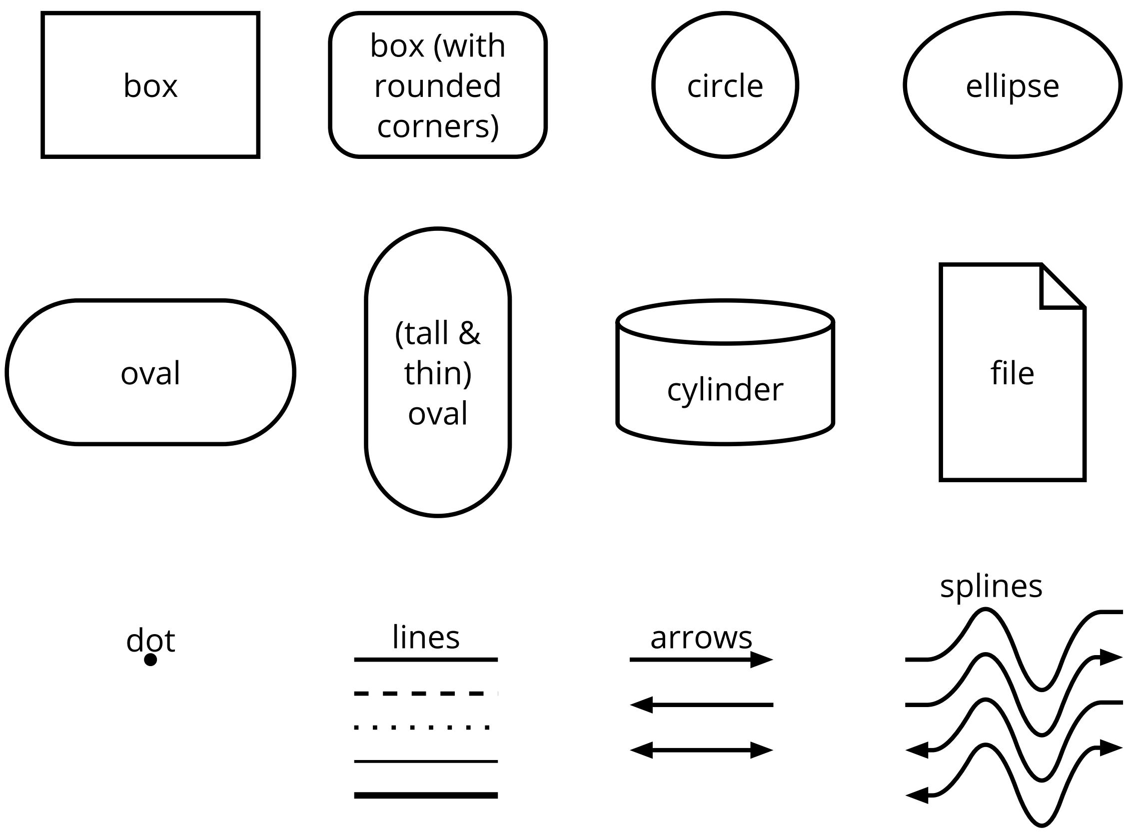
@300
R is Waiting
A is 1
@enduml
```

kroki-symbolator

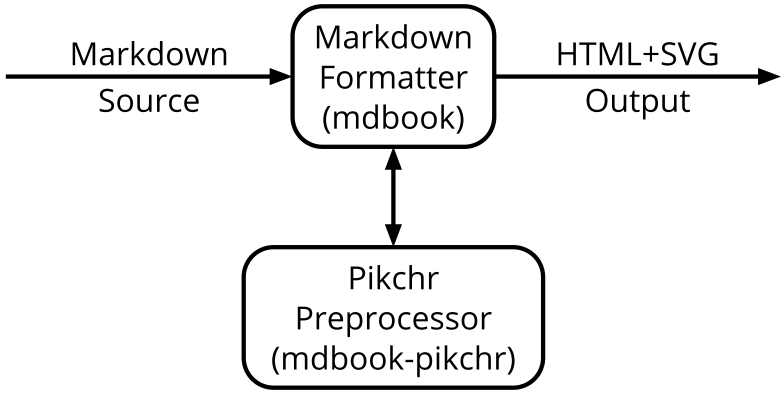
```
module demo_device #(
    //# {{}}
    parameter SIZE = 8,
    parameter RESET_ACTIVE_LEVEL = 1
) (
    //# {{clocks|Clocking}}
    input wire clock,
    //# {{control|Control signals}}
    input wire reset,
    input wire enable,
    //# {{data|Data ports}}
    input wire [SIZE-1:0] data_in,
    output wire [SIZE-1:0] data_out
);
  // ...
endmodule
```

pikchr-demo1

Examples Of Pikchr Objects

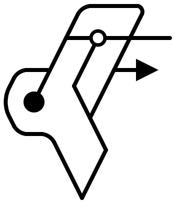


pikchr-demo2



svgbob-demo

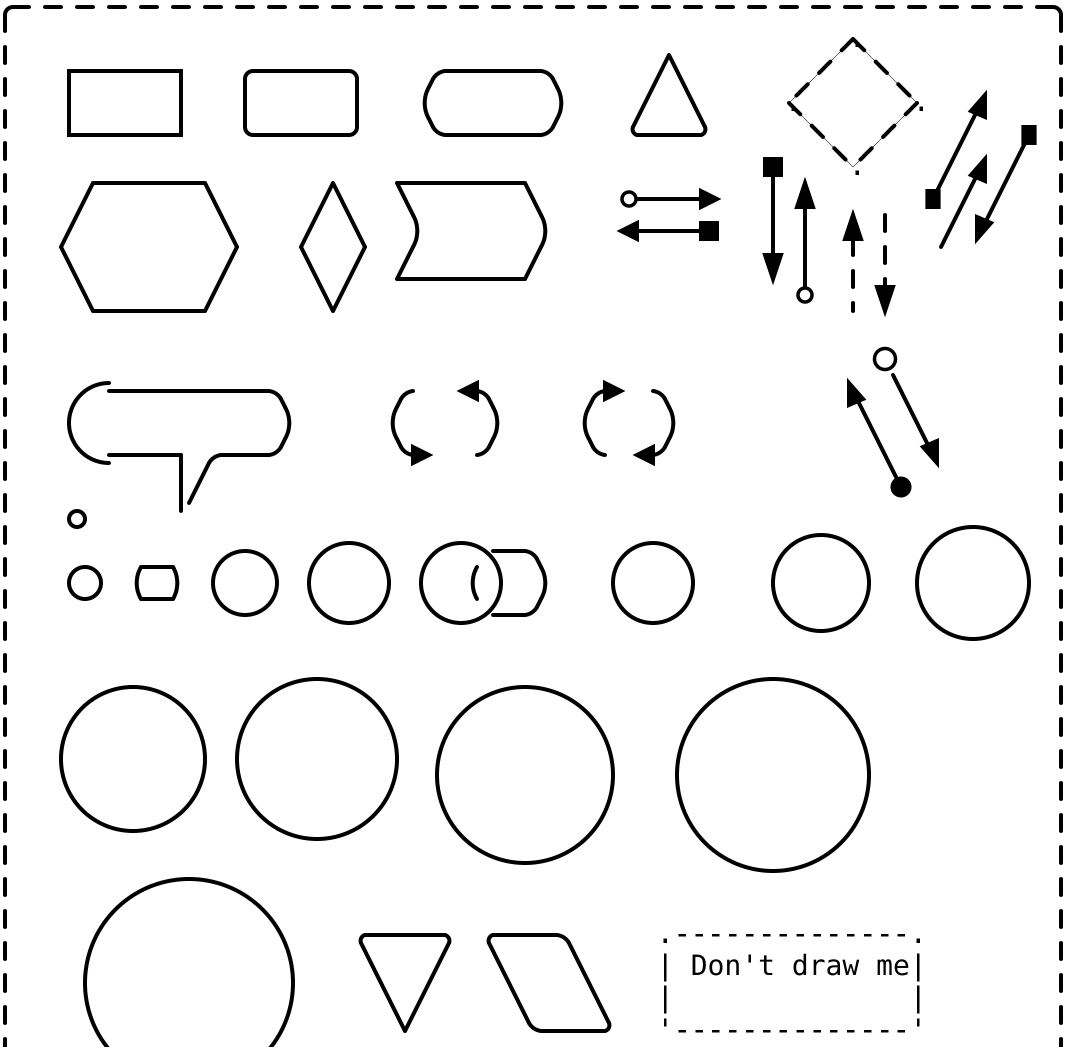
Svgbob is a diagramming model
which uses a set of typing characters
to approximate the intended shape.

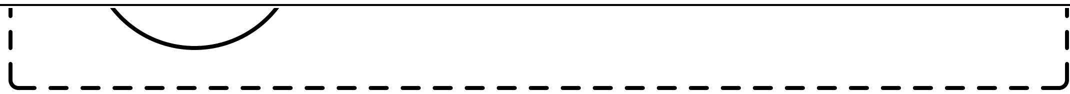


It uses a combination of characters
which are readily available on your keyboards.

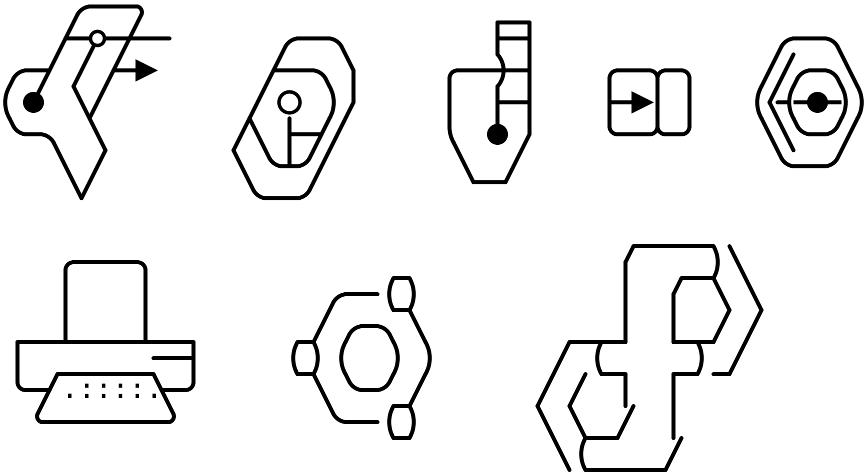
What can it do?

Basic shapes

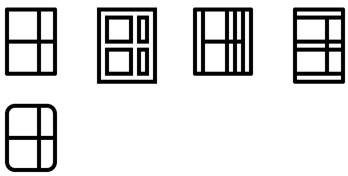




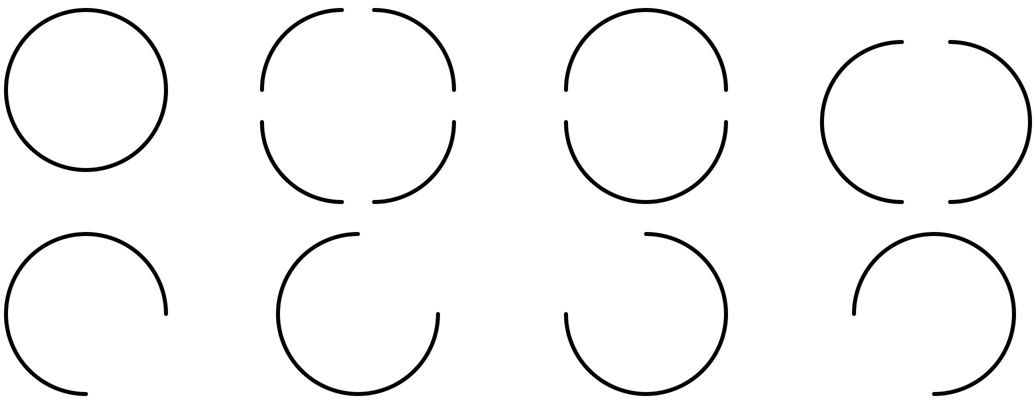
Quick logo scribbles



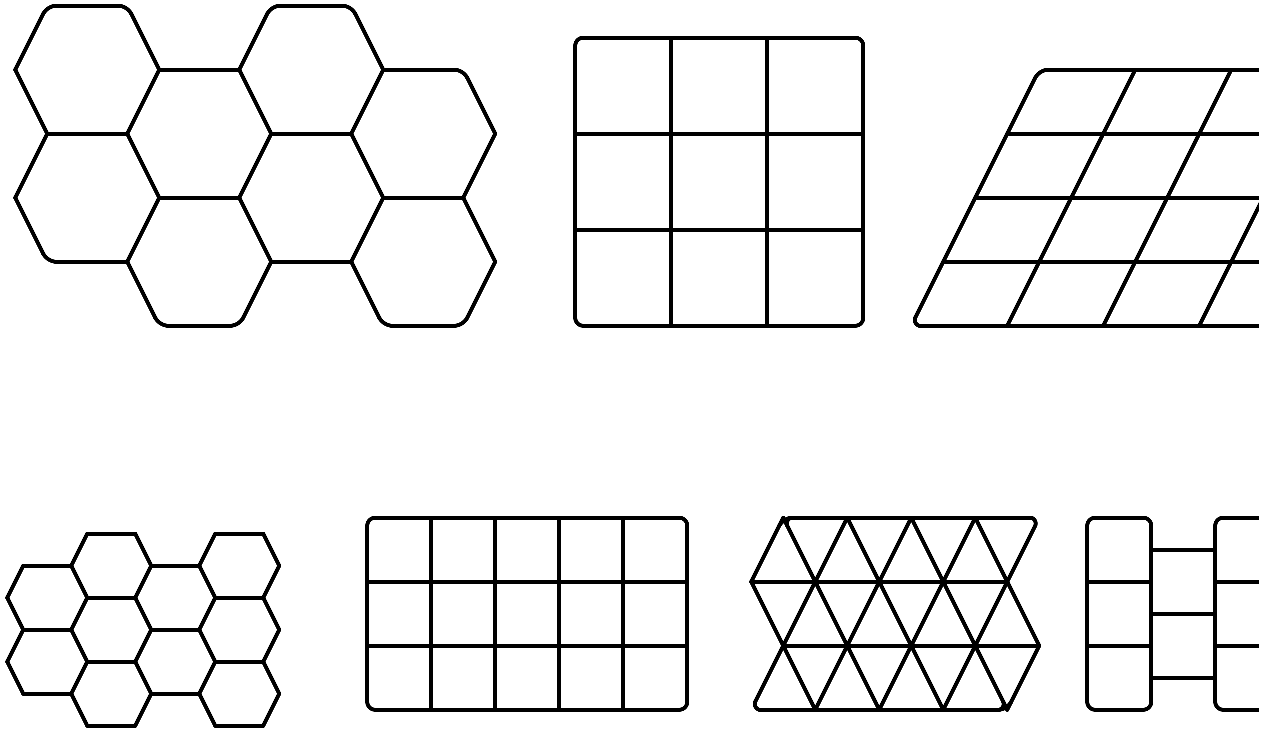
Even unicode box drawing characters are supported



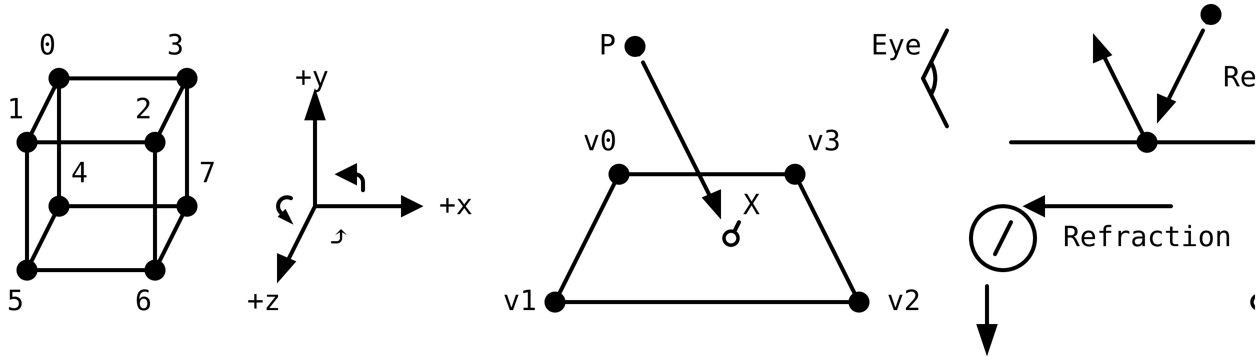
Circle, quarter arcs, half circles, 3/4 quarter arcs



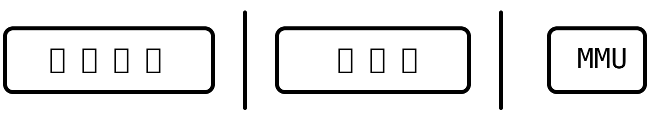
Grids



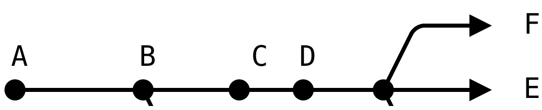
Graphics Diagram

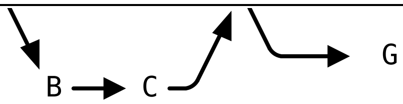


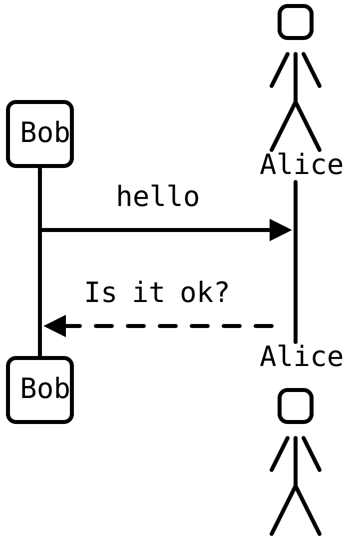
CJK characters

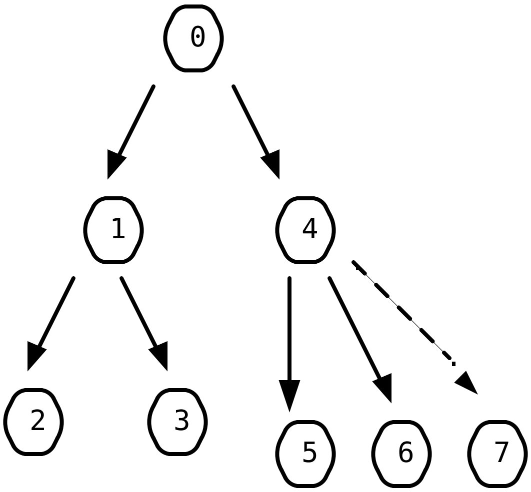


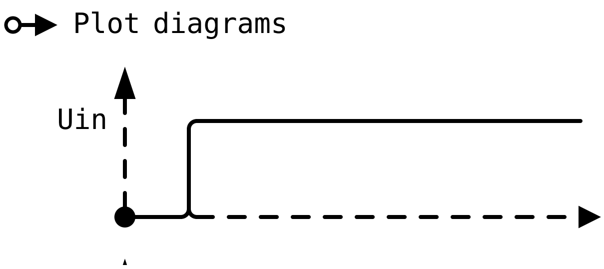
Sequence Diagrams

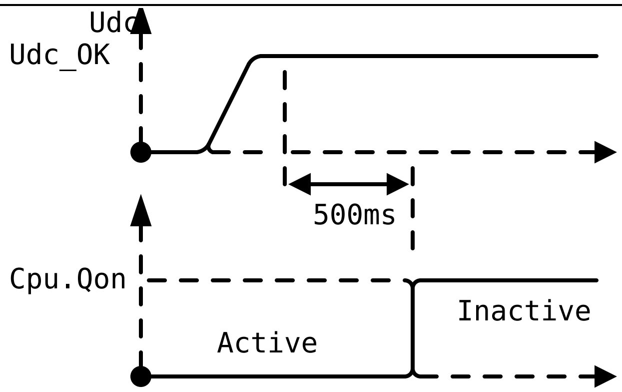



B
C
G

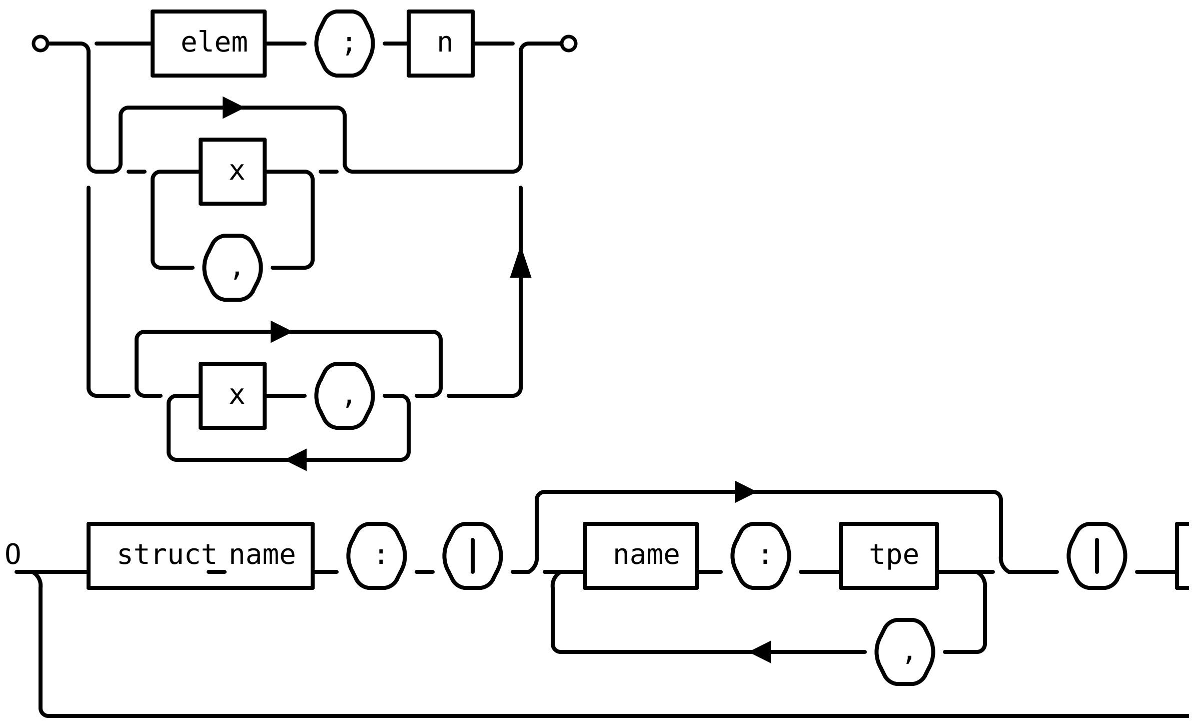

Bob
Alice
hello
Is it ok?
Alice
Bob


0
1
4
2
3
5
6
7

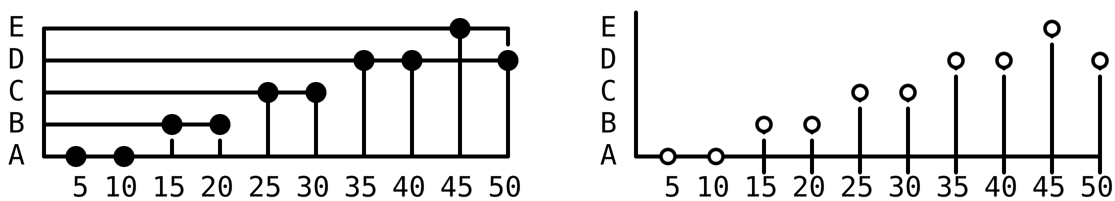

Plot diagrams
Uin

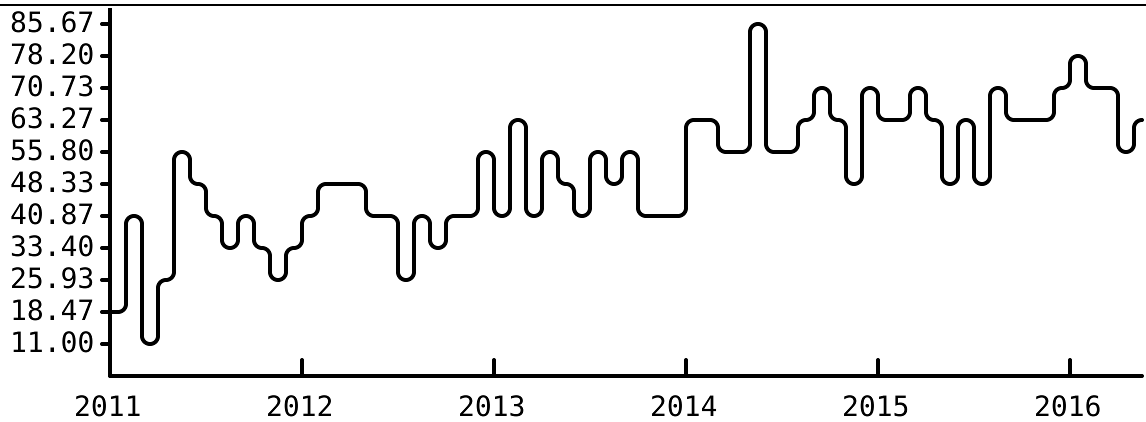


Railroad diagrams

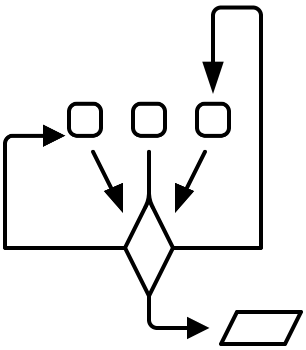


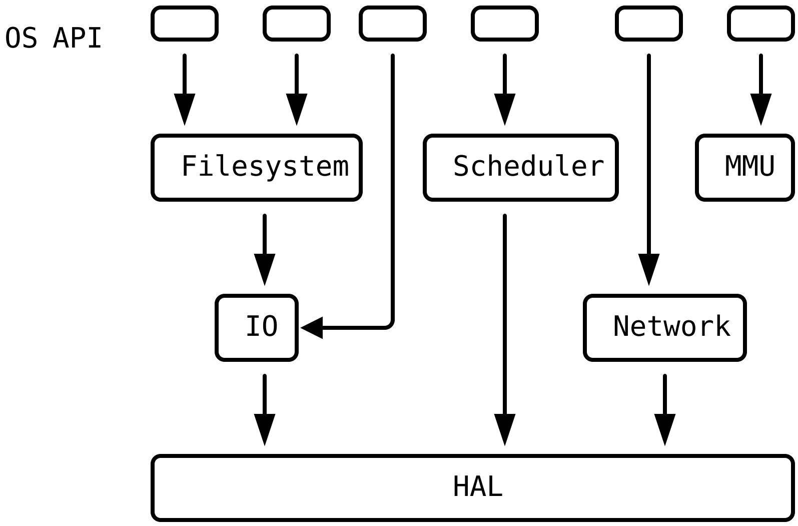
Statistical charts



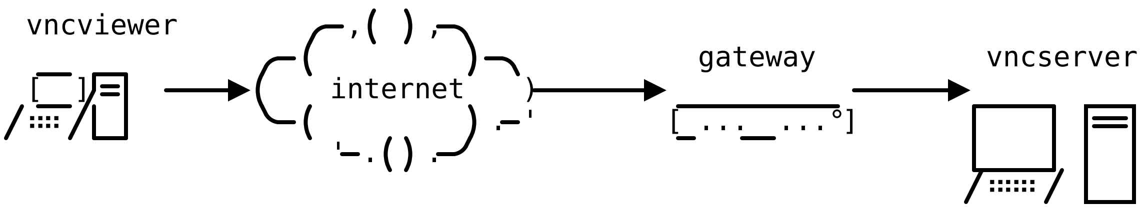


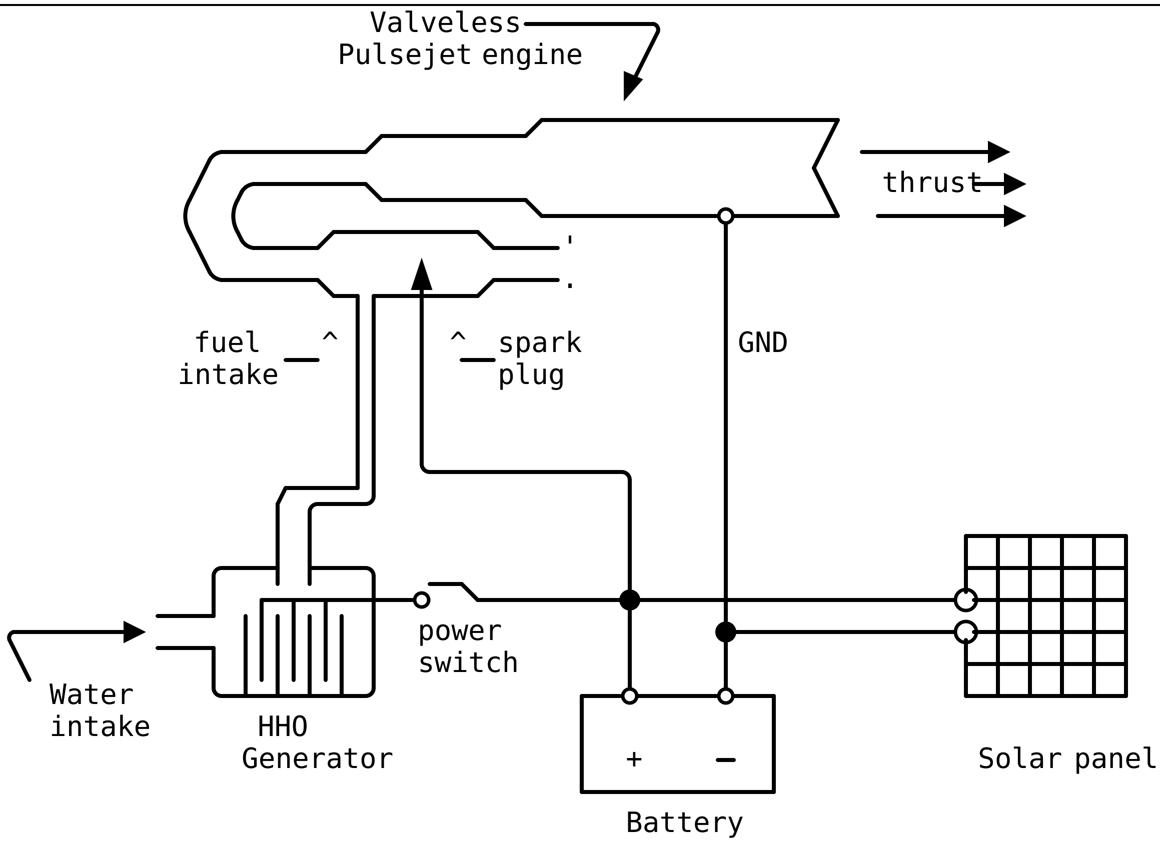
Flow charts

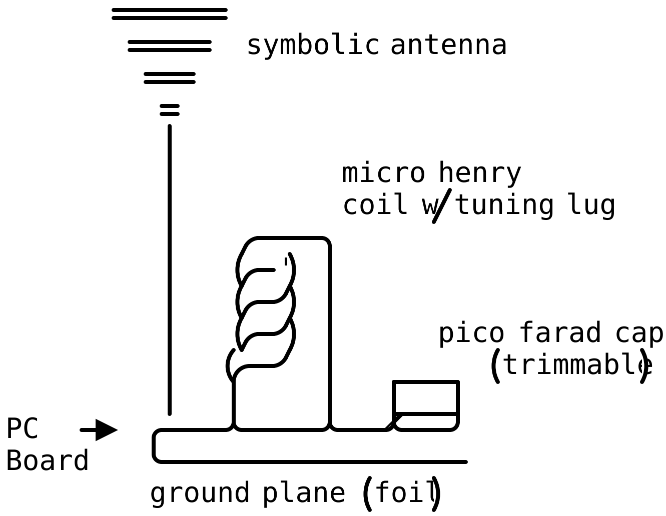




Block diagrams

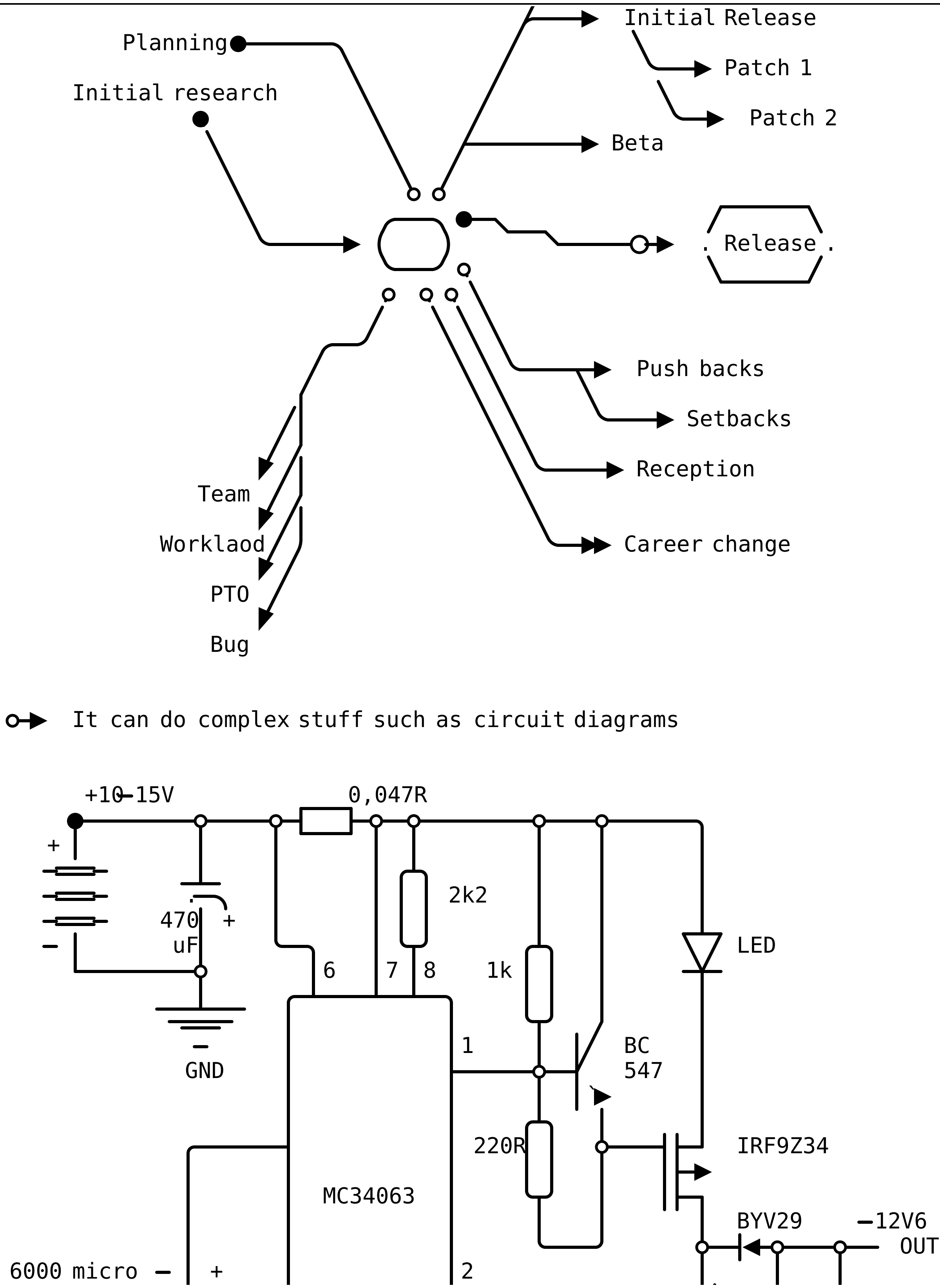


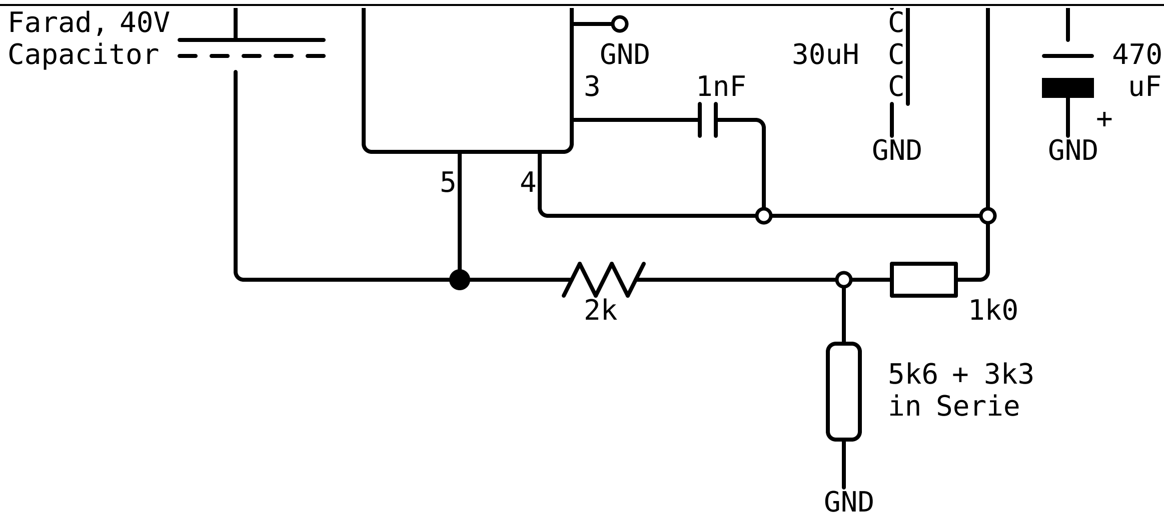




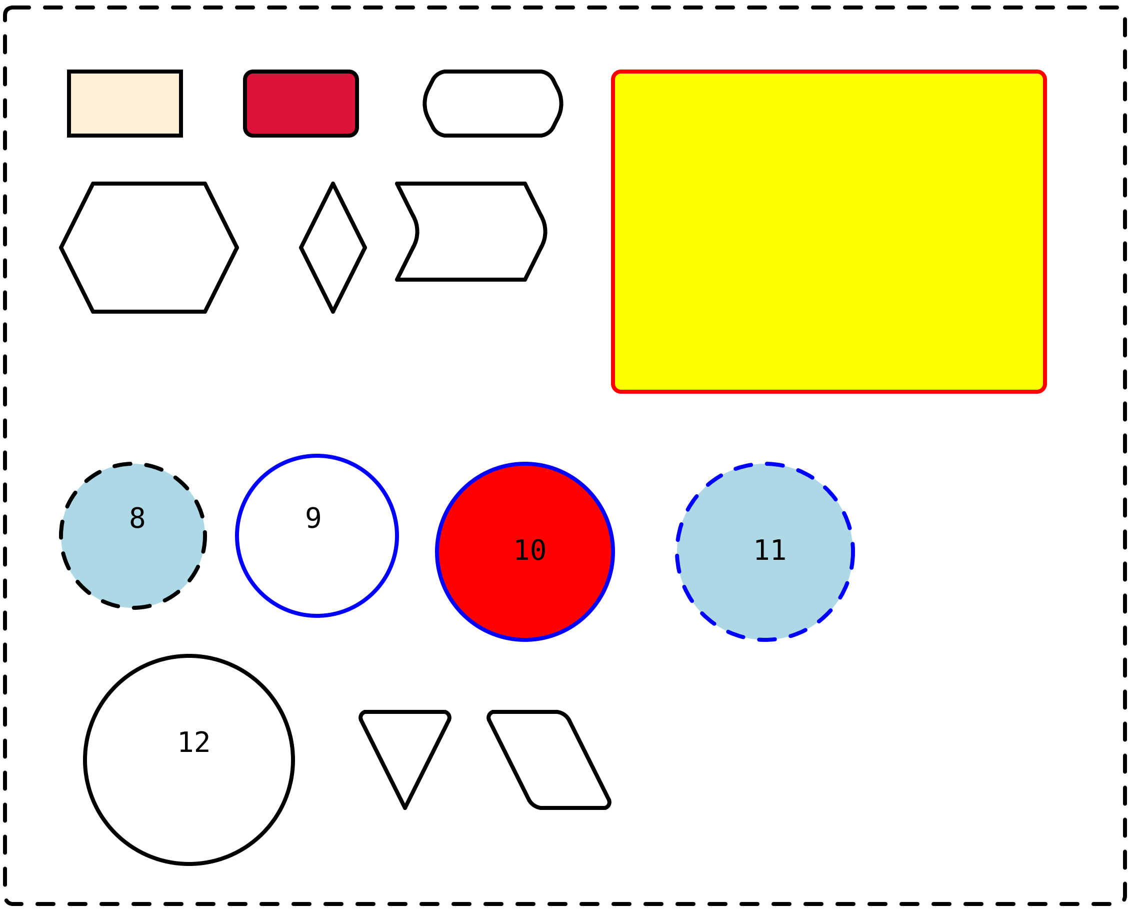
Mindmaps

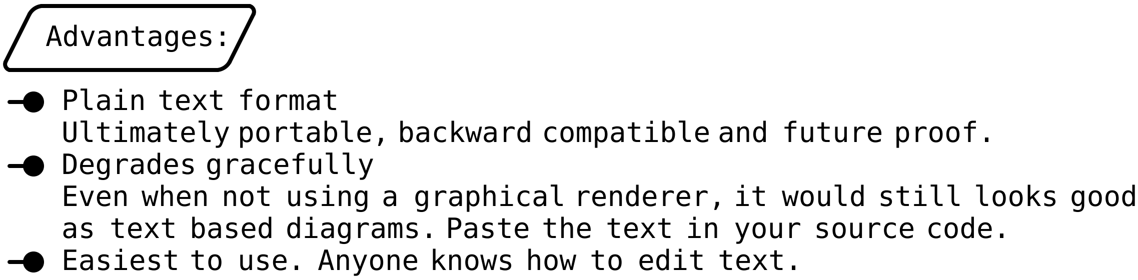
Alpha


Planning
Initial research
Initial Release
Patch 1
Patch 2
Beta
Release
Push backs
Setbacks
Reception
Career change
Team
Worklaod
PTO
Bug
It can do complex stuff such as circuit diagrams
+10-15V
0,047R
+
470
uF
+
2k2
GND
6
7
8
1k
1
BC
547
LED
220R
IRF9Z34
MC34063
BYV29
-12V6
OUT
6000 micro
+
2

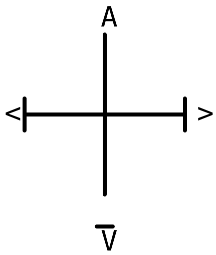


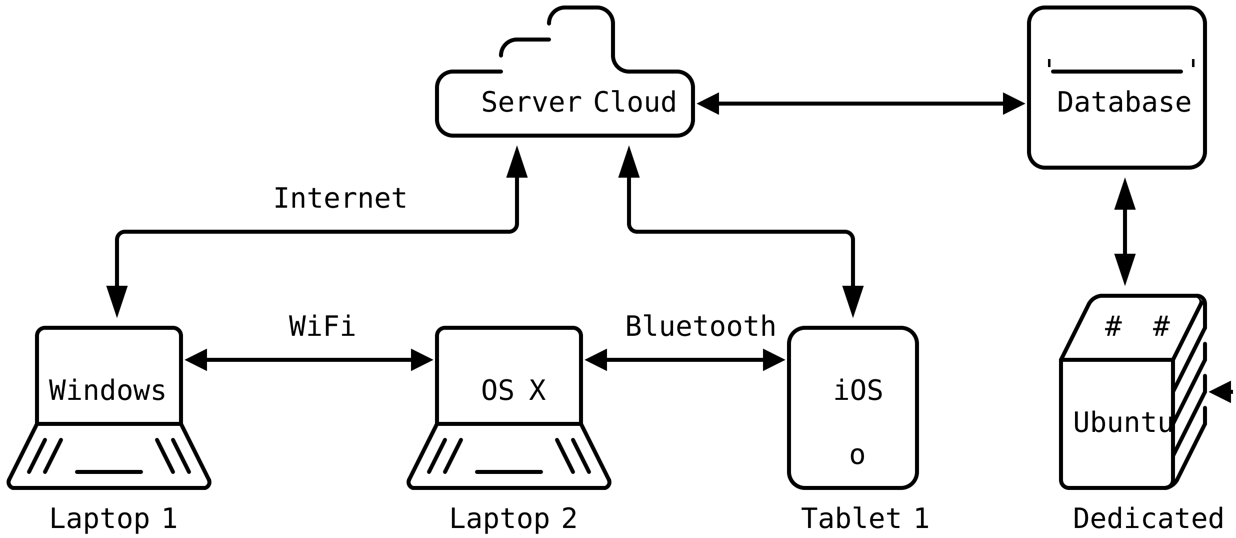
Latest addition: Styling of tagged shapes

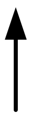


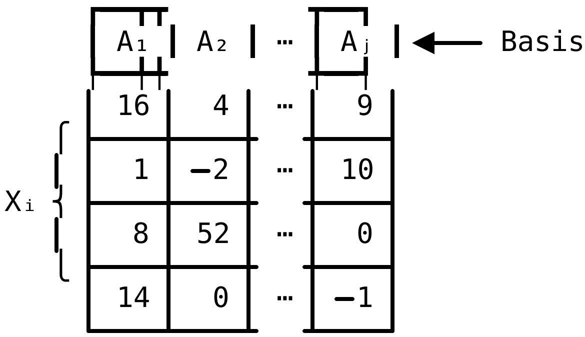


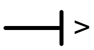
svgbob-otherdemo

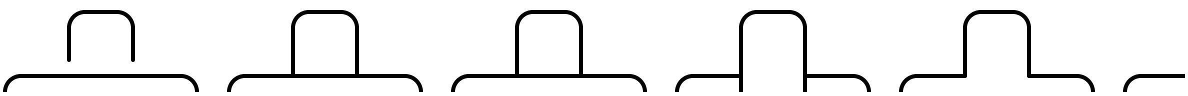

A
<
>
V

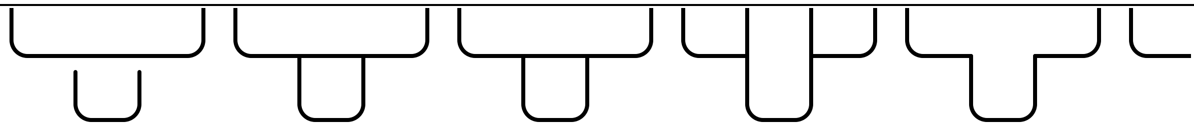

Server Cloud
Database
Internet
WiFi
Bluetooth
Windows
OS X
iOS
o
# #
Ubuntu
Laptop 1
Laptop 2
Tablet 1
Dedicated

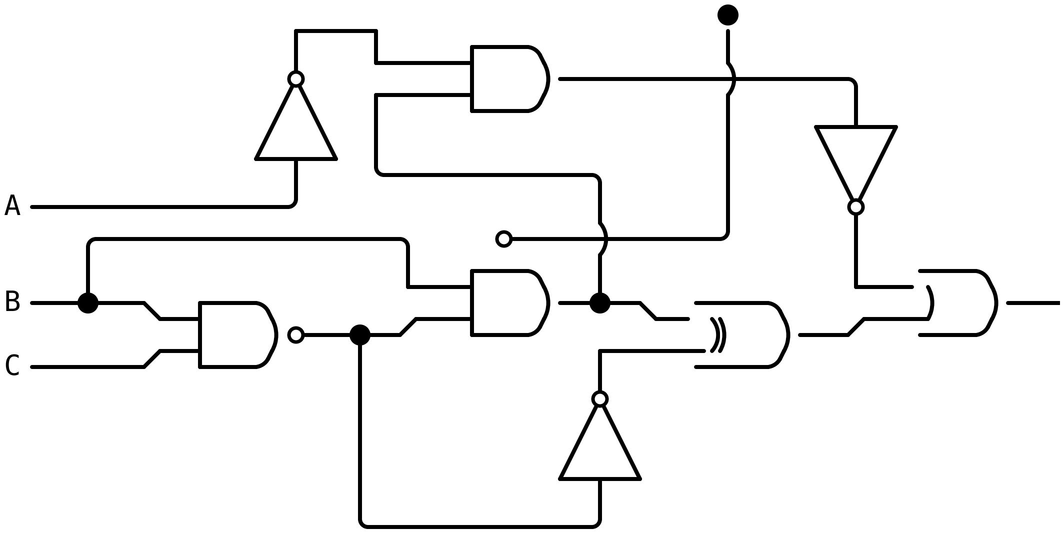


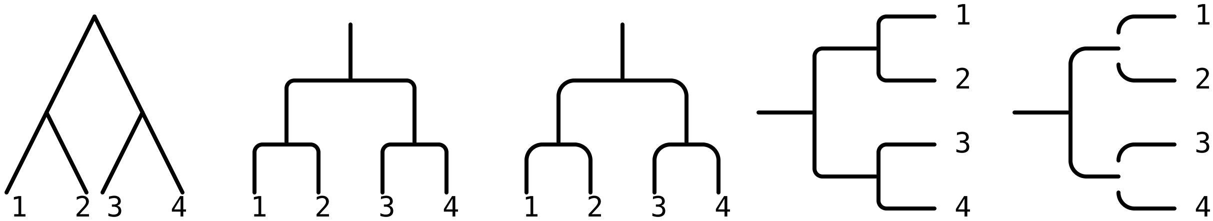

A₁
A₂
⋯
Aⱼ
Basis
16
4
9
Xᵢ
1
−2
10
8
52
0
14
0
−1

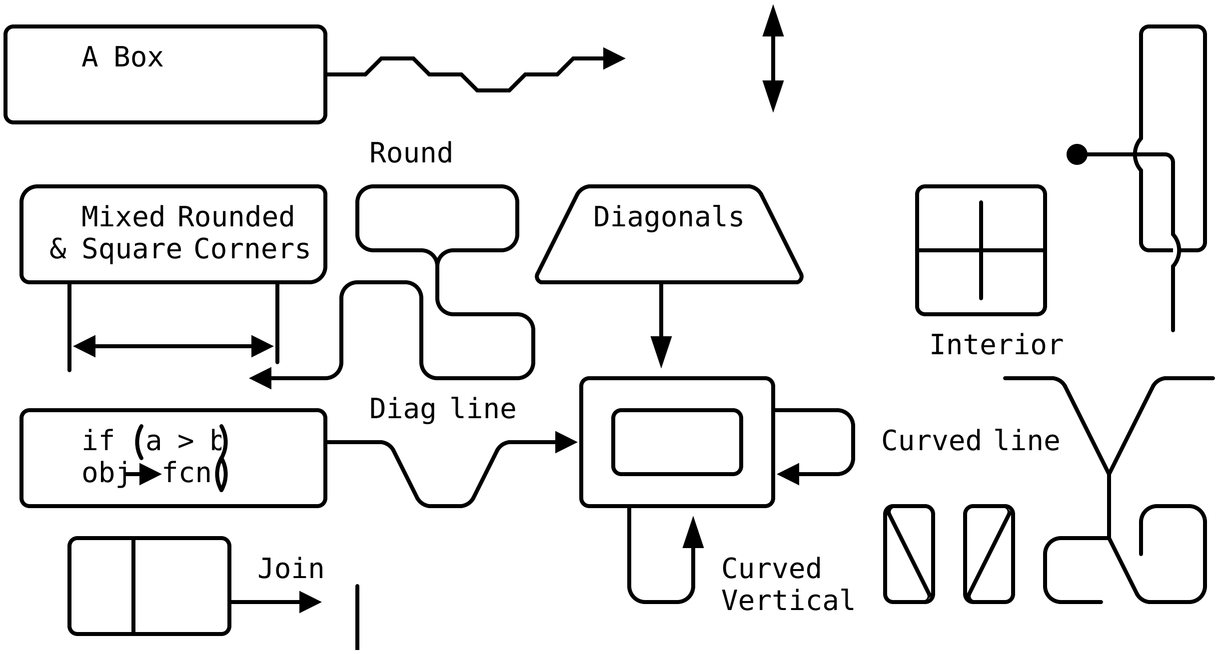

>

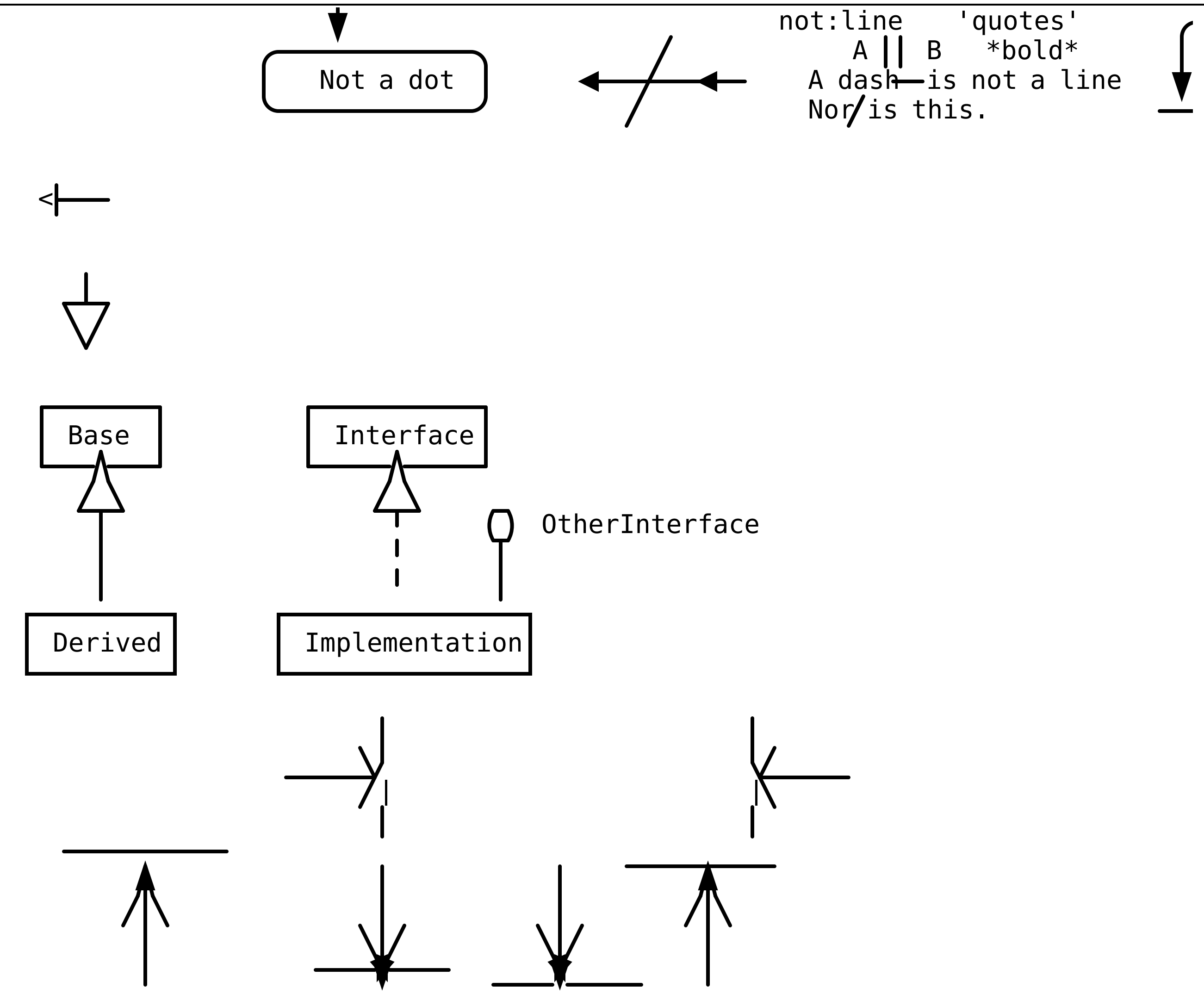





A
B
C

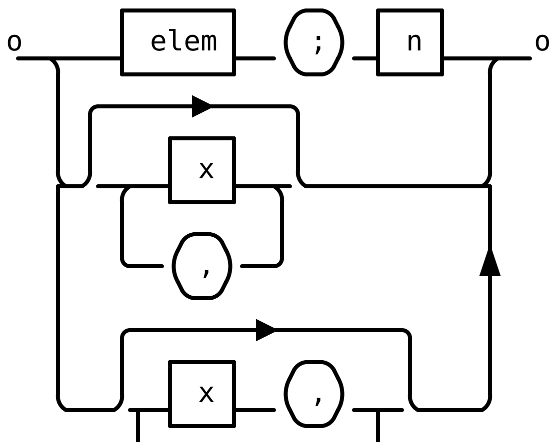

1 2 3 4
1 2 3 4
1 2 3 4
1
2
3
4
1
2
3
4

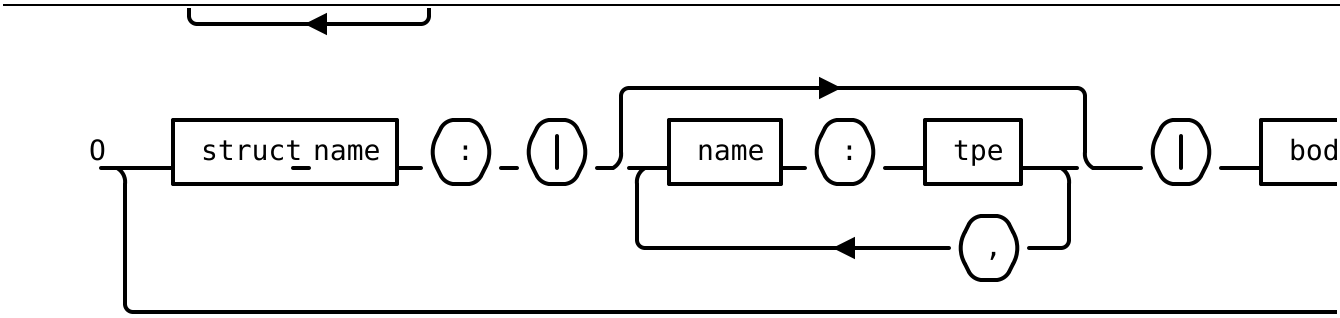

A Box
Round
Mixed Rounded
& Square Corners
Diagonals
Interior
Diag line
if (a > b)
obj→fcn()
Curved line
Join
Curved
Vertical



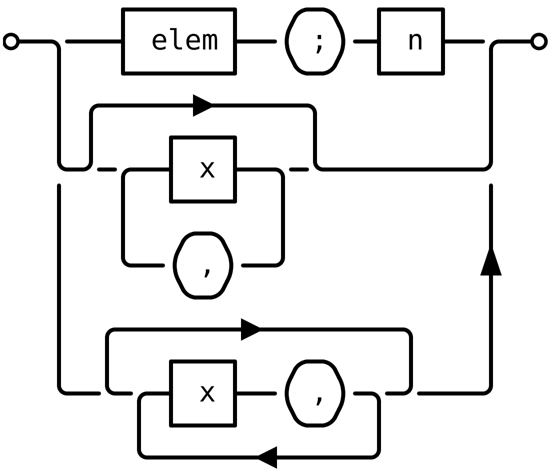
Poor man's rail road

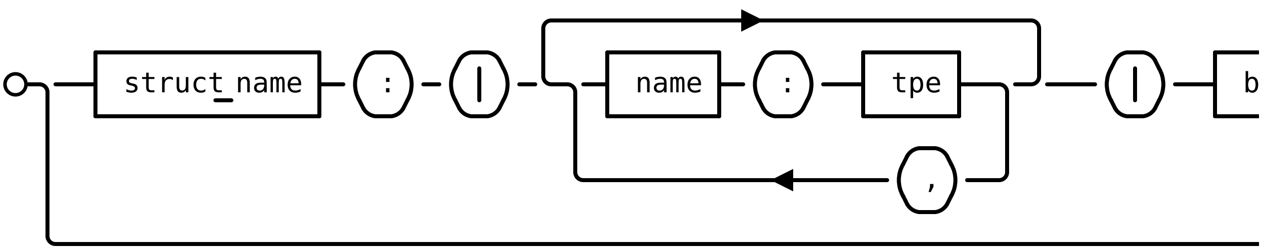
Style #1

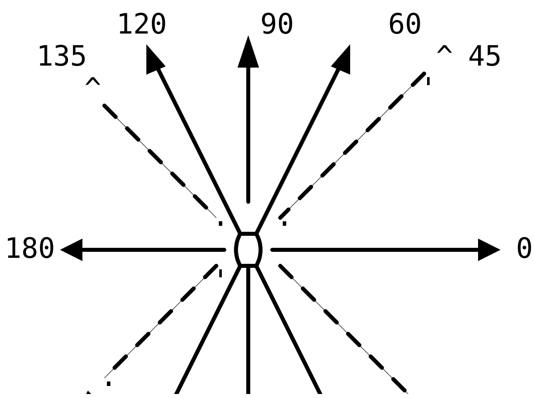


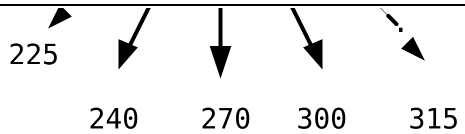

O
struct_name
:
|
name
:
tpe
,
|
bod

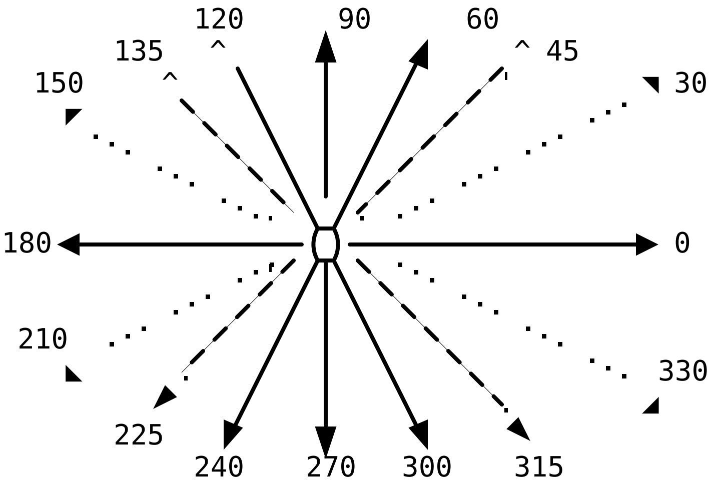
Style #2

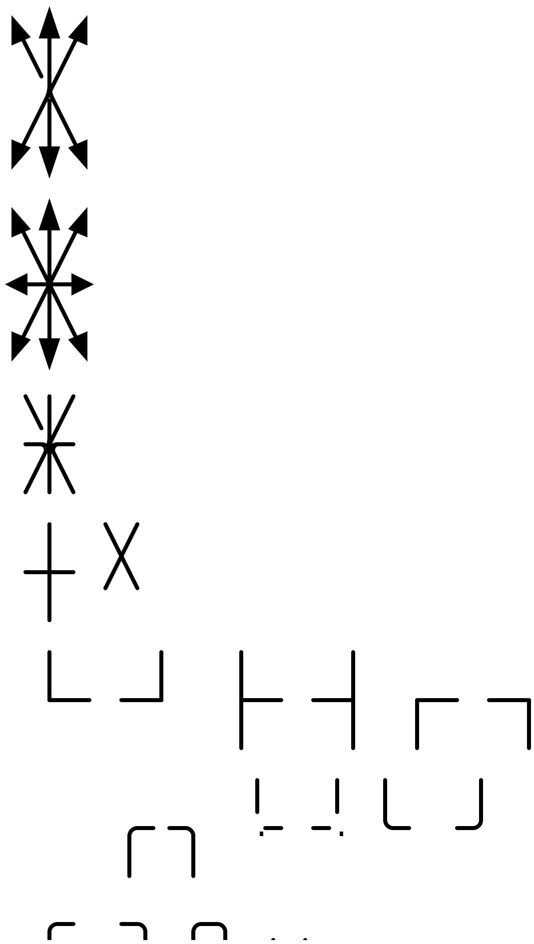

elem
;
n
x
,
x
,

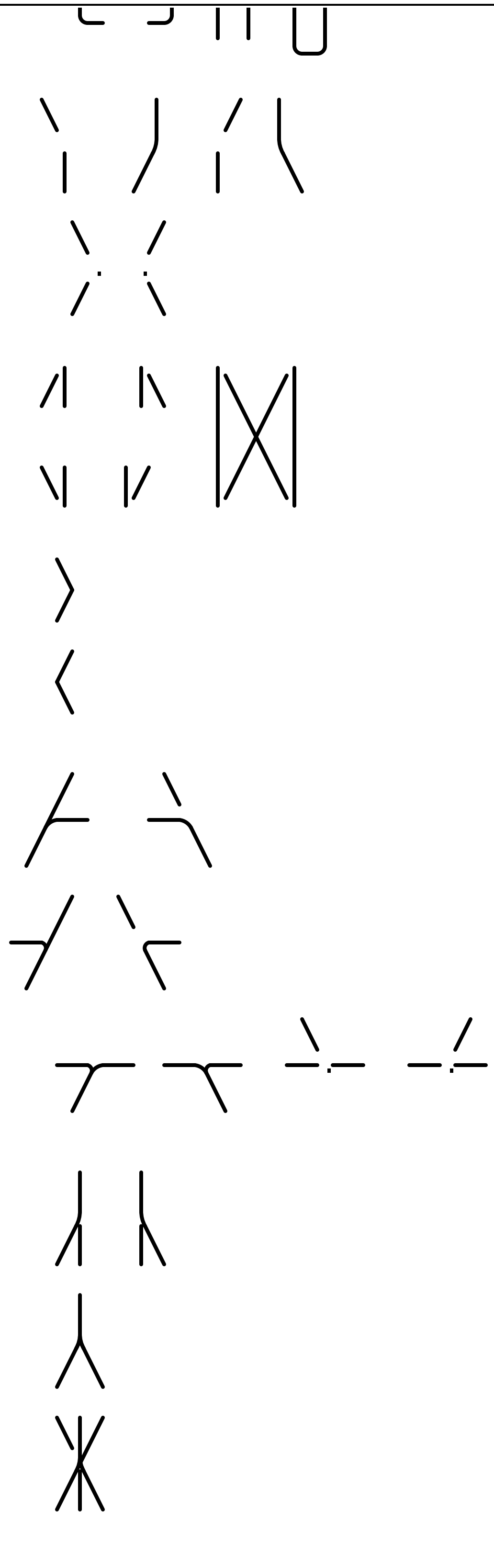

struct_name
:
|
name
:
tpe
,
|
b

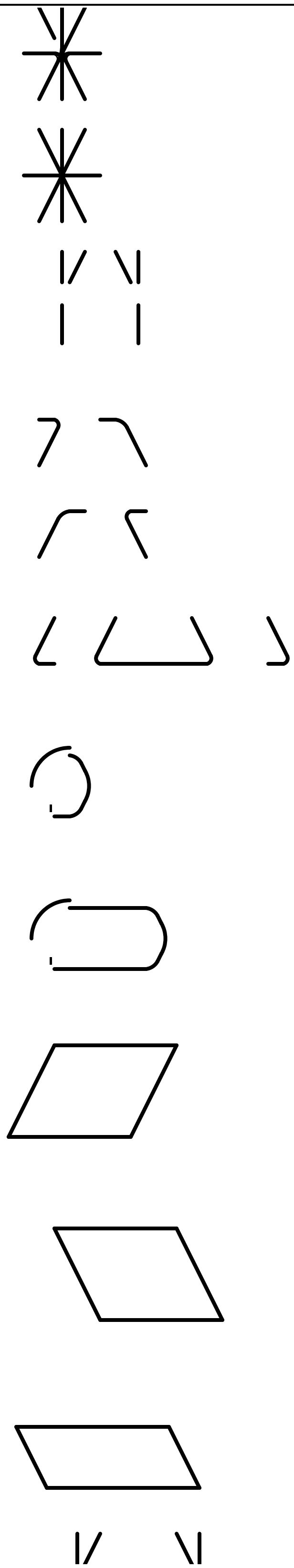

120
90
60
135
^ 45
^
180
0

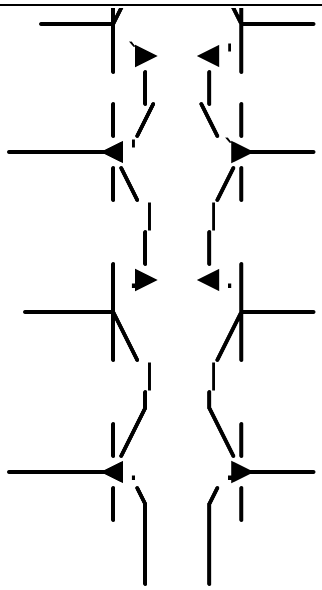

225
240
270
300
315

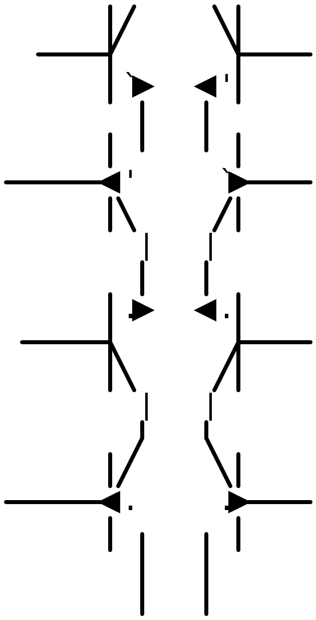

120
90
60
135
45
150
30
180
0
210
330
225
240
270
300
315





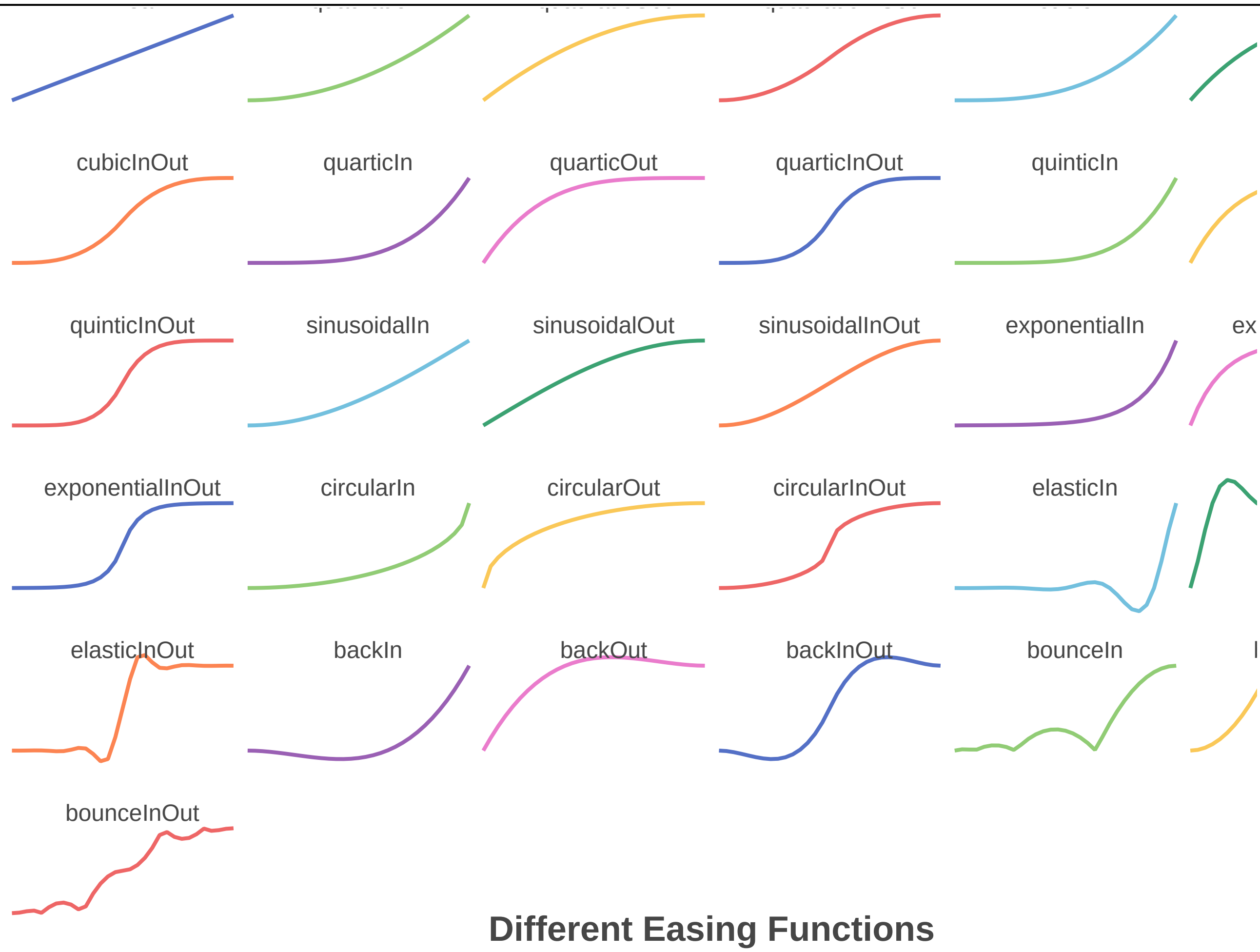




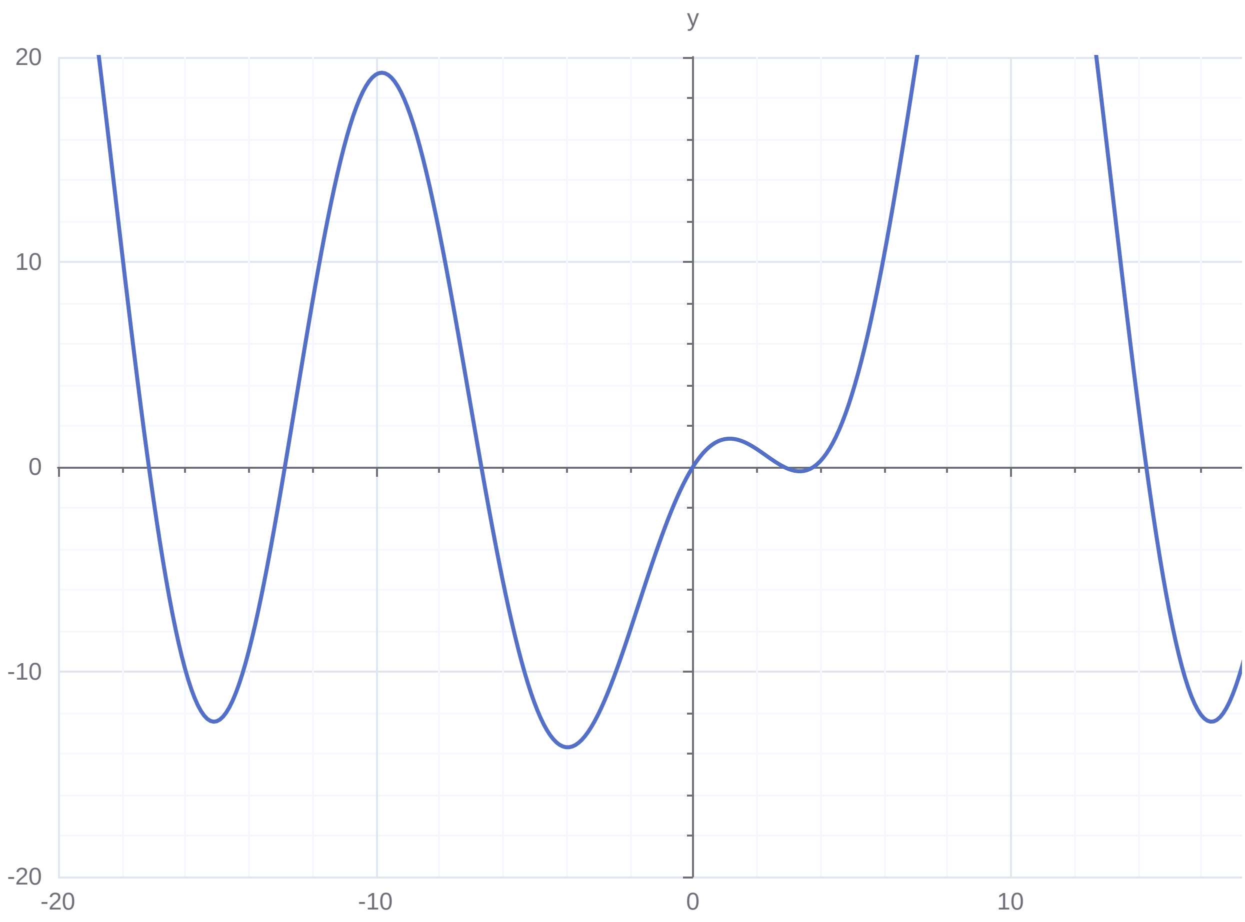


ECharts

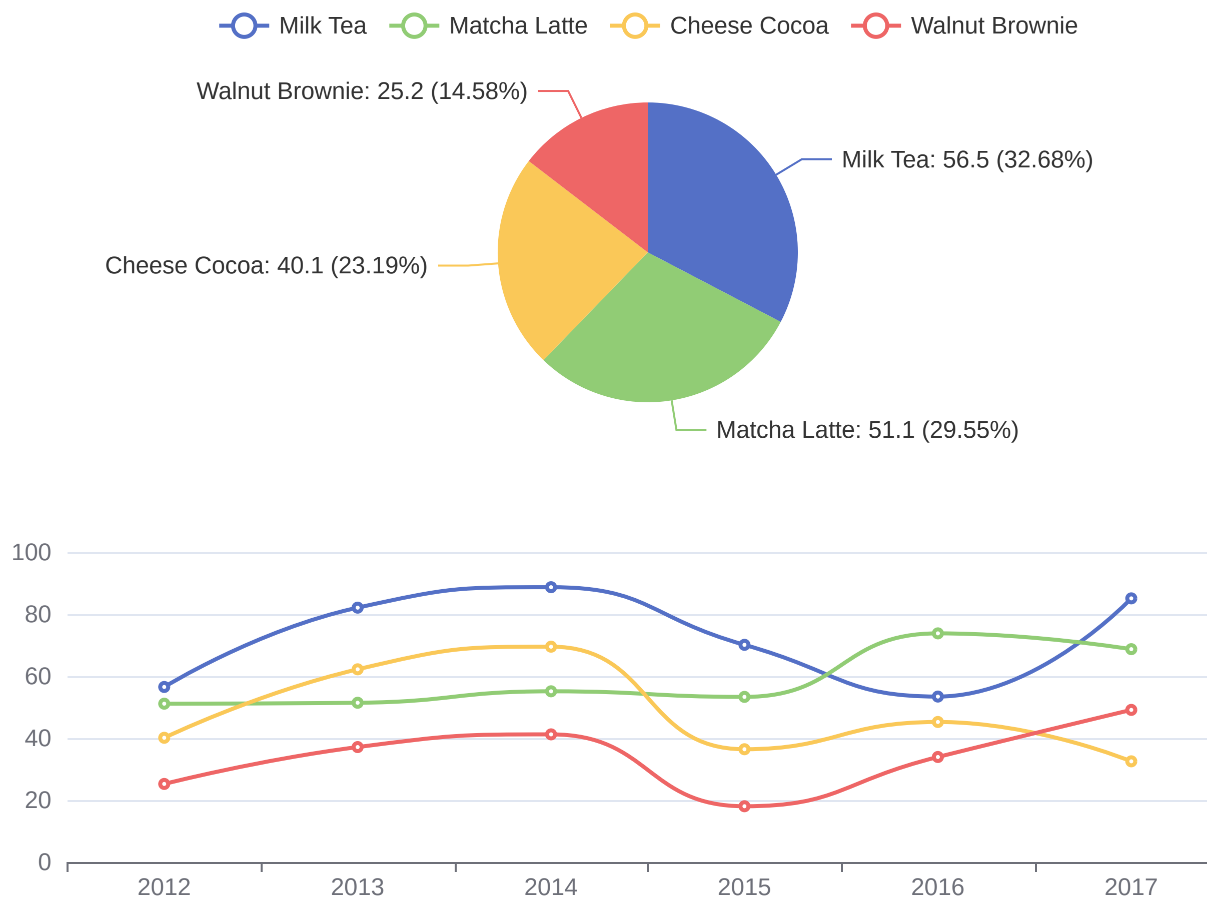
echarts-multi-function


cubicInOut
quarticIn
quarticOut
quarticInOut
quinticIn
quinticInOut
sinusoidalIn
sinusoidalOut
sinusoidalInOut
exponentialIn
ex
exponentialInOut
circularIn
circularOut
circularInOut
elasticIn
elasticInOut
backIn
backOut
backInOut
bounceIn
bounceInOut
Different Easing Functions

echarts-function

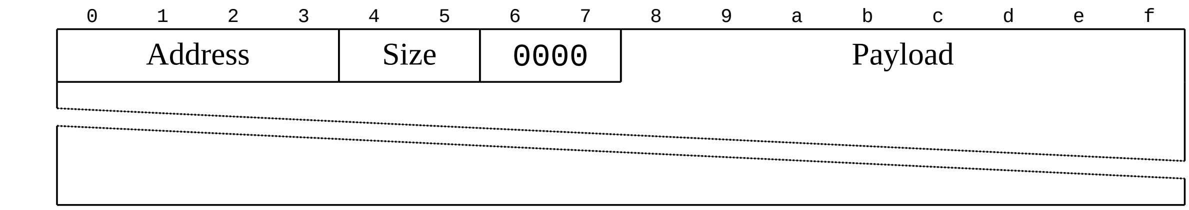


echarts-dataset

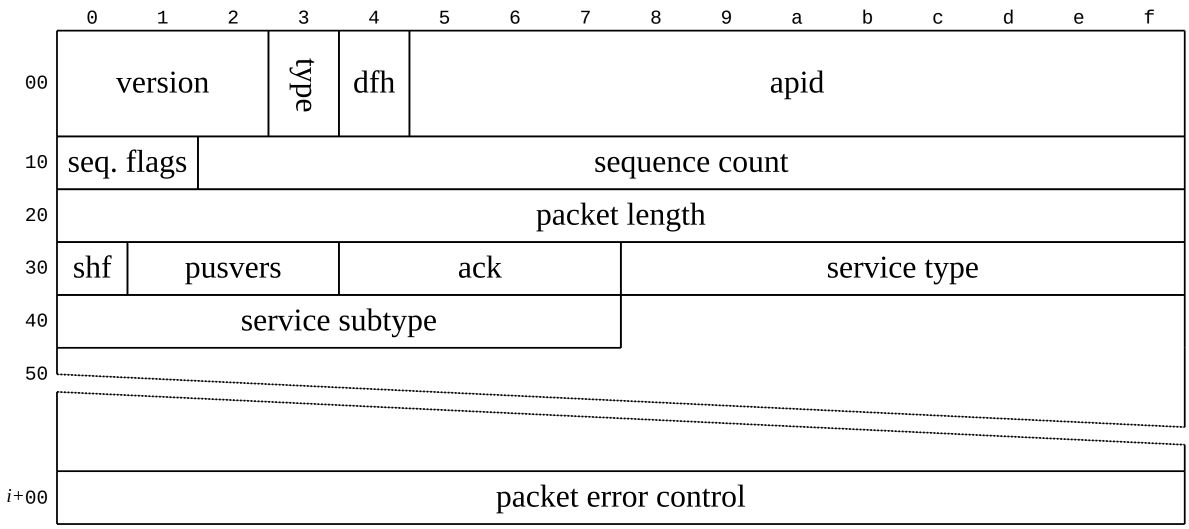


Bytefield

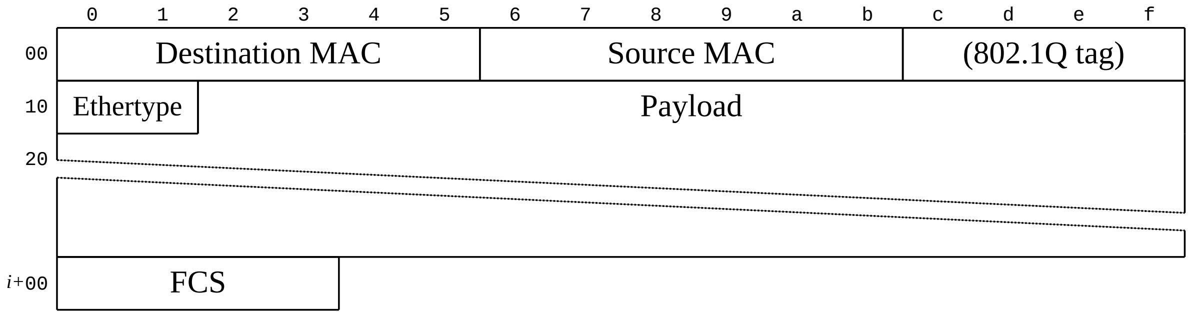
Basic.



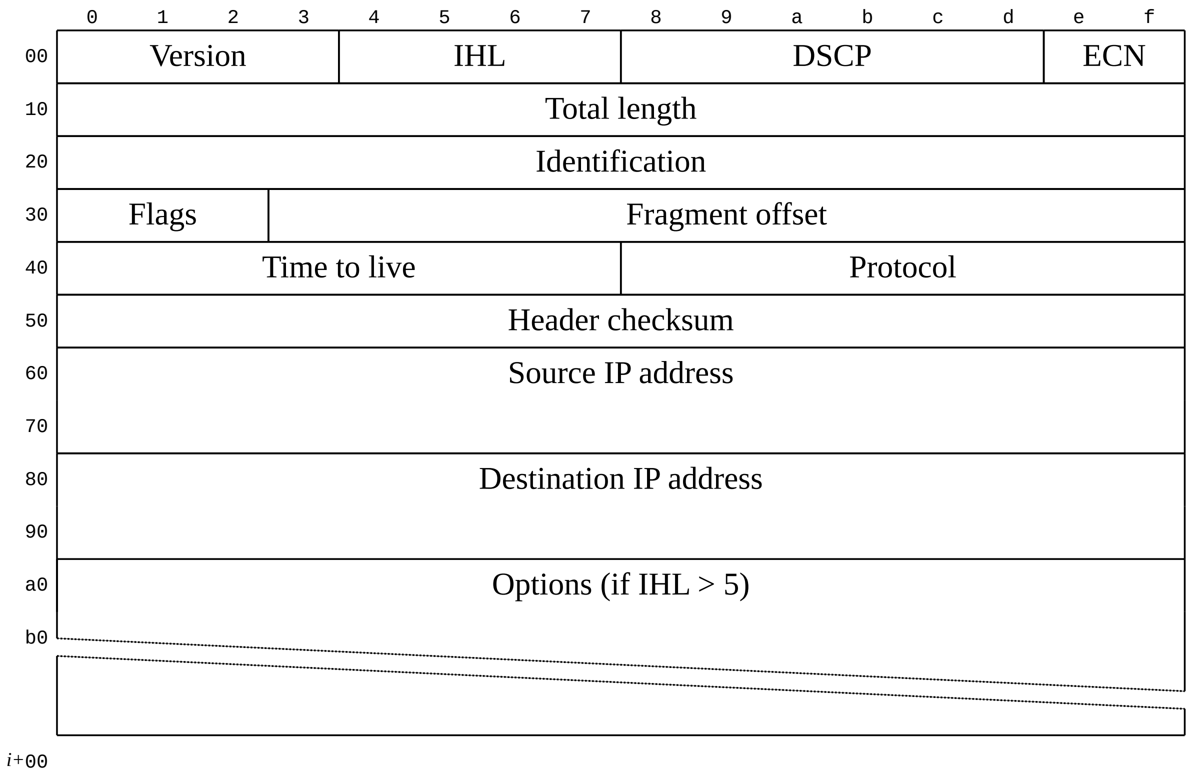
ccsds.edn



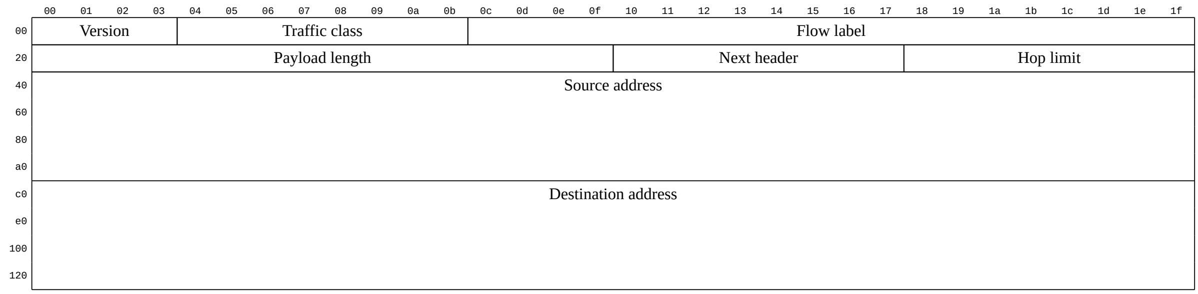
ethernet.edn



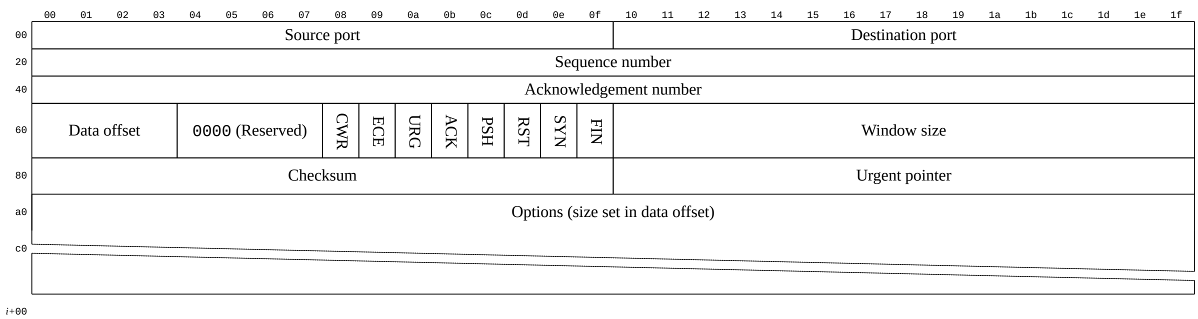
ipv4.edn



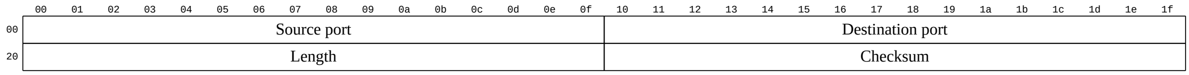
ipv6.edn



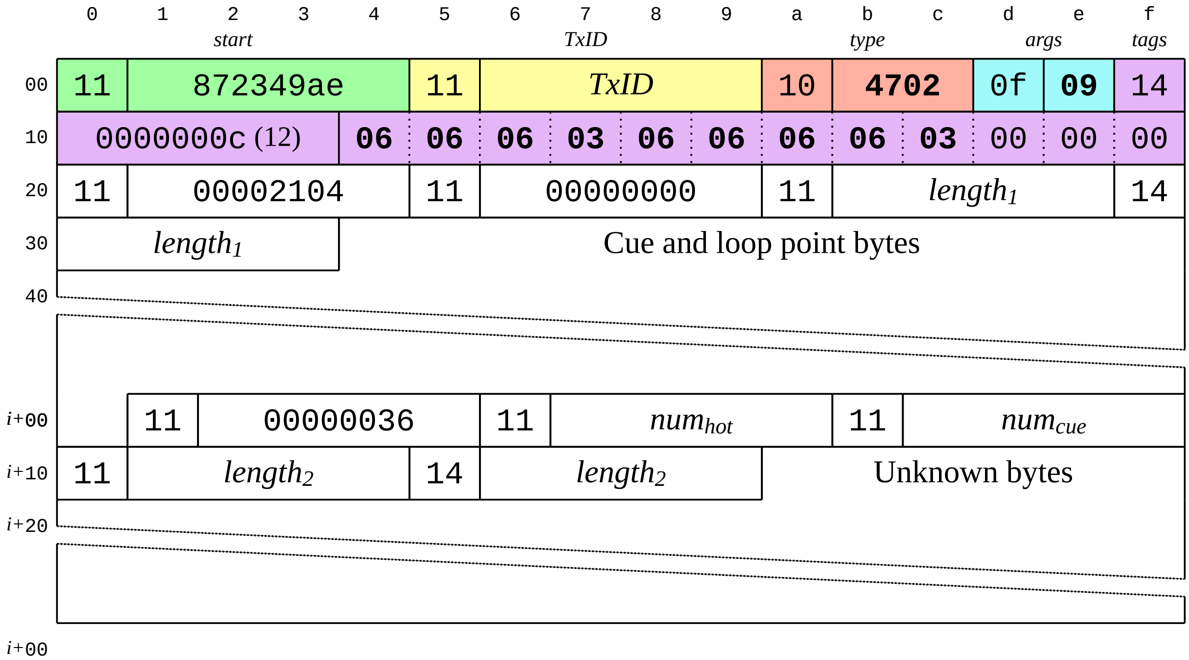
tcp.edn



udp.edn



test.edn



Latex Pictures

```
# node-tikzjax
chemfig
tikz-cd
circuitikz
pgfplots
array
amsmath
amstext
amsfonts
amssymb
tikz-3dplot
```

tikz demo1

```
% demo1
\documentclass[tikz]{standalone}

\usepackage{pgfplots}
\pgfplotsset{compat=1.16}

\begin{document}

\begin{tikzpicture}
\begin{axis}[colormap/viridis]
\addplot3[
        surf,
        samples=18,
        domain=-3:3
]
{exp(-x^2-y^2)*x};
\end{axis}
\end{tikzpicture}

\end{document}
```

tikz demo2

```
% demo2
\documentclass[tikz]{standalone}

\begin{document}
  \begin{tikzpicture}[domain=0:4]
    \draw[very thin,color=gray] (-0.1,-1.1) grid (3.9,3.9);
    \draw[->] (-0.2,0) -- (4.2,0) node[right] {$x$};
    \draw[->] (0,-1.2) -- (0,4.2) node[above] {$f(x)$};
    \draw[color=red]    plot (\x,\x)             node[right] {$f(x) =x$};
    \draw[color=blue]   plot (\x,{sin(\x r)})    node[right] {$f(x) = \sin x$};
    \draw[color=orange] plot (\x,{0.05*exp(\x)}) node[right] {$f(x) = \frac{1}{20} \mathrm e^x$};
  \end{tikzpicture}
\end{document}
```

tikz demo3

```
% demo3
\documentclass[tikz]{standalone}

\usepackage{circuitikz}
\begin{document}

\begin{circuitikz}[american, voltage shift=0.5]
\draw (0,0)
to[isource, l=$I_0$, v=$V_0$] (0,3)
to[short, -*, i=$I_0$] (2,3)
to[R=$R_1$, i>_=$i_1$] (2,0) -- (0,0);
\draw (2,3) -- (4,3)
to[R=$R_2$, i>_=$i_2$]
(4,0) to[short, -*] (2,0);
\end{circuitikz}

\end{document}
```

tikz demo4

```
% demo4
\documentclass[tikz]{standalone}

\usepackage{tikz-cd}

\begin{document}

\begin{tikzcd}
  T
  \arrow[drr, bend left, "x"]
  \arrow[ddr, bend right, "y"]
  \arrow[dr, dotted, "{(x,y)}" description] & & \\
  K & X \times_Z Y \arrow[r, "p"] \arrow[d, "q"]
  & X \arrow[d, "f"] \\
  & Y \arrow[r, "g"]
  & Z
\end{tikzcd}

\quad \quad

\begin{tikzcd}[row sep=2.5em]
A' \arrow[rr,"f'"] \arrow[dr,swap,"a"] \arrow[dd,swap,"g'"] &&
  B' \arrow[dd,swap,"h'" near start] \arrow[dr,"b"] \\
& A \arrow[rr,crossing over,"f" near start] &&
  B \arrow[dd,"h"] \\
C' \arrow[rr,"k'" near end] \arrow[dr,swap,"c"] && D' \arrow[dr,swap,"d"]
\\
& C \arrow[rr,"k"] \arrow[uu,<-,crossing over,"g" near end]&& D
\end{tikzcd}

\end{document}
```

tikz demo5

```
% demo5
\documentclass[tikz]{standalone}

\begin{document}
\begin{tikzpicture}
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=blue] (1) at(0,2){$x_1$};
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=blue] (2) at(0,0){$x_2$};
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=orange] (3) at(2,-1){$a_3^{(2)}$};
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=orange] (4) at(2,1){$a_2^{(2)}$};
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=orange] (5) at(2,3){$a_1^{(2)}$};
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=orange] (6) at(4,-1){$a_3^{(3)}$};
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=orange] (7) at(4,1){$a_2^{(3)}$};
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=orange] (8) at(4,3){$a_1^{(3)}$};
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=purple] (9) at(6,2){$a_1^{(4)}$};
\node[circle,
minimum width =30pt ,
minimum height =30pt ,draw=purple] (10) at(6,0){$a_2^{(4)}$};
\draw[->] (1) --(3);
\draw[->] (1) --(4);
\draw[->] (1) --(5);
\draw[->] (2) --(3);
\draw[->] (2) --(4);
\draw[->] (2) --(5);
\draw[->] (3) --(6);
\draw[->] (3) --(7);
\draw[->] (3) --(8);
```

```
\draw[->] (4) --(6);
\draw[->] (4) --(7);
\draw[->] (4) --(8);
\draw[->] (5) --(6);
\draw[->] (5) --(7);
\draw[->] (5) --(8);
\draw[->] (6) --(9);
\draw[->] (6) --(10);
\draw[->] (7) --(9);
\draw[->] (7) --(10);
\draw[->] (8) --(9);
\draw[->] (8) --(10);
\end{tikzpicture}
\end{document}
```

tikz demo6

```
% demo6
% Author: Till Tantau
% Source: The PGF/TikZ manual
\documentclass[tikz]{standalone}

\begin{document}
\pagestyle{empty}
\begin{tikzpicture}[scale=3,cap=round]
% Local definitions
\def\costhirty{0.8660256}

% Colors
\colorlet{anglecolor}{green!50!black}
\colorlet{sincolor}{red}
\colorlet{tancolor}{orange!80!black}
\colorlet{coscolor}{blue}

% Styles
\tikzstyle{axes}=[]
\tikzstyle{important line}=[very thick]
\tikzstyle{information text}=[rounded corners,fill=red!10,inner sep=1ex]

% The graphic
\draw[style=help lines,step=0.5cm] (-1.4,-1.4) grid (1.4,1.4);

\draw (0,0) circle (1cm);

\begin{scope}[style=axes]
    \draw[->] (-1.5,0) -- (1.5,0) node[right] {$x$};
    \draw[->] (0,-1.5) -- (0,1.5) node[above] {$y$};

    \foreach \x/\xtext in {-1, -.5/-\frac{1}{2}, 1}
    \draw[xshift=\x cm] (0pt,1pt) -- (0pt,-1pt) node[below,fill=white]
            {$\xtext$};

    \foreach \y/\ytext in {-1, -.5/-\frac{1}{2}, .5/\frac{1}{2}, 1}
    \draw[yshift=\y cm] (1pt,0pt) -- (-1pt,0pt) node[left,fill=white]
            {$\ytext$};
\end{scope}

\filldraw[fill=green!20,draw=anglecolor] (0,0) -- (3mm,0pt) arc(0:30:3mm);
\draw (15:2mm) node[anglecolor] {$\alpha$};

\draw[style=important line,sincolor]
```

```
    (30:1cm) -- node[left=1pt,fill=white] {$\sin \alpha$} +(0,-.5);

\draw[style=important line,coscolor]
    (0,0) -- node[below=2pt,fill=white] {$\cos \alpha$} (\costhirty,0);

\draw[style=important line,tancolor] (1,0) --
    node [right=1pt,fill=white]
    {
    $\displaystyle \tan \alpha \color{black}=
    \frac{{\color{sincolor}\sin \alpha}}{\color{coscolor}\cos \alpha}$
    } (intersection of 0,0--30:1cm and 1,0--1,1) coordinate (t);

\draw (0,0) -- (t);

\draw[xshift=1.85cm] node [right,text width=6cm,style=information text]
    {
    The {\color{anglecolor} angle $\alpha$} is $30^\circ$ in the
    example ($\pi/6$ in radians). The {\color{sincolor}sine of
        $\alpha$}, which is the height of the red line, is
    \[
    {\color{sincolor} \sin \alpha} = 1/2.
    \]
    By the Theorem of Pythagoras we have ${\color{coscolor}\cos^2 \alpha}
+
    {\color{sincolor}\sin^2\alpha} =1$. Thus the length of the blue
    line, which is the {\color{coscolor}cosine of $\alpha$}, must be
    \[
    {\color{coscolor}\cos\alpha} = \sqrt{1 - 1/4} = \textstyle
    \frac{1}{2} \sqrt 3.
    \]%
    This shows that {\color{tancolor}$\tan \alpha$}, which is the
    height of the orange line, is
    \[
    {\color{tancolor}\tan\alpha} = \frac{{\color{sincolor}\sin
        \alpha}}{\color{coscolor}\cos \alpha} = 1/\sqrt 3.
    \]%
    };
\end{tikzpicture}

\end{document}
```

tikz demo7

```
% demo7
\documentclass[tikz]{standalone}

\usetikzlibrary {angles,calc,quotes}

\begin{document}
\begin{tikzpicture}[angle radius=.75cm]

\node (A) at (-2,0)     [red,left]   {$A$};
\node (B) at ( 3,.5)    [red,right]  {$B$};
\node (C) at (-2,2)     [blue,left]  {$C$};
\node (D) at ( 3,2.5)   [blue,right] {$D$};
\node (E) at (60:-5mm)  [below]      {$E$};
\node (F) at (60:3.5cm) [above]      {$F$};

\coordinate (X) at (intersection cs:first line={(A)--(B)}, second line=
{(E)--(F)});
\coordinate (Y) at (intersection cs:first line={(C)--(D)}, second line=
{(E)--(F)});

\path
    (A) edge [red, thick]  (B)
    (C) edge [blue, thick] (D)
    (E) edge [thick]       (F)
    pic ["$\alpha$", draw, fill=yellow]   {angle = F--X--A}
    pic ["$\beta$",  draw, fill=green!30] {angle = B--X--F}
    pic ["$\gamma$", draw, fill=yellow]   {angle = E--Y--D}
    pic ["$\delta$", draw, fill=green!30] {angle = C--Y--E};

\node at ($ (D)!.5!(B) $) [right=1cm,text width=6cm,rounded
corners,fill=red!20,inner sep=1ex]
    {
    When we assume that $\color{red}AB$ and $\color{blue}CD$ are
    parallel, i.\,e., ${\color{red}AB} \mathbin{\|} \color{blue}CD$,
    then $\alpha = \gamma$ and $\beta = \delta$.
    };
\end{tikzpicture}
\end{document}
```

tikz demo8

```
% demo8
\documentclass[tikz]{standalone}

\usepackage{chemfig}
\begin{document}

\chemfig{[:-90]HN(-[::-45](-[::-45]R)=[::+45]O)>[::+45]*4(-(=O)-N*5(-(<:(=[::-60]O)-[::+60]OH)-(<[::+0])(<:[::-108])-S>)--)}

\end{document}
```

tikz demo9

```
\documentclass[tikz]{standalone}

\begin{document}

\begin{tikzpicture}

  \node (so32) [align=center] at (-5,-1) {heterotic\\$SO(32)$};
  \node (e8e8) [align=center] at (-3,4) {heterotic\\$E(8) \times E(8)$};
  \node (tiia) [align=center] at (4,3) {Type II A};
  \node (tiib) [align=center] at (5,-2) {Type II B};
  \node (ti) [align=center] at (0,-5) {Type I};

  \draw[bend left,<->] (so32) to node [below right,align=center] {compac-
\\tification} (e8e8);
  \draw[bend left,<->] (e8e8) to node [below left] {M-theory} (tiia);
  \draw[bend left,<->] (tiia) to node [below left] {T-duality} (tiib);
  \draw[bend left,<->] (tiib) to node [above left,align=center]
{orientifold\\action $\Omega$} (ti);
  \draw[bend left,<->] (ti) to node [above right] {S-duality} (so32);

  \begin{scope}
    \clip[bend right]
    (so32.east)
    to (e8e8.south)
    to (tiia.south)
    to (tiib.west)
    to (ti.north)
    to (so32.east);
    \foreach \c in
{so32.east,e8e8.south,tiia.south,tiib.west,ti.north,so32.east}{%
        \foreach \r in {1,...,6}{%
            \draw[DarkBlue] (\c) circle (\r*0.15cm);
          }
      }
  \end{scope}

  \draw[bend right,very thick,gray,fill,fill opacity=0.3] (so32.east) to
(e8e8.south) to (tiia.south) to (tiib.west) to (ti.north) to (so32.east);

  \node (mth) [align=center] at (0,0) {parameter space of\\[2ex]{\Large
\textbf{M-Theory}}};

\end{tikzpicture}
```

```
\end{document}
```

TexNote.

```
# latex    c/c++/rust
\#      ==> \\\#
\$      ==> \\\$
\%      ==> \\\%
\&      ==> \\\&
\_      ==> \\\_
\{      ==> \\\{
\}      ==> \\\}
\~{}    ==> \\\~\{\}
\^{}    ==> \\\^\{\}
\       ==> \\
\<      ==> \\\<
\>      ==> \\\>
```

Latex Document.

Latex demo

test in margin here

LATEX.js Showcase

made with ♥ by Michael Brade

2017–2021

Abstract

This document will show most of the features of LATEX.js while at the same time being a gentle introduction to LATEX. In the appendix, the API as well as the format of custom macro definitions in JavaScript will be explained.

1 Characters

It is possible to input any UTF-8 character either directly or by character code using one of the following:

- `\symbol{"00A9}`: ©
- `\char"A9`: ©
- `^^A9 or ^^^^00A9`: © or ©

Special characters, like those:

$ & % # _ { } ~ ^ \

have to be escaped. More than 200 symbols are accessible through macros. For instance: 30 ℃ is 86 °F. See also Section 11.

2 Spaces and Comments

Spaces and comments, of course, work just as they do in LATEX. This is an example: Supercalifragilisticexpialidocious It does not matter whether you enter one or several spaces after a word, it will always be typeset as one space—unless you force several spaces, like now. New TEXusers may miss whitespaces after a command. Experienced TEX users are TEXperts, and know how to use whitespaces. Longer comments can be embedded in the `comment` environment: This is another example for embedding comments in your document.

3 Dashes and Hyphens

LaTeX knows four kinds of dashes. Access three of them with different numbers of consecutive dashes. The fourth sign is actually not a dash at all—it is the mathematical minus sign:

> daughter-in-law, X-rated
> pages 13–67
> yes—or no?
> 0, 1 and −1

The names for these dashes are: '-' hyphen, '–' en-dash, '—' em-dash, and '−' minus sign. LaTeX.js outputs the actual true unicode character for those instead of using the hypen-minus.

4 Text and Paragraphs, Ligatures

An empty line starts a new paragraph, and so does `\par`.

Like this. A new line usually starts automatically when the previous one is full. However, using `\newline` or `\\`,
one can force
to start a new line. Ligatures are supported as well, for instance:

fi, fl, ff, ffi, ffl ... instead of fi, fl, ffl ...

Use `\/` to prevent a ligature.

4.1 Multicolumns

The multi-column layout, using the `multicols` environment, allows easy definition of multiple columns of text—just like in newspapers. The first and mandatoriy argument specifies the number of columns the text should be divided into. It is often convenient to spread some text over all columns, just before the multicolumn output. In LaTeX, this was needed to prevent any page break in between. To achieve this, the `multicols` environment has an optional second argument which can be used for this purpose. For instance, this text you are reading now was started with the argument `\subsection{Multicolumns}`.

5 Boxes

LaTeX.js supports most of the standard LaTeX boxes.

`\mbox{`*text*`}`

We already know one of them: it's called `\mbox`. It simply packs up a series of boxes into another one, and can be used to prevent LaTeX.js from breaking two words. As boxes can be put inside boxes, these horizontal box packers give you ultimate flexibility.

`\makebox[`*width*`][`*pos*`]{`*text*`}`

`\framebox[`*width*`][`*pos*`]{`*text*`}`

width defines the width of the resulting box as seen from the outside. The *pos* parameter takes a one letter value: **c**enter, flush**l**eft, or flush**r**ight. **s**pread is not really working in HTML. The command `framebox` works exactly the same as `makebox`, but it draws a box around the text. The following example shows you some things you could do with the `makebox` and `framebox` commands.

c e **n** t r a l

Guess I'm framed now!

Bummer, I am too wide

never mind, so am I Can you read this?

`\parbox[`*pos*`][`*height*`][`*inner-pos*`]{`*width*`}{`*text*`}`

The `\parbox` command produces a box the contents of which are created in paragraph mode. However, only small pieces of text should be used, paragraph-making environments shouldn't be used inside a `\parbox` argument. For larger pieces of text, including ones containing a paragraph-making environment, you should use a `minipage` environment. By default LaTeX will position vertically a parbox so its center lines up with the center of the surrounding text line. When the optional position argument is present and equal either to '`t`' or '`b`', this allows you respectively to align either the top or bottom line in the parbox with the baseline of the surrounding text. You may also specify '`m`' for position to get the default behaviour. The optional *height* argument overrides the natural height of the box. The *inner-pos* argument controls the placement of the text inside the box, as follows; if it is not specified, *pos* is used. `t` text is placed at the top of the box. `c` text is centered in the box. `b` text is placed at the bottom of the box. `s` is not supported in HTML

The following examples demonstrate simple positioning:

- simple alignment: Some text parbox default alignment, parbox test text some text parbox top alignment, text parbox test text some text parbox bottom alignment, text parbox test text some text.
- alignment with a given height: Some text parbox default alignment, parbox test text some text parbox top alignment, text parbox test text some text parbox bottom alignment, text parbox test text some text.

The following examples demonstrate all explicit *pos*/*inner-pos* combinations:

- center alignment: Some text parbox default alignment, parbox test text some text

  parbox top alignment, text parbox test text some text parbox bottom alignment, text parbox test text some text.

- top alignment: Some text parbox default alignment, parbox test text some text

  parbox top alignment, text parbox test text some text parbox bottom alignment, text parbox test text some text.

- bottom alignment: Some text parbox default alignment, parbox test text some text

  parbox top alignment, text parbox test text some text parbox bottom alignment, text parbox test text some text.

- top/bottom in one line: Some text parbox default alignment, parbox test text some text parbox top alignment, text parbox test text some text.

5.1 Low-level box-interface

LaTeX.js supports the following low-level TeX commands: `\llap{`*text*`}`, `\rlap{`*text*`}`, and `\smash{`*text*`}`, as well as `\hphantom{`*text*`}`, `\vphantom{`*text*`}`, and `\phantom{`*text*`}`. A phantom looks like this: yes, now the phantom is gone. `\llap` could be used to put something in the margin. However, there are better alternatives for that. See `\marginpar`.

6 Spacing

The following horizontal spaces are supported:

Negative thin space: ||
No space (natural): ||
Thin space: | | or | |
Normal space: | |
Normal space: | |
Non-break space: | |
en-space: | |
em-space: | |
2x em-space: | |
3cm horizontal space: | |

7 Environments

7.1 Lists: Itemize, Enumerate, and Description

The `itemize` environment is suitable for simple lists, the `enumerate` environment for enumerated lists, and the `description` environment for descriptions.

1. You can nest the list environments to your taste:
    - But it might start to look silly.

- With a dash.

2. Therefore remember:

   Stupid things will not become smart because they are in a list.

   Smart things, though, can be presented beautifully in a list.

important Technical note: Viewing this in Chrome, however, will show too much vertical space at the end of a nested environment (see above). On top of that, margin collapsing for inline-block boxes is not allowed. Maybe using `dl` elements is too complicated for this and a simple nested `div` should be used instead.

Lists can be deeply nested:

- list text, level one
  - list text, level two
    - list text, level three And a new paragraph can be started, too.
      - list text, level four And a new paragraph can be started, too. This is the maximum level.
      - list text, level four
    - list text, level three
  - list text, level two
- list text, level one
- list text, level one

7.2 Flushleft, Flushright, and Center

The `flushleft` environment:

This text is
left-aligned. LaTeX is not trying to make each line the same length.

The `flushright` environment:

This text is right-aligned. LaTeX is not trying to make each line the same length.

And the `center` environment:

At the centre
of the earth

7.3 Quote, Quotation, and Verse

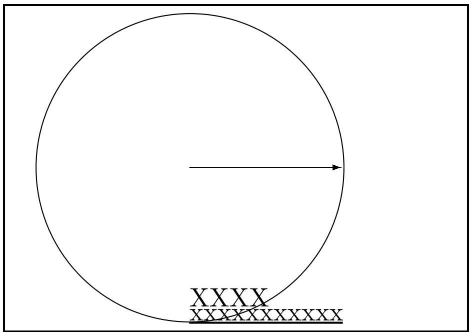
The `quote` environment is useful for quotes, important phrases and examples. A typographical rule of thumb for the line length is:

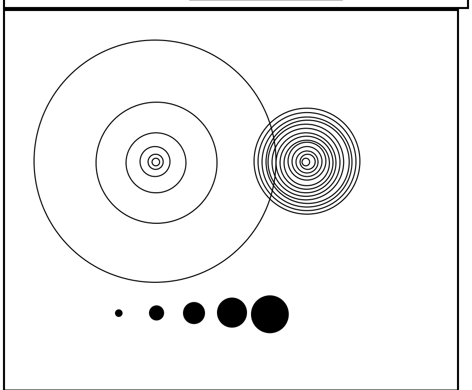
> On average, no line should be longer than 66 characters.

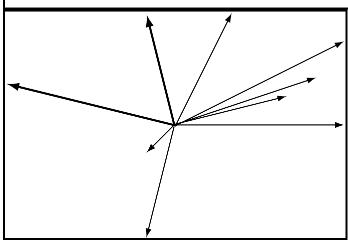
There are two similar environments: the `quotation` and the `verse` environments. The `quotation` environment is useful for longer quotes going over several paragraphs, because it indents the first line of each paragraph. The `verse` environment is useful for poems where the line breaks are important. The lines are separated by issuing a \\ at the end of a line and an empty line after each verse.

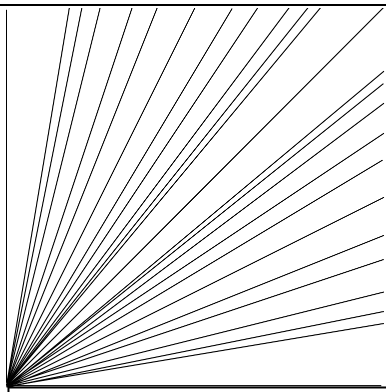
> Humpty Dumpty sat on a wall:
> Humpty Dumpty had a great fall.
> All the King's horses and all the King's men
> Couldn't put Humpty together again. —J.W. Elliott

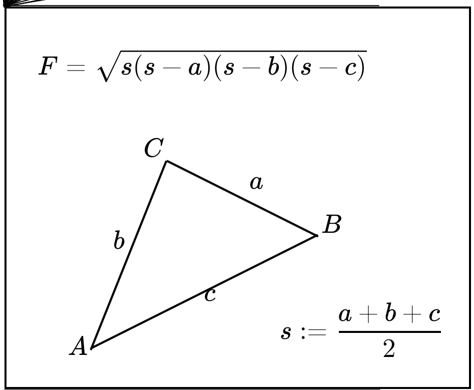
7.4 Picture

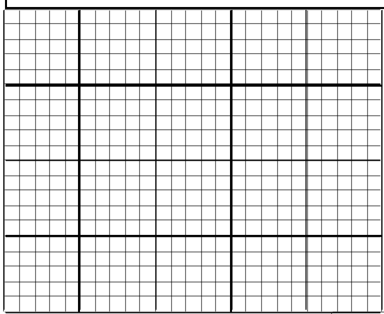


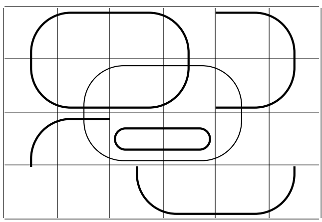


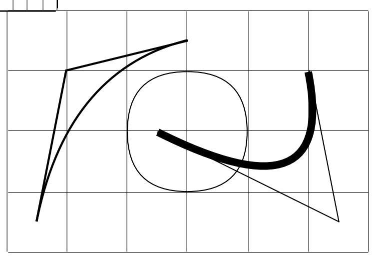












8 Labels and References

Labels can be attached to parts, chapters, sections, items of enumerations, footnotes, tables and figures. For instance: item 2 was important, and regarding fonts, read Section 12.1. And below, we can reference item 1(a)iB and 1b.

1. list text, level one
    (a) list text, level two

i. list text, level three And a new paragraph can be started, too.
   A. list text, level four And a new paragraph can be started, too. This is the maximum level.
   B. list text, level four
ii. list text, level three

(b) list text, level two

2. list text, level one
3. list text, level one

9 Mathematical Formulae

Math is typeset using KaTeX. Inline math: $f(x) = \int_{-\infty}^{\infty} \hat{f}(\xi)\, e^{2\pi i \xi x}\, d\xi$ as well as display math is supported:

$$f(n) = \begin{cases} \frac{n}{2}, & \text{if } n \text{ is even} \\ 3n+1, & \text{if } n \text{ is odd} \end{cases}$$

10 Groups

Today is Monday, April 28, 2025. Actually, what about some groups? They are nice.

11 Symbols

lowercase greek letters: αβγδεζηϑικλμνξοπρστυφχψωuppercase greek letters: ΑΒΓΔΕΖΗΘΙΚΛΜΝΞΟΠΡΣΤΥΦΧΨΩcurrencies: €¢££₿₵¤₫ƒ₤₦₱₩¥old-style numerals: 0123456789math: ‰ ‰ ‱$\frac{131}{244}$/÷×−±√¬∗^{123}arrows: ←↑→↓misc: ✓ ※ ªºꝏ‹|›‖¦ ◯ ⓒⓒⓟⓡ℠™ № ℞℮ ♪ ⁒non-ASCII: ÆæĲĳŒœÞþẞßĐđØøÐđŁłȷŊŋ

12 Fonts

Usually, LaTeX.js chooses the right font—just like LaTeX. In some cases, one might like to change fonts and sizes by hand. To do this, use the standard commands. The actual size of each font is a design issue and depends on the document class (in this case on the CSS file). The small and **bold** Romans ruled all of great big *Italy. You can also* emphasize *text if it is set in italics,* in a *sans-serif* font, or in `typewriter` style. The environment form of the font commands is available, too:

This whole paragraph is emphasized, for instance.

12.1 An advice

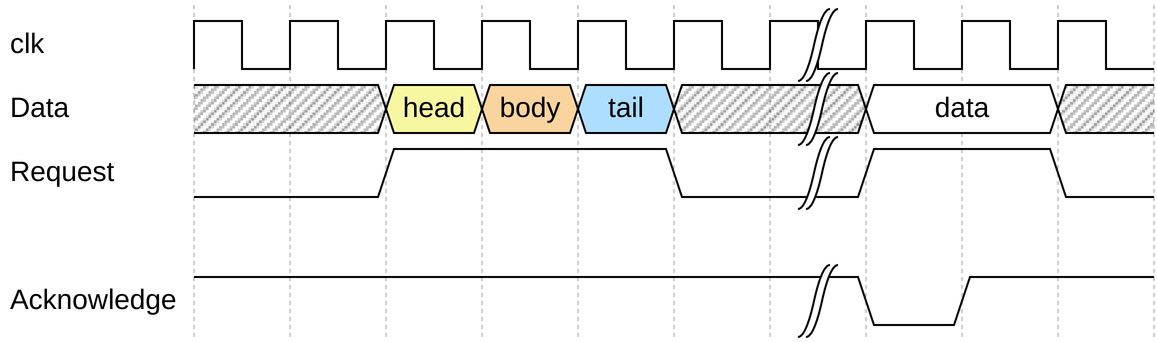
Remember! *The* MORE fonts you use in a `document`, *the* more READABLE and *beautiful it becomes*.

A Source

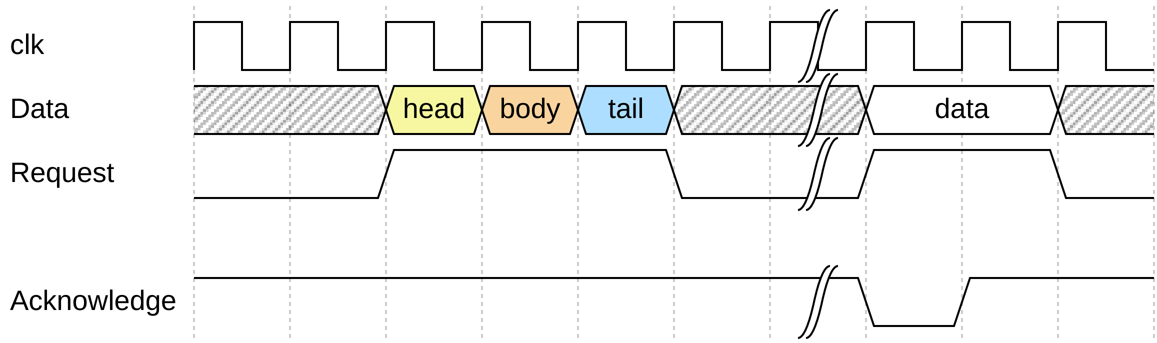
The source of LaTeX.js is here on GitHub: https://github.com/michael-brade/LaTeX.js

WaveDrom

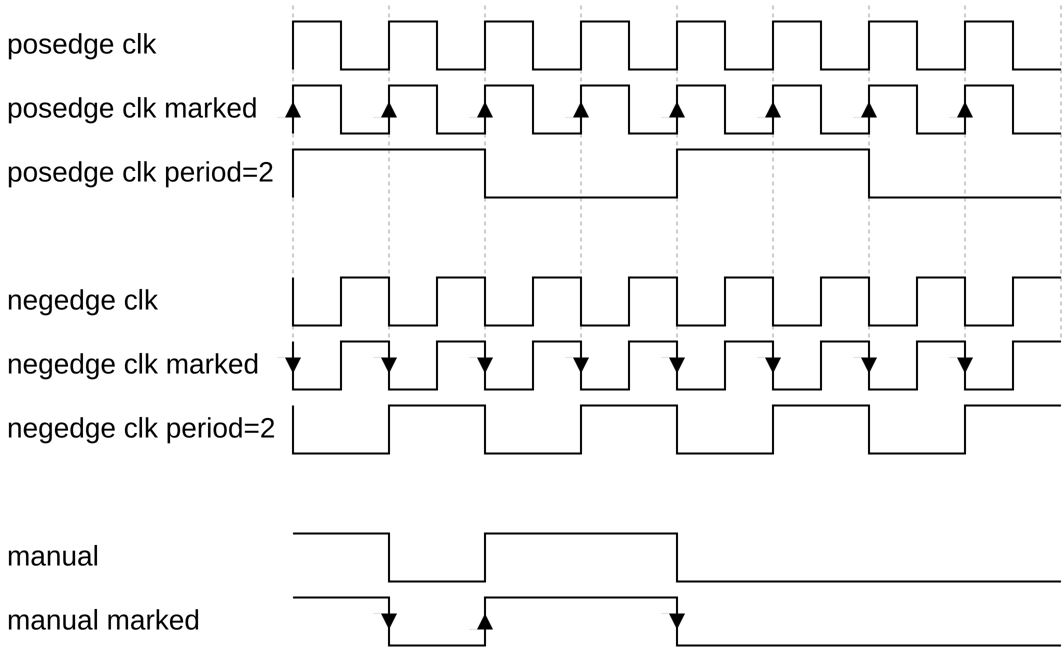
basic1



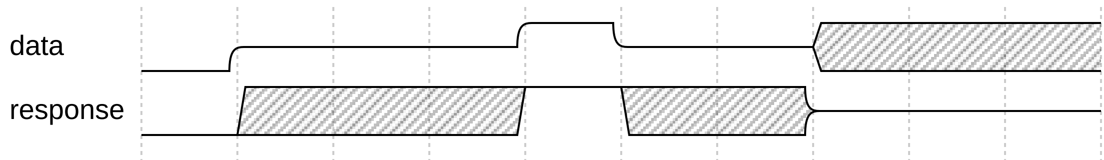
basic2



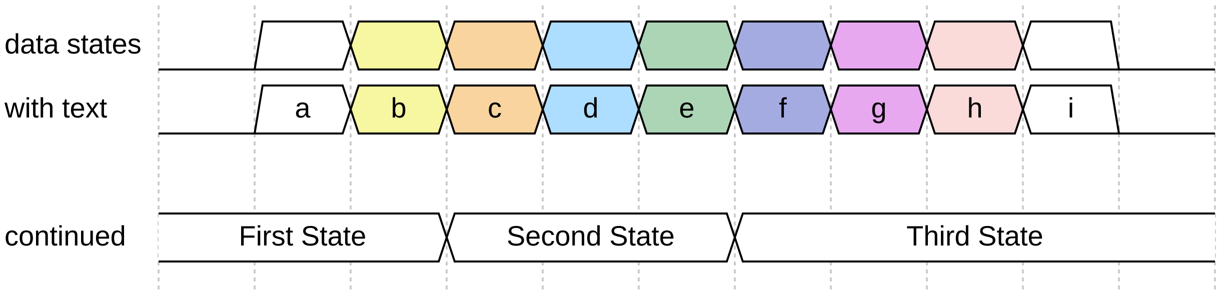
clock



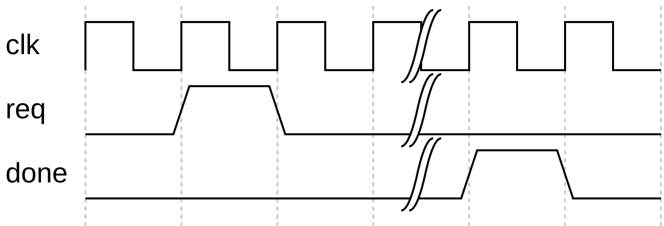
High Impedance & Undefined



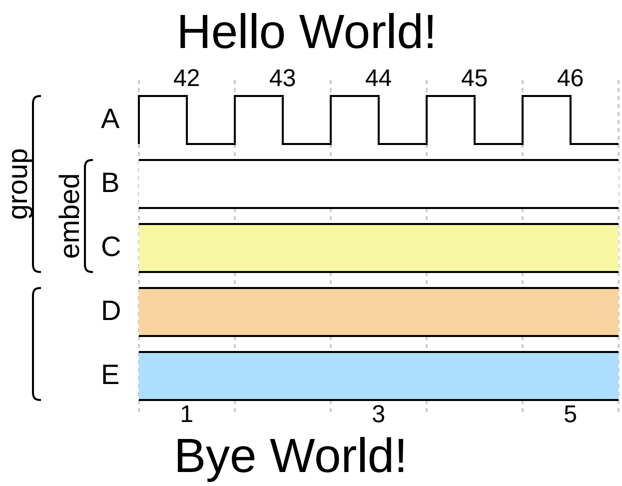
Data types



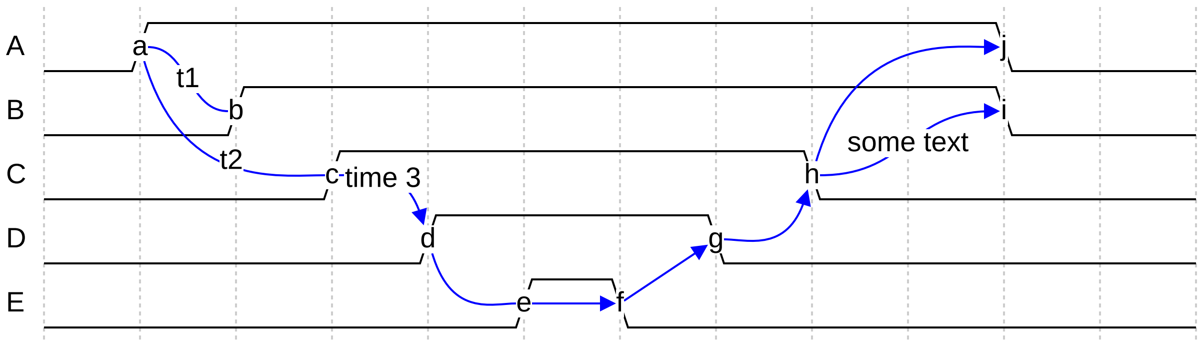
Graps



Groups



Edges



emoji-cheat-sheet

This cheat sheet is automatically generated from GitHub Emoji API and Unicode Full Emoji List.

Table of Contents

- Smileys & Emotion
- People & Body
- Animals & Nature
- Food & Drink
- Travel & Places
- Activities
- Objects
- Symbols
- Flags
- GitHub Custom Emoji

Smileys & Emotion

- Face Smiling
- Face Affection
- Face Tongue
- Face Hand
- Face Neutral Skeptical
- Face Sleepy
- Face Unwell
- Face Hat
- Face Glasses
- Face Concerned

- Face Negative
- Face Costume
- Cat Face
- Monkey Face
- Heart
- Emotion

Face Smiling

ico	shortcode	ico	shortcode
😀	`grinning`	😃	`smiley`
😄	smile	😁	grin
😆	`laughing` `satisfied`	😅	`sweat_smile`
🤣	rofl	😂	joy
🙂	`slightly_smiling_face`	🙃	`upside_down_face`
😉	wink	😊	blush
😇	`innocent`		

Face Affection

ico	shortcode	ico	shortcode
🥰	`smiling_face_with_three_hearts`	😍	`heart_eyes`
🤩	star_struck	😘	kissing_heart
😗	`kissing`	☺️	`relaxed`
😚	kissing_closed_eyes	😙	kissing_smiling_eyes
🥲	`smiling_face_with_tear`		

Face Tongue

ico	shortcode	ico	shortcode
😋	`yum`	😛	`stuck_out_tongue`
😜	`stuck_out_tongue_winking_eye`	🤪	`zany_face`
😝	`stuck_out_tongue_closed_eyes`	🤑	`money_mouth_face`

Face Hand

ico	shortcode	ico	shortcode
🤗	`hugs`	🤭	`hand_over_mouth`
🤫	`shushing_face`	🤔	`thinking`

Face Neutral Skeptical

ico	shortcode	ico	shortcode
🤐	`zipper_mouth_face`	🤨	`raised_eyebrow`
😐	`neutral_face`	😑	`expressionless`
😶	`no_mouth`	😶‍🌫️	`face_in_clouds`
😏	`smirk`	😒	`unamused`
🙄	`roll_eyes`	😬	`grimacing`
😮‍💨	`face_exhaling`	🤥	`lying_face`

Face Sleepy

ico	shortcode	ico	shortcode
😌	`relieved`	😔	`pensive`
😪	`sleepy`	🤤	`drooling_face`
😴	`sleeping`		

Face Unwell

ico	shortcode	ico	shortcode
😷	`mask`	🤒	`face_with_thermometer`
🤕	`face_with_head_bandage`	🤢	`nauseated_face`
🤮	`vomiting_face`	🤧	`sneezing_face`
🥵	`hot_face`	🥶	`cold_face`
🥴	`woozy_face`	😵	`dizzy_face`
😵‍💫	`face_with_spiral_eyes`	🤯	`exploding_head`

Face Hat

ico	shortcode	ico	shortcode
🤠	`cowboy_hat_face`	🥳	`partying_face`
🥸	`disguised_face`		

Face Glasses

ico	shortcode	ico	shortcode
😎	`sunglasses`	🤓	`nerd_face`
🧐	`monocle_face`		

Face Concerned

ico	shortcode	ico	shortcode
😕	`confused`	😟	`worried`
🙁	`slightly_frowning_face`	☹️	`frowning_face`
😮	`open_mouth`	😯	`hushed`
😲	`astonished`	😳	`flushed`

ico	shortcode	ico	shortcode
🥺	`pleading_face`	😦	`frowning`
😧	`anguished`	😨	`fearful`
😰	`cold_sweat`	😥	`disappointed_relieved`
😢	`cry`	😭	`sob`
😱	`scream`	😖	`confounded`
😣	`persevere`	😞	`disappointed`
😓	`sweat`	😩	`weary`
😫	`tired_face`	🥱	`yawning_face`

Face Negative

ico	shortcode	ico	shortcode
😤	`triumph`	😡	`pout` `rage`
😠	`angry`	🤬	`cursing_face`
😈	`smiling_imp`	👿	`imp`
💀	`skull`	☠️	`skull_and_crossbones`

Face Costume

ico	shortcode	ico	shortcode

💩	`hankey`
	`poop`
	`shit`
		🤡	`clown_face`

👹	`japanese_ogre`	👺	`japanese_goblin`
👻	`ghost`	👽	`alien`
👾	`space_invader`	🤖	`robot`

Cat Face

ico	shortcode	ico	shortcode
😺	`smiley_cat`	😸	`smile_cat`
😹	`joy_cat`	😻	`heart_eyes_cat`
😼	`smirk_cat`	😽	`kissing_cat`
🙀	`scream_cat`	😿	`crying_cat_face`
😾	`pouting_cat`		

Monkey Face

ico	shortcode	ico	shortcode
🙈	`see_no_evil`	🙉	`hear_no_evil`
🙊	`speak_no_evil`		

Heart

ico	shortcode	ico	shortcode
💌	`love_letter`	💘	`cupid`
💝	`gift_heart`	💖	`sparkling_heart`
💗	`heartpulse`	💓	`heartbeat`
💞	`revolving_hearts`	💕	`two_hearts`
💟	`heart_decoration`	❣️	`heavy_heart_exclamation`
💔	`broken_heart`	❤️‍🔥	`heart_on_fire`
❤️‍🩹	`mending_heart`	❤️	`heart`
🧡	`orange_heart`	💛	`yellow_heart`
💚	`green_heart`	💙	`blue_heart`
💜	`purple_heart`	🤎	`brown_heart`
🖤	`black_heart`	🤍	`white_heart`

Emotion

ico	shortcode	ico	shortcode
💋	`kiss`	💯	`100`
💢	anger	💥	boom collision
💫	`dizzy`	💦	`sweat_drops`
💨	dash	🕳️	hole
💬	`speech_balloon`	👁️‍🗨️	`eye_speech_bubble`
🗨️	left_speech_bubble	🗯️	right_anger_bubble
💭	`thought_balloon`	💤	`zzz`

People & Body

- Hand Fingers Open
- Hand Fingers Partial
- Hand Single Finger
- Hand Fingers Closed
- Hands
- Hand Prop
- Body Parts
- Person
- Person Gesture
- Person Role
- Person Fantasy
- Person Activity
- Person Sport
- Person Resting
- Family
- Person Symbol

Hand Fingers Open

ico	shortcode	ico	shortcode
👋	`wave`	🤚	`raised_back_of_hand`
🖐️	`raised_hand_with_fingers_splayed`	✋	`hand` `raised_hand`
🖖	`vulcan_salute`		

Hand Fingers Partial

ico	shortcode	ico	shortcode
👌	`ok_hand`	🤌	`pinched_fingers`
🤏	`pinching_hand`	✌️	`v`
🤞	`crossed_fingers`	🤟	`love_you_gesture`
🤘	`metal`	🤙	`call_me_hand`

Hand Single Finger

ico	shortcode	ico	shortcode
👈	`point_left`	👉	`point_right`
👆	`point_up_2`	🖕	`fu` `middle_finger`
👇	`point_down`	☝️	`point_up`

Hand Fingers Closed

ico	shortcode	ico	shortcode
👍	`+1` `thumbsup`	👎	`-1` `thumbsdown`

ico	shortcode	ico	shortcode
✊	fist fist_raised	👊	facepunch fist_oncoming punch
🤛	fist_left	🤜	fist_right

Hands

ico	shortcode	ico	shortcode
👏	clap	🙌	raised_hands
👐	open_hands	🤲	palms_up_together
🤝	handshake	🙏	pray

Hand Prop

ico	shortcode	ico	shortcode
✍️	writing_hand	💅	nail_care
🤳	selfie		

Body Parts

ico	shortcode	ico	shortcode
💪	muscle	🦾	mechanical_arm
🦿	mechanical_leg	🦵	leg
🦶	foot	👂	ear
🦻	ear_with_hearing_aid	👃	nose
🧠	brain	🫀	anatomical_heart
🫁	lungs	🦷	tooth
🦴	bone	👀	eyes

ico	shortcode	ico	shortcode
👁️	`eye`	👅	`tongue`
👄	`lips`		

Person

ico	shortcode	ico	shortcode
👶	`baby`	🧒	`child`
👦	`boy`	👧	`girl`
🧑	`adult`	👱	`blond_haired_person`
👨	`man`	🧔	`bearded_person`
🧔‍♂️	`man_beard`	🧔‍♀️	`woman_beard`
👨‍🦰	`red_haired_man`	👨‍🦱	`curly_haired_man`
👨‍🦳	`white_haired_man`	👨‍🦲	`bald_man`
👩	`woman`	👩‍🦰	`red_haired_woman`
🧑‍🦰	`person_red_hair`	👩‍🦱	`curly_haired_woman`
🧑‍🦱	`person_curly_hair`	👩‍🦳	`white_haired_woman`
🧑‍🦳	`person_white_hair`	👩‍🦲	`bald_woman`
🧑‍🦲	`person_bald`	👱‍♀️	`blond_haired_woman` `blonde_woman`
👱‍♂️	`blond_haired_man`	🧓	`older_adult`
👴	`older_man`	👵	`older_woman`

Person Gesture

ico	shortcode	ico	shortcode
🙍	`frowning_person`	🙍‍♂️	`frowning_man`
🙍‍♀️	`frowning_woman`	🙎	`pouting_face`
🙎‍♂️	`pouting_man`	🙎‍♀️	`pouting_woman`

ico	shortcode	ico	shortcode
🙅	`no_good`	🙅‍♂️	`ng_man` `no_good_man`
🙅‍♀️	`ng_woman` `no_good_woman`	🙆	`ok_person`
🙆‍♂️	`ok_man`	🙆‍♀️	`ok_woman`
💁	`information_desk_person` `tipping_hand_person`	💁‍♂️	`sassy_man` `tipping_hand_man`
💁‍♀️	`sassy_woman` `tipping_hand_woman`	🙋	`raising_hand`
🙋‍♂️	`raising_hand_man`	🙋‍♀️	`raising_hand_woman`
🧏	`deaf_person`	🧏‍♂️	`deaf_man`
🧏‍♀️	`deaf_woman`	🙇	`bow`
🙇‍♂️	`bowing_man`	🙇‍♀️	`bowing_woman`
🤦	`facepalm`	🤦‍♂️	`man_facepalming`
🤦‍♀️	`woman_facepalming`	🤷	`shrug`
🤷‍♂️	`man_shrugging`	🤷‍♀️	`woman_shrugging`

Person Role

ico	shortcode	ico	shortcode
🧑‍⚕️	`health_worker`	👨‍⚕️	`man_health_worker`
👩‍⚕️	`woman_health_worker`	🧑‍🎓	`student`
👨‍🎓	`man_student`	👩‍🎓	`woman_student`
🧑‍🏫	`teacher`	👨‍🏫	`man_teacher`
👩‍🏫	`woman_teacher`	🧑‍⚖️	`judge`
👨‍⚖️	`man_judge`	👩‍⚖️	`woman_judge`
🧑‍🌾	`farmer`	👨‍🌾	`man_farmer`
👩‍🌾	`woman_farmer`	🧑‍🍳	`cook`

ico	shortcode	ico	shortcode
👨‍🍳	`man_cook`	👩‍🍳	`woman_cook`
🧑‍🔧	`mechanic`	👨‍🔧	`man_mechanic`
👩‍🔧	`woman_mechanic`	🧑‍🏭	`factory_worker`
👨‍🏭	`man_factory_worker`	👩‍🏭	`woman_factory_worker`
🧑‍💼	`office_worker`	👨‍💼	`man_office_worker`
👩‍💼	`woman_office_worker`	🧑‍🔬	`scientist`
👨‍🔬	`man_scientist`	👩‍🔬	`woman_scientist`
🧑‍💻	`technologist`	👨‍💻	`man_technologist`
👩‍💻	`woman_technologist`	🧑‍🎤	`singer`
👨‍🎤	`man_singer`	👩‍🎤	`woman_singer`
🧑‍🎨	`artist`	👨‍🎨	`man_artist`
👩‍🎨	`woman_artist`	🧑‍✈️	`pilot`
👨‍✈️	`man_pilot`	👩‍✈️	`woman_pilot`
🧑‍🚀	`astronaut`	👨‍🚀	`man_astronaut`
👩‍🚀	`woman_astronaut`	🧑‍🚒	`firefighter`
👨‍🚒	`man_firefighter`	👩‍🚒	`woman_firefighter`
👮	`cop` `police_officer`	👮‍♂️	`policeman`
👮‍♀️	`policewoman`	🕵️	`detective`
🕵️‍♂️	`male_detective`	🕵️‍♀️	`female_detective`
💂	`guard`	💂‍♂️	`guardsman`
💂‍♀️	`guardswoman`	🥷	`ninja`
👷	`construction_worker`	👷‍♂️	`construction_worker_man`
👷‍♀️	`construction_worker_woman`	🤴	`prince`
👸	`princess`	👳	`person_with_turban`
👳‍♂️	`man_with_turban`	👳‍♀️	`woman_with_turban`
👲	`man_with_gua_pi_mao`	🧕	`woman_with_headscarf`

ico	shortcode	ico	shortcode
🤵	`person_in_tuxedo`	🤵‍♂️	`man_in_tuxedo`
🤵‍♀️	`woman_in_tuxedo`	👰	`person_with_veil`
👰‍♂️	`man_with_veil`	👰‍♀️	`bride_with_veil` `woman_with_veil`
🤰	`pregnant_woman`	🤱	`breast_feeding`
👩‍🍼	`woman_feeding_baby`	👨‍🍼	`man_feeding_baby`
🧑‍🍼	`person_feeding_baby`		

Person Fantasy

ico	shortcode	ico	shortcode
👼	`angel`	🎅	`santa`
🤶	`mrs_claus`	🧑‍🎄	`mx_claus`
🦸	`superhero`	🦸‍♂️	`superhero_man`
🦸‍♀️	`superhero_woman`	🦹	`supervillain`
🦹‍♂️	`supervillain_man`	🦹‍♀️	`supervillain_woman`
🧙	`mage`	🧙‍♂️	`mage_man`
🧙‍♀️	`mage_woman`	🧚	`fairy`
🧚‍♂️	`fairy_man`	🧚‍♀️	`fairy_woman`
🧛	`vampire`	🧛‍♂️	`vampire_man`
🧛‍♀️	`vampire_woman`	🧜	`merperson`
🧜‍♂️	`merman`	🧜‍♀️	`mermaid`
🧝	`elf`	🧝‍♂️	`elf_man`
🧝‍♀️	`elf_woman`	🧞	`genie`
🧞‍♂️	`genie_man`	🧞‍♀️	`genie_woman`
🧟	`zombie`	🧟‍♂️	`zombie_man`
🧟‍♀️	`zombie_woman`		

Person Activity

ico	shortcode	ico	shortcode
💆	`massage`	💆‍♂️	`massage_man`
💆‍♀️	`massage_woman`	💇	`haircut`
💇‍♂️	`haircut_man`	💇‍♀️	`haircut_woman`
🚶	`walking`	🚶‍♂️	`walking_man`
🚶‍♀️	`walking_woman`	🧍	`standing_person`
🧍‍♂️	`standing_man`	🧍‍♀️	`standing_woman`
🧎	`kneeling_person`	🧎‍♂️	`kneeling_man`
🧎‍♀️	`kneeling_woman`	🧑‍🦯	`person_with_probing_car`
👨‍🦯	`man_with_probing_cane`	👩‍🦯	`woman_with_probing_cane`
🧑‍🦼	`person_in_motorized_wheelchair`	👨‍🦼	`man_in_motorized_wheelc`
👩‍🦼	`woman_in_motorized_wheelchair`	🧑‍🦽	`person_in_manual_wheelc`
👨‍🦽	`man_in_manual_wheelchair`	👩‍🦽	`woman_in_manual_wheelch`
🏃	`runner` `running`	🏃‍♂️	`running_man`
🏃‍♀️	`running_woman`	💃	`dancer` `woman_dancing`
🕺	`man_dancing`	🕴️	`business_suit_levitatin`
👯	`dancers`	👯‍♂️	`dancing_men`
👯‍♀️	`dancing_women`	🧖	`sauna_person`
🧖‍♂️	`sauna_man`	🧖‍♀️	`sauna_woman`
🧗	`climbing`	🧗‍♂️	`climbing_man`
🧗‍♀️	`climbing_woman`		

Person Sport

ico	shortcode	ico	shortcode
🤺	`person_fencing`	🏇	`horse_racing`
⛷️	`skier`	🏂	`snowboarder`
🏌️	`golfing`	🏌️‍♂️	`golfing_man`
🏌️‍♀️	`golfing_woman`	🏄	`surfer`
🏄‍♂️	`surfing_man`	🏄‍♀️	`surfing_woman`
🚣	`rowboat`	🚣‍♂️	`rowing_man`
🚣‍♀️	`rowing_woman`	🏊	`swimmer`
🏊‍♂️	`swimming_man`	🏊‍♀️	`swimming_woman`
⛹️	`bouncing_ball_person`	⛹️‍♂️	`basketball_man` `bouncing_ball_man`
⛹️‍♀️	`basketball_woman` `bouncing_ball_woman`	🏋️	`weight_lifting`
🏋️‍♂️	`weight_lifting_man`	🏋️‍♀️	`weight_lifting_woman`
🚴	`bicyclist`	🚴‍♂️	`biking_man`
🚴‍♀️	`biking_woman`	🚵	`mountain_bicyclist`
🚵‍♂️	`mountain_biking_man`	🚵‍♀️	`mountain_biking_woman`
🤸	`cartwheeling`	🤸‍♂️	`man_cartwheeling`
🤸‍♀️	`woman_cartwheeling`	🤼	`wrestling`
🤼‍♂️	`men_wrestling`	🤼‍♀️	`women_wrestling`
🤽	`water_polo`	🤽‍♂️	`man_playing_water_polo`
🤽‍♀️	`woman_playing_water_polo`	🤾	`handball_person`
🤾‍♂️	`man_playing_handball`	🤾‍♀️	`woman_playing_handball`
🤹	`juggling_person`	🤹‍♂️	`man_juggling`
🤹‍♀️	`woman_juggling`		

Person Resting

ico	shortcode	ico	shortcode
🧘	`lotus_position`	🧘‍♂️	`lotus_position_man`
🧘‍♀️	`lotus_position_woman`	🛀	`bath`
🛌	`sleeping_bed`		

Family

ico	shortcode	ico	shortcode
🧑‍🤝‍🧑	`people_holding_hands`	👭	`two_women_holding_hands`
👫	`couple`	👬	`two_men_holding_hands`
💏	`couplekiss`	👩‍❤️‍💋‍👨	`couplekiss_man_woman`
👨‍❤️‍💋‍👨	`couplekiss_man_man`	👩‍❤️‍💋‍👩	`couplekiss_woman_woman`
💑	`couple_with_heart`	👩‍❤️‍👨	`couple_with_heart_woman_m`
👨‍❤️‍👨	`couple_with_heart_man_man`	👩‍❤️‍👩	`couple_with_heart_woman_w`
👨‍👩‍👦	`family_man_woman_boy`	👨‍👩‍👧	`family_man_woman_girl`
👨‍👩‍👧‍👦	`family_man_woman_girl_boy`	👨‍👩‍👦‍👦	`family_man_woman_boy_boy`
👨‍👩‍👧‍👧	`family_man_woman_girl_girl`	👨‍👨‍👦	`family_man_man_boy`
👨‍👨‍👧	`family_man_man_girl`	👨‍👨‍👧‍👦	`family_man_man_girl_boy`
👨‍👨‍👦‍👦	`family_man_man_boy_boy`	👨‍👨‍👧‍👧	`family_man_man_girl_girl`
👩‍👩‍👦	`family_woman_woman_boy`	👩‍👩‍👧	`family_woman_woman_girl`
👩‍👩‍👧‍👦	`family_woman_woman_girl_boy`	👩‍👩‍👦‍👦	`family_woman_woman_boy_bo`
👩‍👩‍👧‍👧	`family_woman_woman_girl_girl`	👨‍👦	`family_man_boy`
👨‍👦‍👦	`family_man_boy_boy`	👨‍👧	`family_man_girl`
👨‍👧‍👦	`family_man_girl_boy`	👨‍👧‍👧	`family_man_girl_girl`
👩‍👦	`family_woman_boy`	👩‍👦‍👦	`family_woman_boy_boy`
👩‍👧	`family_woman_girl`	👩‍👧‍👦	`family_woman_girl_boy`

ico	shortcode	ico	shortcode
👩‍👧‍👧	`family_woman_girl_girl`		

Person Symbol

ico	shortcode	ico	shortcode
🗣️	`speaking_head`	👤	`bust_in_silhouette`
👥	`busts_in_silhouette`	🫂	`people_hugging`
👪	`family`	👣	`footprints`

Animals & Nature

- Animal Mammal
- Animal Bird
- Animal Amphibian
- Animal Reptile
- Animal Marine
- Animal Bug
- Plant Flower
- Plant Other

Animal Mammal

ico	shortcode	ico	shortcode
🐵	`monkey_face`	🐒	`monkey`
🦍	`gorilla`	🦧	`orangutan`
🐶	`dog`	🐕	`dog2`
🦮	`guide_dog`	🐕‍🦺	`service_dog`
🐩	`poodle`	🐺	`wolf`
🦊	`fox_face`	🦝	`raccoon`

ico	shortcode	ico	shortcode
🐱	`cat`	🐈	`cat2`
🐈‍⬛	`black_cat`	🦁	`lion`
🐯	`tiger`	🐅	`tiger2`
🐆	`leopard`	🐴	`horse`
🐎	`racehorse`	🦄	`unicorn`
🦓	`zebra`	🦌	`deer`
🦬	`bison`	🐮	`cow`
🐂	`ox`	🐃	`water_buffalo`
🐄	`cow2`	🐷	`pig`
🐖	`pig2`	🐗	`boar`
🐽	`pig_nose`	🐏	`ram`
🐑	`sheep`	🐐	`goat`
🐪	`dromedary_camel`	🐫	`camel`
🦙	`llama`	🦒	`giraffe`
🐘	`elephant`	🦣	`mammoth`
🦏	`rhinoceros`	🦛	`hippopotamus`
🐭	`mouse`	🐁	`mouse2`
🐀	`rat`	🐹	`hamster`
🐰	`rabbit`	🐇	`rabbit2`
🐿️	`chipmunk`	🦫	`beaver`
🦔	`hedgehog`	🦇	`bat`
🐻	`bear`	🐻‍❄️	`polar_bear`
🐨	`koala`	🐼	`panda_face`
🦥	`sloth`	🦦	`otter`
🦨	`skunk`	🦘	`kangaroo`
🦡	`badger`	🐾	`feet` `paw_prints`

Animal Bird

ico	shortcode	ico	shortcode
🦃	`turkey`	🐔	`chicken`
🐓	`rooster`	🐣	`hatching_chick`
🐤	`baby_chick`	🐥	`hatched_chick`
🐦	`bird`	🐧	`penguin`
🕊️	`dove`	🦅	`eagle`
🦆	`duck`	🦢	`swan`
🦉	`owl`	🦤	`dodo`
🪶	`feather`	🦩	`flamingo`
🦚	`peacock`	🦜	`parrot`

Animal Amphibian

ico	shortcode
🐸	`frog`

Animal Reptile

ico	shortcode	ico	shortcode
🐊	`crocodile`	🐢	`turtle`
🦎	`lizard`	🐍	`snake`
🐲	`dragon_face`	🐉	`dragon`
🦕	`sauropod`	🦖	`t-rex`

Animal Marine

ico	shortcode	ico	shortcode
🐳	`whale`	🐋	`whale2`
🐬	`dolphin` `flipper`	🦭	`seal`
🐟	`fish`	🐠	`tropical_fish`
🐡	`blowfish`	🦈	`shark`
🐙	`octopus`	🐚	`shell`

Animal Bug

ico	shortcode	ico	shortcode
🐌	`snail`	🦋	`butterfly`
🐛	`bug`	🐜	`ant`
🐝	`bee` `honeybee`	🪲	`beetle`
🐞	`lady_beetle`	🦗	`cricket`
🪳	`cockroach`	🕷️	`spider`
🕸️	`spider_web`	🦂	`scorpion`
🦟	`mosquito`	🪰	`fly`
🪱	`worm`	🦠	`microbe`

Plant Flower

ico	shortcode	ico	shortcode
💐	`bouquet`	🌸	`cherry_blossom`
💮	`white_flower`	🏵️	`rosette`
🌹	`rose`	🥀	`wilted_flower`

ico	shortcode	ico	shortcode
🌺	hibiscus	🌻	sunflower
🌼	blossom	🌷	tulip

Plant Other

ico	shortcode	ico	shortcode
🌱	seedling	🪴	potted_plant
🌲	evergreen_tree	🌳	deciduous_tree
🌴	palm_tree	🌵	cactus
🌾	ear_of_rice	🌿	herb
☘️	shamrock	🍀	four_leaf_clover
🍁	maple_leaf	🍂	fallen_leaf
🍃	leaves	🍄	mushroom

Food & Drink

- Food Fruit
- Food Vegetable
- Food Prepared
- Food Asian
- Food Marine
- Food Sweet
- Drink
- Dishware

Food Fruit

ico	shortcode	ico	shortcode
🍇	grapes	🍈	melon

ico	shortcode	ico	shortcode
🍉	watermelon	🍊	mandarin orange tangerine
🍋	lemon	🍌	banana
🍍	pineapple	🥭	mango
🍎	apple	🍏	green_apple
🍐	pear	🍑	peach
🍒	cherries	🍓	strawberry
🫐	blueberries	🥝	kiwi_fruit
🍅	tomato	🫒	olive
🥥	coconut		

Food Vegetable

ico	shortcode	ico	shortcode
🥑	avocado	🍆	eggplant
🥔	potato	🥕	carrot
🌽	corn	🌶️	hot_pepper
🫑	bell_pepper	🥒	cucumber
🥬	leafy_green	🥦	broccoli
🧄	garlic	🧅	onion
🥜	peanuts	🌰	chestnut

Food Prepared

ico	shortcode	ico	shortcode
🍞	bread	🥐	croissant
🥖	baguette_bread	🫓	flatbread

ico	shortcode	ico	shortcode
🥨	`pretzel`	🥯	`bagel`
🥞	`pancakes`	🧇	`waffle`
🧀	`cheese`	🍖	`meat_on_bone`
🍗	`poultry_leg`	🥩	`cut_of_meat`
🥓	`bacon`	🍔	`hamburger`
🍟	`fries`	🍕	`pizza`
🌭	`hotdog`	🥪	`sandwich`
🌮	`taco`	🌯	`burrito`
🫔	`tamale`	🥙	`stuffed_flatbread`
🧆	`falafel`	🥚	`egg`
🍳	`fried_egg`	🥘	`shallow_pan_of_food`
🍲	`stew`	🫕	`fondue`
🥣	`bowl_with_spoon`	🥗	`green_salad`
🍿	`popcorn`	🧈	`butter`
🧂	`salt`	🥫	`canned_food`

Food Asian

ico	shortcode	ico	shortcode
🍱	`bento`	🍘	`rice_cracker`
🍙	`rice_ball`	🍚	`rice`
🍛	`curry`	🍜	`ramen`
🍝	`spaghetti`	🍠	`sweet_potato`
🍢	`oden`	🍣	`sushi`
🍤	`fried_shrimp`	🍥	`fish_cake`
🥮	`moon_cake`	🍡	`dango`
🥟	`dumpling`	🥠	`fortune_cookie`

ico	shortcode	ico	shortcode
🥡	`takeout_box`		

Food Marine

ico	shortcode	ico	shortcode
🦀	`crab`	🦞	`lobster`
🦐	`shrimp`	🦑	`squid`
🦪	`oyster`		

Food Sweet

ico	shortcode	ico	shortcode
🍦	`icecream`	🍧	`shaved_ice`
🍨	`ice_cream`	🍩	`doughnut`
🍪	`cookie`	🎂	`birthday`
🍰	`cake`	🧁	`cupcake`
🥧	`pie`	🍫	`chocolate_bar`
🍬	`candy`	🍭	`lollipop`
🍮	`custard`	🍯	`honey_pot`

Drink

ico	shortcode	ico	shortcode
🍼	`baby_bottle`	🥛	`milk_glass`
☕	`coffee`	🫖	`teapot`
🍵	`tea`	🍶	`sake`
🍾	`champagne`	🍷	`wine_glass`
🍸	`cocktail`	🍹	`tropical_drink`

ico	shortcode	ico	shortcode
🍺	`beer`	🍻	`beers`
🥂	`clinking_glasses`	🥃	`tumbler_glass`
🥤	`cup_with_straw`	🧋	`bubble_tea`
🧃	`beverage_box`	🧉	`mate`
🧊	`ice_cube`		

Dishware

ico	shortcode	ico	shortcode
🥢	`chopsticks`	🍽️	`plate_with_cutlery`
🍴	`fork_and_knife`	🥄	`spoon`
🔪	`hocho` `knife`	🏺	`amphora`

Travel & Places

- Place Map
- Place Geographic
- Place Building
- Place Religious
- Place Other
- Transport Ground
- Transport Water
- Transport Air
- Hotel
- Time
- Sky & Weather

Place Map

ico	shortcode	ico	shortcode
🌍	`earth_africa`	🌎	`earth_americas`
🌏	`earth_asia`	🌐	`globe_with_meridians`
🗺️	`world_map`	🗾	`japan`
🧭	`compass`		

Place Geographic

ico	shortcode	ico	shortcode
🏔️	`mountain_snow`	⛰️	`mountain`
🌋	`volcano`	🗻	`mount_fuji`
🏕️	`camping`	🏖️	`beach_umbrella`
🏜️	`desert`	🏝️	`desert_island`
🏞️	`national_park`		

Place Building

ico	shortcode	ico	shortcode
🏟️	`stadium`	🏛️	`classical_building`
🏗️	`building_construction`	🧱	`bricks`
🪨	`rock`	🪵	`wood`
🛖	`hut`	🏘️	`houses`
🏚️	`derelict_house`	🏠	`house`
🏡	`house_with_garden`	🏢	`office`
🏣	`post_office`	🏤	`european_post_office`
🏥	`hospital`	🏦	`bank`
🏨	`hotel`	🏩	`love_hotel`

ico	shortcode	ico	shortcode
🏪	`convenience_store`	🏫	`school`
🏬	`department_store`	🏭	`factory`
🏯	`japanese_castle`	🏰	`european_castle`
💒	`wedding`	🗼	`tokyo_tower`
🗽	`statue_of_liberty`		

Place Religious

ico	shortcode	ico	shortcode
⛪	`church`	🕌	`mosque`
🛕	`hindu_temple`	🕍	`synagogue`
⛩️	`shinto_shrine`	🕋	`kaaba`

Place Other

ico	shortcode	ico	shortcode
⛲	`fountain`	⛺	`tent`
🌁	`foggy`	🌃	`night_with_stars`
🏙️	`cityscape`	🌄	`sunrise_over_mountains`
🌅	`sunrise`	🌆	`city_sunset`
🌇	`city_sunrise`	🌉	`bridge_at_night`
♨️	`hotsprings`	🎠	`carousel_horse`
🎡	`ferris_wheel`	🎢	`roller_coaster`
💈	`barber`	🎪	`circus_tent`

Transport Ground

ico	shortcode	ico	shortcode
🚂	`steam_locomotive`	🚃	`railway_car`
🚄	`bullettrain_side`	🚅	`bullettrain_front`
🚆	`train2`	🚇	`metro`
🚈	`light_rail`	🚉	`station`
🚊	`tram`	🚝	`monorail`
🚞	`mountain_railway`	🚋	`train`
🚌	`bus`	🚍	`oncoming_bus`
🚎	`trolleybus`	🚐	`minibus`
🚑	`ambulance`	🚒	`fire_engine`
🚓	`police_car`	🚔	`oncoming_police_car`
🚕	`taxi`	🚖	`oncoming_taxi`
🚗	`car` `red_car`	🚘	`oncoming_automobile`
🚙	`blue_car`	🛻	`pickup_truck`
🚚	`truck`	🚛	`articulated_lorry`
🚜	`tractor`	🏎️	`racing_car`
🏍️	`motorcycle`	🛵	`motor_scooter`
🦽	`manual_wheelchair`	🦼	`motorized_wheelchair`
🛺	`auto_rickshaw`	🚲	`bike`
🛴	`kick_scooter`	🛹	`skateboard`
🛼	`roller_skate`	🚏	`busstop`
🛣️	`motorway`	🛤️	`railway_track`
🛢️	`oil_drum`	⛽	`fuelpump`
🚨	`rotating_light`	🚥	`traffic_light`
🚦	`vertical_traffic_light`	🛑	`stop_sign`

ico	shortcode	ico	shortcode
🚧	`construction`		

Transport Water

ico	shortcode	ico	shortcode
⚓	`anchor`	⛵	`boat` `sailboat`
🛶	`canoe`	🚤	`speedboat`
🛳️	`passenger_ship`	⛴️	`ferry`
🛥️	`motor_boat`	🚢	`ship`

Transport Air

ico	shortcode	ico	shortcode
✈️	`airplane`	🛩️	`small_airplane`
🛫	`flight_departure`	🛬	`flight_arrival`
🪂	`parachute`	💺	`seat`
🚁	`helicopter`	🚟	`suspension_railway`
🚠	`mountain_cableway`	🚡	`aerial_tramway`
🛰️	`artificial_satellite`	🚀	`rocket`
🛸	`flying_saucer`		

Hotel

ico	shortcode	ico	shortcode
🛎️	`bellhop_bell`	🧳	`luggage`

Time

ico	shortcode	ico	shortcode
⌛	`hourglass`	⏳	`hourglass_flowing_sand`
⌚	watch	⏰	alarm_clock
⏱️	`stopwatch`	⏲️	`timer_clock`
🕰️	mantelpiece_clock	🕛	clock12
🕧	`clock1230`	🕐	`clock1`
🕜	clock130	🕑	clock2
🕝	`clock230`	🕒	`clock3`
🕞	clock330	🕓	clock4
🕟	`clock430`	🕔	`clock5`
🕠	clock530	🕕	clock6
🕡	`clock630`	🕖	`clock7`
🕢	clock730	🕗	clock8
🕣	`clock830`	🕘	`clock9`
🕤	clock930	🕙	clock10
🕥	`clock1030`	🕚	`clock11`
🕦	clock1130		

Sky & Weather

ico	shortcode	ico	shortcode
🌑	`new_moon`	🌒	`waxing_crescent_moon`
🌓	first_quarter_moon	🌔	moon waxing_gibbous_moon
🌕	`full_moon`	🌖	`waning_gibbous_moon`
🌗	last_quarter_moon	🌘	waning_crescent_moon
🌙	`crescent_moon`	🌚	`new_moon_with_face`

ico	shortcode	ico	shortcode
🌛	`first_quarter_moon_with_face`	🌜	`last_quarter_moon_with_fa`
🌡️	`thermometer`	☀️	`sunny`
🌝	`full_moon_with_face`	🌞	`sun_with_face`
🪐	`ringed_planet`	⭐	`star`
🌟	`star2`	🌠	`stars`
🌌	`milky_way`	☁️	`cloud`
⛅	`partly_sunny`	⛈️	`cloud_with_lightning_and_`
🌤️	`sun_behind_small_cloud`	🌥️	`sun_behind_large_cloud`
🌦️	`sun_behind_rain_cloud`	🌧️	`cloud_with_rain`
🌨️	`cloud_with_snow`	🌩️	`cloud_with_lightning`
🌪️	`tornado`	🌫️	`fog`
🌬️	`wind_face`	🌀	`cyclone`
🌈	`rainbow`	🌂	`closed_umbrella`
☂️	`open_umbrella`	☔	`umbrella`
⛱️	`parasol_on_ground`	⚡	`zap`
❄️	`snowflake`	☃️	`snowman_with_snow`
⛄	`snowman`	☄️	`comet`
🔥	`fire`	💧	`droplet`
🌊	`ocean`		

Activities

- Event
- Award Medal
- Sport
- Game
- Arts & Crafts

Event

ico	shortcode	ico	shortcode
🎃	`jack_o_lantern`	🎄	`christmas_tree`
🎆	`fireworks`	🎇	`sparkler`
🧨	`firecracker`	✨	`sparkles`
🎈	`balloon`	🎉	`tada`
🎊	`confetti_ball`	🎋	`tanabata_tree`
🎍	`bamboo`	🎎	`dolls`
🎏	`flags`	🎐	`wind_chime`
🎑	`rice_scene`	🧧	`red_envelope`
🎀	`ribbon`	🎁	`gift`
🎗️	`reminder_ribbon`	🎟️	`tickets`
🎫	`ticket`		

Award Medal

ico	shortcode	ico	shortcode
🎖️	`medal_military`	🏆	`trophy`
🏅	`medal_sports`	🥇	`1st_place_medal`
🥈	`2nd_place_medal`	🥉	`3rd_place_medal`

Sport

ico	shortcode	ico	shortcode
⚽	`soccer`	⚾	`baseball`
🥎	`softball`	🏀	`basketball`
🏐	`volleyball`	🏈	`football`
🏉	`rugby_football`	🎾	`tennis`

ico	shortcode	ico	shortcode
🥏	`flying_disc`	🎳	`bowling`
🏏	cricket_game	🏑	field_hockey
🏒	`ice_hockey`	🥍	`lacrosse`
🏓	ping_pong	🏸	badminton
🥊	`boxing_glove`	🥋	`martial_arts_uniform`
🥅	goal_net	⛳	golf
⛸️	`ice_skate`	🎣	`fishing_pole_and_fish`
🤿	diving_mask	🎽	running_shirt_with_sash
🎿	`ski`	🛷	`sled`
🥌	curling_stone		

Game

ico	shortcode	ico	shortcode
🎯	`dart`	🪀	`yo_yo`
🪁	kite	🔫	gun
🎱	`8ball`	🔮	`crystal_ball`
🪄	magic_wand	🎮	video_game
🕹️	`joystick`	🎰	`slot_machine`
🎲	game_die	🧩	jigsaw
🧸	`teddy_bear`	🪅	`pinata`
🪆	nesting_dolls	♠️	spades
♥️	`hearts`	♦️	`diamonds`
♣️	clubs	♟️	chess_pawn
🃏	`black_joker`	🀄	`mahjong`
🎴	flower_playing_cards		

Arts & Crafts

ico	shortcode	ico	shortcode
🎭	`performing_arts`	🖼️	`framed_picture`
🎨	`art`	🧵	`thread`
🪡	`sewing_needle`	🧶	`yarn`
🪢	`knot`		

Objects

- Clothing
- Sound
- Music
- Musical Instrument
- Phone
- Computer
- Light & Video
- Book Paper
- Money
- Mail
- Writing
- Office
- Lock
- Tool
- Science
- Medical
- Household
- Other Object

Clothing

ico	shortcode	ico	shortcode
👓	`eyeglasses`	🕶️	`dark_sunglasses`
🥽	`goggles`	🥼	`lab_coat`
🦺	`safety_vest`	👔	`necktie`
👕	`shirt` `tshirt`	👖	`jeans`
🧣	`scarf`	🧤	`gloves`
🧥	`coat`	🧦	`socks`
👗	`dress`	👘	`kimono`
🥻	`sari`	🩱	`one_piece_swimsuit`
🩲	`swim_brief`	🩳	`shorts`
👙	`bikini`	👚	`womans_clothes`
👛	`purse`	👜	`handbag`
👝	`pouch`	🛍️	`shopping`
🎒	`school_satchel`	🩴	`thong_sandal`
👞	`mans_shoe` `shoe`	👟	`athletic_shoe`
🥾	`hiking_boot`	🥿	`flat_shoe`
👠	`high_heel`	👡	`sandal`
🩰	`ballet_shoes`	👢	`boot`
👑	`crown`	👒	`womans_hat`
🎩	`tophat`	🎓	`mortar_board`
🧢	`billed_cap`	🪖	`military_helmet`
⛑️	`rescue_worker_helmet`	📿	`prayer_beads`
💄	`lipstick`	💍	`ring`
💎	`gem`		

Sound

ico	shortcode	ico	shortcode
🔇	`mute`	🔈	`speaker`
🔉	`sound`	🔊	`loud_sound`
📢	`loudspeaker`	📣	`mega`
📯	`postal_horn`	🔔	`bell`
🔕	`no_bell`		

Music

ico	shortcode	ico	shortcode
🎼	`musical_score`	🎵	`musical_note`
🎶	`notes`	🎙️	`studio_microphone`
🎚️	`level_slider`	🎛️	`control_knobs`
🎤	`microphone`	🎧	`headphones`
📻	`radio`		

Musical Instrument

ico	shortcode	ico	shortcode
🎷	`saxophone`	🪗	`accordion`
🎸	`guitar`	🎹	`musical_keyboard`
🎺	`trumpet`	🎻	`violin`
🪕	`banjo`	🥁	`drum`
🪘	`long_drum`		

Phone

ico	shortcode	ico	shortcode
📱	`iphone`	📲	`calling`
☎️	`phone` `telephone`	📞	`telephone_receiver`
📟	`pager`	📠	`fax`

Computer

ico	shortcode	ico	shortcode
🔋	`battery`	🔌	`electric_plug`
💻	`computer`	🖥️	`desktop_computer`
🖨️	`printer`	⌨️	`keyboard`
🖱️	`computer_mouse`	🖲️	`trackball`
💽	`minidisc`	💾	`floppy_disk`
💿	`cd`	📀	`dvd`
🧮	`abacus`		

Light & Video

ico	shortcode	ico	shortcode
🎥	`movie_camera`	🎞️	`film_strip`
📽️	`film_projector`	🎬	`clapper`
📺	`tv`	📷	`camera`
📸	`camera_flash`	📹	`video_camera`
📼	`vhs`	🔍	`mag`
🔎	`mag_right`	🕯️	`candle`
💡	`bulb`	🔦	`flashlight`

ico	shortcode	ico	shortcode
🏮	izakaya_lantern lantern	🪔	diya_lamp

Book Paper

ico	shortcode	ico	shortcode
📔	`notebook_with_decorative_cover`	📕	`closed_book`
📖	book open_book	📗	green_book
📘	`blue_book`	📙	`orange_book`
📚	books	📓	notebook
📒	`ledger`	📃	`page_with_curl`
📜	scroll	📄	page_facing_up
📰	`newspaper`	🗞️	`newspaper_roll`
📑	bookmark_tabs	🔖	bookmark
🏷️	`label`		

Money

ico	shortcode	ico	shortcode
💰	`moneybag`	🪙	`coin`
💴	yen	💵	dollar
💶	`euro`	💷	`pound`
💸	money_with_wings	💳	credit_card
🧾	`receipt`	💹	`chart`

Mail

ico	shortcode	ico	shortcode
✉️	`envelope`	📧	`e-mail` `email`
📨	`incoming_envelope`	📩	`envelope_with_arrow`
📤	`outbox_tray`	📥	`inbox_tray`
📦	`package`	📫	`mailbox`
📪	`mailbox_closed`	📬	`mailbox_with_mail`
📭	`mailbox_with_no_mail`	📮	`postbox`
🗳️	`ballot_box`		

Writing

ico	shortcode	ico	shortcode
✏️	`pencil2`	✒️	`black_nib`
🖋️	`fountain_pen`	🖊️	`pen`
🖌️	`paintbrush`	🖍️	`crayon`
📝	`memo` `pencil`		

Office

ico	shortcode	ico	shortcode
💼	`briefcase`	📁	`file_folder`
📂	`open_file_folder`	🗂️	`card_index_dividers`
📅	`date`	📆	`calendar`
🗒️	`spiral_notepad`	🗓️	`spiral_calendar`
📇	`card_index`	📈	`chart_with_upwards_trend`

ico	shortcode	ico	shortcode
📉	chart_with_downwards_trend	📊	bar_chart
📋	clipboard	📌	pushpin
📍	round_pushpin	📎	paperclip
🖇️	paperclips	📏	straight_ruler
📐	triangular_ruler	✂️	scissors
🗃️	card_file_box	🗄️	file_cabinet
🗑️	wastebasket		

Lock

ico	shortcode	ico	shortcode
🔒	lock	🔓	unlock
🔏	lock_with_ink_pen	🔐	closed_lock_with_key
🔑	key	🗝️	old_key

Tool

ico	shortcode	ico	shortcode
🔨	hammer	🪓	axe
⛏️	pick	⚒️	hammer_and_pick
🛠️	hammer_and_wrench	🗡️	dagger
⚔️	crossed_swords	💣	bomb
🪃	boomerang	🏹	bow_and_arrow
🛡️	shield	🪚	carpentry_saw
🔧	wrench	🪛	screwdriver
🔩	nut_and_bolt	⚙️	gear
🗜️	clamp	⚖️	balance_scale

ico	shortcode	ico	shortcode
🦯	probing_cane	🔗	link
⛓️	chains	🪝	hook
🧰	toolbox	🧲	magnet
🪜	ladder		

Science

ico	shortcode	ico	shortcode
⚗️	alembic	🧪	test_tube
🧫	petri_dish	🧬	dna
🔬	microscope	🔭	telescope
📡	satellite		

Medical

ico	shortcode	ico	shortcode
💉	syringe	🩸	drop_of_blood
💊	pill	🩹	adhesive_bandage
🩺	stethoscope		

Household

ico	shortcode	ico	shortcode
🚪	door	🛗	elevator
🪞	mirror	🪟	window
🛏️	bed	🛋️	couch_and_lamp
🪑	chair	🚽	toilet
🪠	plunger	🚿	shower

ico	shortcode	ico	shortcode
🛁	bathtub	🪤	mouse_trap
🪒	razor	🧴	lotion_bottle
🧷	safety_pin	🧹	broom
🧺	basket	🧻	roll_of_paper
🪣	bucket	🧼	soap
🪥	toothbrush	🧽	sponge
🧯	fire_extinguisher	🛒	shopping_cart

Other Object

ico	shortcode	ico	shortcode
🚬	smoking	⚰️	coffin
🪦	headstone	⚱️	funeral_urn
🧿	nazar_amulet	🗿	moyai
🪧	placard		

Symbols

- Transport Sign
- Warning
- Arrow
- Religion
- Zodiac
- Av Symbol
- Gender
- Math
- Punctuation
- Currency
- Other Symbol

- Keycap
- Alphanum
- Geometric

Transport Sign

ico	shortcode	ico	shortcode
🏧	`atm`	🚮	`put_litter_in_its_place`
🚰	`potable_water`	♿	`wheelchair`
🚹	`mens`	🚺	`womens`
🚻	`restroom`	🚼	`baby_symbol`
🚾	`wc`	🛂	`passport_control`
🛃	`customs`	🛄	`baggage_claim`
🛅	`left_luggage`		

Warning

ico	shortcode	ico	shortcode
⚠️	`warning`	🚸	`children_crossing`
⛔	`no_entry`	🚫	`no_entry_sign`
🚳	`no_bicycles`	🚭	`no_smoking`
🚯	`do_not_litter`	🚱	`non-potable_water`
🚷	`no_pedestrians`	📵	`no_mobile_phones`
🔞	`underage`	☢️	`radioactive`
☣️	`biohazard`		

Arrow

ico	shortcode	ico	shortcode
⬆️	`arrow_up`	↗️	`arrow_upper_right`
➡️	`arrow_right`	↘️	`arrow_lower_right`
⬇️	`arrow_down`	↙️	`arrow_lower_left`
⬅️	`arrow_left`	↖️	`arrow_upper_left`
↕️	`arrow_up_down`	↔️	`left_right_arrow`
↩️	`leftwards_arrow_with_hook`	↪️	`arrow_right_hook`
⤴️	`arrow_heading_up`	⤵️	`arrow_heading_down`
🔃	`arrows_clockwise`	🔄	`arrows_counterclockwise`
🔙	`back`	🔚	`end`
🔛	`on`	🔜	`soon`
🔝	`top`		

Religion

ico	shortcode	ico	shortcode
🛐	`place_of_worship`	⚛️	`atom_symbol`
🕉️	`om`	✡️	`star_of_david`
☸️	`wheel_of_dharma`	☯️	`yin_yang`
✝️	`latin_cross`	☦️	`orthodox_cross`
☪️	`star_and_crescent`	☮️	`peace_symbol`
🕎	`menorah`	🔯	`six_pointed_star`

Zodiac

ico	shortcode	ico	shortcode
♈	`aries`	♉	`taurus`

ico	shortcode	ico	shortcode
♊	gemini	♋	cancer
♌	leo	♍	virgo
♎	libra	♏	scorpius
♐	sagittarius	♑	capricorn
♒	aquarius	♓	pisces
⛎	ophiuchus		

Av Symbol

ico	shortcode	ico	shortcode
🔀	twisted_rightwards_arrows	🔁	repeat
🔂	repeat_one	▶️	arrow_forward
⏩	fast_forward	⏭️	next_track_button
⏯️	play_or_pause_button	◀️	arrow_backward
⏪	rewind	⏮️	previous_track_button
🔼	arrow_up_small	⏫	arrow_double_up
🔽	arrow_down_small	⏬	arrow_double_down
⏸️	pause_button	⏹️	stop_button
⏺️	record_button	⏏️	eject_button
🎦	cinema	🔅	low_brightness
🔆	high_brightness	📶	signal_strength
📳	vibration_mode	📴	mobile_phone_off

Gender

ico	shortcode	ico	shortcode
♀️	female_sign	♂️	male_sign

ico	shortcode	ico	shortcode
⚧️	transgender_symbol		

Math

ico	shortcode	ico	shortcode
✖️	`heavy_multiplication_x`	➕	`heavy_plus_sign`
➖	heavy_minus_sign	➗	heavy_division_sign
♾️	`infinity`		

Punctuation

ico	shortcode	ico	shortcode
‼️	`bangbang`	⁉️	`interrobang`
❓	question	❔	grey_question
❕	`grey_exclamation`	❗	`exclamation` `heavy_exclamation_mark`
〰️	wavy_dash		

Currency

ico	shortcode	ico	shortcode
💱	`currency_exchange`	💲	`heavy_dollar_sign`

Other Symbol

ico	shortcode	ico	shortcode
⚕️	`medical_symbol`	♻️	`recycle`
⚜️	fleur_de_lis	🔱	trident

ico	shortcode	ico	shortcode
📛	`name_badge`	🔰	`beginner`
⭕	`o`	✅	`white_check_mark`
☑️	`ballot_box_with_check`	✔️	`heavy_check_mark`
❌	`x`	❎	`negative_squared_cross_mark`
➰	`curly_loop`	➿	`loop`
〽️	`part_alternation_mark`	✳️	`eight_spoked_asterisk`
✴️	`eight_pointed_black_star`	❇️	`sparkle`
©️	`copyright`	®️	`registered`
™️	`tm`		

Keycap

ico	shortcode	ico	shortcode
#️⃣	`hash`	*️⃣	`asterisk`
0️⃣	`zero`	1️⃣	`one`
2️⃣	`two`	3️⃣	`three`
4️⃣	`four`	5️⃣	`five`
6️⃣	`six`	7️⃣	`seven`
8️⃣	`eight`	9️⃣	`nine`
🔟	`keycap_ten`		

Alphanum

ico	shortcode	ico	shortcode
🔠	`capital_abcd`	🔡	`abcd`
🔢	`1234`	🔣	`symbols`
🔤	`abc`	🅰️	`a`

ico	shortcode	ico	shortcode
🆎	ab	🅱️	b
🆑	`cl`	🆒	`cool`
🆓	free	ℹ️	information_source
🆔	`id`	Ⓜ️	`m`
🆕	new	🆖	ng
🅾️	`o2`	🆗	`ok`
🅿️	parking	🆘	sos
🆙	`up`	🆚	`vs`
🈁	koko	🈂️	sa
🈷️	`u6708`	🈶	`u6709`
🈯	u6307	🉐	ideograph_advantage
🈹	`u5272`	🈚	`u7121`
🈲	u7981	🉑	accept
🈸	`u7533`	🈴	`u5408`
🈳	u7a7a	㊗️	congratulations
㊙️	`secret`	🈺	`u55b6`
🈵	u6e80		

Geometric

ico	shortcode	ico	shortcode
🔴	`red_circle`	🟠	`orange_circle`
🟡	yellow_circle	🟢	green_circle
🔵	`large_blue_circle`	🟣	`purple_circle`
🟤	brown_circle	⚫	black_circle
⚪	`white_circle`	🟥	`red_square`
🟧	orange_square	🟨	yellow_square

ico	shortcode	ico	shortcode
🟩	`green_square`	🟦	`blue_square`
🟪	`purple_square`	🟫	`brown_square`
⬛	`black_large_square`	⬜	`white_large_square`
◼️	`black_medium_square`	◻️	`white_medium_square`
◾	`black_medium_small_square`	◽	`white_medium_small_squ`
▪️	`black_small_square`	▫️	`white_small_square`
🔶	`large_orange_diamond`	🔷	`large_blue_diamond`
🔸	`small_orange_diamond`	🔹	`small_blue_diamond`
🔺	`small_red_triangle`	🔻	`small_red_triangle_dov`
💠	`diamond_shape_with_a_dot_inside`	🔘	`radio_button`
🔳	`white_square_button`	🔲	`black_square_button`

Flags

- Flag
- Country Flag
- Subdivision Flag

Flag

ico	shortcode	ico	shortcode
🏁	`checkered_flag`	🚩	`triangular_flag_on_post`
🎌	`crossed_flags`	🏴	`black_flag`
🏳️	`white_flag`	🏳️‍🌈	`rainbow_flag`
🏳️‍⚧️	`transgender_flag`	🏴‍☠️	`pirate_flag`

Country Flag

ico	shortcode	ico	shortc
🇦🇨	ascension_island	🇦🇩	andorra
🇦🇪	united_arab_emirates	🇦🇫	afghanistan
🇦🇬	antigua_barbuda	🇦🇮	anguilla
🇦🇱	albania	🇦🇲	armenia
🇦🇴	angola	🇦🇶	antarctica
🇦🇷	argentina	🇦🇸	american_samoa
🇦🇹	austria	🇦🇺	australia
🇦🇼	aruba	🇦🇽	aland_islands
🇦🇿	azerbaijan	🇧🇦	bosnia_herzegovi
🇧🇧	barbados	🇧🇩	bangladesh
🇧🇪	belgium	🇧🇫	burkina_faso
🇧🇬	bulgaria	🇧🇭	bahrain
🇧🇮	burundi	🇧🇯	benin
🇧🇱	st_barthelemy	🇧🇲	bermuda
🇧🇳	brunei	🇧🇴	bolivia
🇧🇶	caribbean_netherlands	🇧🇷	brazil
🇧🇸	bahamas	🇧🇹	bhutan
🇧🇻	bouvet_island	🇧🇼	botswana
🇧🇾	belarus	🇧🇿	belize
🇨🇦	canada	🇨🇨	cocos_islands
🇨🇩	congo_kinshasa	🇨🇫	central_african_
🇨🇬	congo_brazzaville	🇨🇭	switzerland
🇨🇮	cote_divoire	🇨🇰	cook_islands
🇨🇱	chile	🇨🇲	cameroon
🇨🇳	cn	🇨🇴	colombia

ico	shortcode	ico	shortc
🇨🇵	clipperton_island	🇨🇷	costa_rica
🇨🇺	cuba	🇨🇻	cape_verde
🇨🇼	curacao	🇨🇽	christmas_island
🇨🇾	cyprus	🇨🇿	czech_republic
🇩🇪	de	🇩🇬	diego_garcia
🇩🇯	djibouti	🇩🇰	denmark
🇩🇲	dominica	🇩🇴	dominican_republ
🇩🇿	algeria	🇪🇦	ceuta_melilla
🇪🇨	ecuador	🇪🇪	estonia
🇪🇬	egypt	🇪🇭	western_sahara
🇪🇷	eritrea	🇪🇸	es
🇪🇹	ethiopia	🇪🇺	eu european_union
🇫🇮	finland	🇫🇯	fiji
🇫🇰	falkland_islands	🇫🇲	micronesia
🇫🇴	faroe_islands	🇫🇷	fr
🇬🇦	gabon	🇬🇧	gb uk
🇬🇩	grenada	🇬🇪	georgia
🇬🇫	french_guiana	🇬🇬	guernsey
🇬🇭	ghana	🇬🇮	gibraltar
🇬🇱	greenland	🇬🇲	gambia
🇬🇳	guinea	🇬🇵	guadeloupe
🇬🇶	equatorial_guinea	🇬🇷	greece
🇬🇸	south_georgia_south_sandwich_islands	🇬🇹	guatemala
🇬🇺	guam	🇬🇼	guinea_bissau
🇬🇾	guyana	🇭🇰	hong_kong

ico	shortcode	ico	shortc
🇭🇲	heard_mcdonald_islands	🇭🇳	honduras
🇭🇷	croatia	🇭🇹	haiti
🇭🇺	hungary	🇮🇨	canary_islands
🇮🇩	indonesia	🇮🇪	ireland
🇮🇱	israel	🇮🇲	isle_of_man
🇮🇳	india	🇮🇴	british_indian_oc
🇮🇶	iraq	🇮🇷	iran
🇮🇸	iceland	🇮🇹	it
🇯🇪	jersey	🇯🇲	jamaica
🇯🇴	jordan	🇯🇵	jp
🇰🇪	kenya	🇰🇬	kyrgyzstan
🇰🇭	cambodia	🇰🇮	kiribati
🇰🇲	comoros	🇰🇳	st_kitts_nevis
🇰🇵	north_korea	🇰🇷	kr
🇰🇼	kuwait	🇰🇾	cayman_islands
🇰🇿	kazakhstan	🇱🇦	laos
🇱🇧	lebanon	🇱🇨	st_lucia
🇱🇮	liechtenstein	🇱🇰	sri_lanka
🇱🇷	liberia	🇱🇸	lesotho
🇱🇹	lithuania	🇱🇺	luxembourg
🇱🇻	latvia	🇱🇾	libya
🇲🇦	morocco	🇲🇨	monaco
🇲🇩	moldova	🇲🇪	montenegro
🇲🇫	st_martin	🇲🇬	madagascar
🇲🇭	marshall_islands	🇲🇰	macedonia
🇲🇱	mali	🇲🇲	myanmar
🇲🇳	mongolia	🇲🇴	macau

ico	shortcode	ico	shortc
🇲🇵	northern_mariana_islands	🇲🇶	martinique
🇲🇷	mauritania	🇲🇸	montserrat
🇲🇹	malta	🇲🇺	mauritius
🇲🇻	maldives	🇲🇼	malawi
🇲🇽	mexico	🇲🇾	malaysia
🇲🇿	mozambique	🇳🇦	namibia
🇳🇨	new_caledonia	🇳🇪	niger
🇳🇫	norfolk_island	🇳🇬	nigeria
🇳🇮	nicaragua	🇳🇱	netherlands
🇳🇴	norway	🇳🇵	nepal
🇳🇷	nauru	🇳🇺	niue
🇳🇿	new_zealand	🇴🇲	oman
🇵🇦	panama	🇵🇪	peru
🇵🇫	french_polynesia	🇵🇬	papua_new_guinea
🇵🇭	philippines	🇵🇰	pakistan
🇵🇱	poland	🇵🇲	st_pierre_miquelo
🇵🇳	pitcairn_islands	🇵🇷	puerto_rico
🇵🇸	palestinian_territories	🇵🇹	portugal
🇵🇼	palau	🇵🇾	paraguay
🇶🇦	qatar	🇷🇪	reunion
🇷🇴	romania	🇷🇸	serbia
🇷🇺	ru	🇷🇼	rwanda
🇸🇦	saudi_arabia	🇸🇧	solomon_islands
🇸🇨	seychelles	🇸🇩	sudan
🇸🇪	sweden	🇸🇬	singapore
🇸🇭	st_helena	🇸🇮	slovenia
🇸🇯	svalbard_jan_mayen	🇸🇰	slovakia

ico	shortcode	ico	shortc
🇸🇱	`sierra_leone`	🇸🇲	`san_marino`
🇸🇳	`senegal`	🇸🇴	`somalia`
🇸🇷	`suriname`	🇸🇸	`south_sudan`
🇸🇹	`sao_tome_principe`	🇸🇻	`el_salvador`
🇸🇽	`sint_maarten`	🇸🇾	`syria`
🇸🇿	`swaziland`	🇹🇦	`tristan_da_cunha`
🇹🇨	`turks_caicos_islands`	🇹🇩	`chad`
🇹🇫	`french_southern_territories`	🇹🇬	`togo`
🇹🇭	`thailand`	🇹🇯	`tajikistan`
🇹🇰	`tokelau`	🇹🇱	`timor_leste`
🇹🇲	`turkmenistan`	🇹🇳	`tunisia`
🇹🇴	`tonga`	🇹🇷	`tr`
🇹🇹	`trinidad_tobago`	🇹🇻	`tuvalu`
🇹🇼	`taiwan`	🇹🇿	`tanzania`
🇺🇦	`ukraine`	🇺🇬	`uganda`
🇺🇲	`us_outlying_islands`	🇺🇳	`united_nations`
🇺🇸	`us`	🇺🇾	`uruguay`
🇺🇿	`uzbekistan`	🇻🇦	`vatican_city`
🇻🇨	`st_vincent_grenadines`	🇻🇪	`venezuela`
🇻🇬	`british_virgin_islands`	🇻🇮	`us_virgin_island`
🇻🇳	`vietnam`	🇻🇺	`vanuatu`
🇼🇫	`wallis_futuna`	🇼🇸	`samoa`
🇽🇰	`kosovo`	🇾🇪	`yemen`
🇾🇹	`mayotte`	🇿🇦	`south_africa`
🇿🇲	`zambia`	🇿🇼	`zimbabwe`

Subdivision Flag

ico	shortcode	ico	shortcode
🏴󠁧󠁢󠁥󠁮󠁧󠁿	`england`	🏴󠁧󠁢󠁳󠁣󠁴󠁿	`scotland`
🏴󠁧󠁢󠁷󠁬󠁳󠁿	`wales`		

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// demo0
#import "@preview/cetz:0.3.4": canvas, draw
#import draw: line, content, rect, circle, bezier, group

#set page(width: auto, height: auto, margin: 15pt)

#canvas({
  // Core constants and reusable values
  let plot_width = 24
  let timeline_width = 16
  let time_tick_spacing = 1.0
  let step_width = time_tick_spacing
  let step_gap = 0.1
  let box_width_factor = 0.9
  let rect_height = 0.3
  let border_radius = 0.05
  let gap = 0.15
  let strategy_y_positions = (11, 6.5, 2.0) // Further compressed y-
positions

  // Colors - more pastel version for better text readability
  // Grays
  let dark_gray = rgb("#5D6B7A")
  let cpu_color = rgb("#8899AA")
  let gpu_color = rgb("#6A7A8A")
  let section_bg = rgb("#F7FAFC")
  // Define a consistent CPU light gray color for all strategies
  let cpu_light_gray = cpu_color.lighten(70%)

  // Blues with pastel tones
  let light_blue = rgb("#BBDEFB")
  let mid_blue = rgb("#90CAF9")
  let dark_blue = rgb("#64B5F6")

  // Greens with pastel tones
  let light_green = rgb("#C8E6C9")
  let mid_green = rgb("#A5D6A7")
  let dark_green = rgb("#81C784")

  // Oranges with pastel tones
  let light_orange = rgb("#FFE0B2")
  let mid_orange = rgb("#FFCC80")
  let dark_orange = rgb("#FFB74D")
```

```
  // Reds with pastel tones
  let light_red = rgb("#FFCDD2")
  let mid_red = rgb("#EF9A9A")
  let dark_red = rgb("#E57373")

  // Helper functions
  let draw_sim_block(x_pos, y_pos, width, color, label, task_label: "",
opacity: 100%) = {
    rect(
      (x_pos, y_pos - rect_height / 2),
      (x_pos + width, y_pos + rect_height / 2),
      fill: color.transparentize(100% - opacity),
      stroke: color.darken(10%),
      radius: border_radius,
      name: "block-" + label,
    )
    if task_label != "" {
      content((x_pos + width / 2, y_pos), text(size: 8pt)[#task_label],
name: "label-" + label)
    }
  }

  let draw_structure_rect(x_pos, y_pos, width, color, label, struct_name)
= {
    rect(
      (x_pos, y_pos),
      (x_pos + 0.95 * width, y_pos + 0.1),
      fill: color.transparentize(20%),
      stroke: none,
      name: "block-" + struct_name,
    )
    content((x_pos + width / 2, y_pos + 0.15), text(size: 8pt)[#label],
name: "label-" + struct_name, anchor: "south")
  }

  let draw_util_bar(x_pos, y_pos, percentage, label, is_bad: false) = {
    // Bar showing CPU or GPU utilization - consolidated function
    let width = 2.8
    let height = 0.4

    // Background bar
    rect(
      (x_pos, y_pos - height / 2),
      (x_pos + width, y_pos + height / 2),
      fill: rgb("#E2E8F0"),
```

```
      stroke: 0.5pt,
      radius: 0.1,
      name: "bar-bg-" + label,
    )

    // Filled portion with appropriate color
    let fill_width = width * percentage / 100

    // Determine color based on percentage and is_bad flag
    let fill_color = if is_bad {
      light_red.darken(20%) // Use darker red for explicitly marked "bad"
utilization
    } else if percentage < 30 {
      light_red
    } else if percentage < 70 {
      light_orange
    } else {
      light_green
    }

    rect(
      (x_pos, y_pos - height / 2),
      (x_pos + fill_width, y_pos + height / 2),
      fill: fill_color,
      stroke: none,
      radius: (north-west: 0.1, south-west: 0.1),
      name: "bar-fill-" + label,
    )

    // Label
    content(
      (rel: (0.03, 0), to: "bar-bg-" + label),
      text(size: 8pt, weight: "bold")[#label #percentage%],
      name: "bar-label-" + label,
      anchor: "center",
    )
  }

  // Title, subtitle and legend
  let title_y = 15.0
  content(
    (plot_width / 2, title_y),
    text(weight: "bold", size: 16pt)[GPU Batching Strategies for Atomistic
Simulations],
    name: "main-title",
```

```
  )
  content(
    (rel: (0, -1), to: "main-title"),
    text(size: 12pt)[Comparison of Unbatched vs. BinningAutoBatcher vs.
InFlightAutoBatcher],
    name: "subtitle",
  )


  // Add section backgrounds and scenario setups
  let strategies = (
    (strategy_y_positions.at(0), "Unbatched\nSimulations", (80, 5)),
    (strategy_y_positions.at(1), "Binning\nAutoBatcher", (40, 60)),
    (strategy_y_positions.at(2), "InFlight\nAutoBatcher", (60, 90)),
  )

  for (idx, (y_pos, label, (cpu_util, gpu_util))) in
strategies.enumerate() {
    // Background
    rect(
      (0.5, y_pos - 2.0),
      (plot_width - 0.5, y_pos + 2.0),
      fill: section_bg,
      stroke: none,
      radius: border_radius * 3,
      name: "section-bg-" + str(idx),
    )

    // Scenario label
    content((2, y_pos + 1.0), text(weight: "bold", size: 12pt)[#label],
name: "scenario-label-" + str(idx))

    // Utilization meters
    draw_util_bar(.8, y_pos - 0.1, cpu_util, "CPU Utilization")

    // GPU utilization - mark as bad for unbatched simulations (idx == 0)
    draw_util_bar(.8, y_pos - 1, gpu_util, "GPU Utilization", is_bad: idx
== 0)

    // CPU/GPU labels
    content(
      (4.6, y_pos + 1.3),
      text(fill: cpu_color, size: 10pt, weight: "bold")[CPU],
      anchor: "east",
      name: "cpu-label-" + str(idx),
```

```
    )
    content(
      (4.6, y_pos - 0.5),
      text(fill: gpu_color, size: 10pt, weight: "bold")[GPU],
      anchor: "east",
      name: "gpu-label-" + str(idx),
    )

    // Timeline axis with ticks
    line(
      (4.8, y_pos - 1.5),
      (4.8 + timeline_width, y_pos - 1.5),
      stroke: 0.8pt,
      mark: (end: "stealth", fill: black, scale: 0.5),
      name: "timeline-" + str(idx),
    )

    for t in range(1, 17) {
      let x_pos = 4 + t * time_tick_spacing
      if x_pos <= 4 + timeline_width {
        line(
          (x_pos, y_pos - 1.5 - 0.05),
          (x_pos, y_pos - 1.5 + 0.05),
          stroke: 0.8pt,
          name: "tick-" + str(idx) + "-" + str(t),
        )
        content(
          (rel: (0, -0.2), to: "tick-" + str(idx) + "-" + str(t)),
          text(size: 8pt)[t=#t],
          anchor: "north",
          name: "tick-label-" + str(idx) + "-" + str(t),
        )
      }
    }

    // CPU/GPU separator
    let separator_y = if idx == 0 { y_pos + 0.6 } else { y_pos + 0.95 }
    line(
      (4, separator_y),
      (4 + timeline_width, separator_y),
      stroke: (dash: "dotted", paint: dark_gray.lighten(30%)),
      name: "separator-" + str(idx),
    )
  }
```

```
  // 1. Unbatched Simulations
  let unbatched_cpu_y = strategy_y_positions.at(0) + 1.4
  let unbatched_gpu_y = strategy_y_positions.at(0) - 0.5

  // Structure rectangles with consistent pattern
  let unbatched_structures = (
    (5.0, 2.0, mid_blue, "Structure 1"),
    (7.0, 2.3, mid_green, "Structure 2"),
    (9.3, 1.8, mid_orange, "Structure 3"),
    (11.1, 2.1, mid_red, "Structure 4"),
    (13.2, 1.8, dark_green, "Structure 5"),
    (15.0, 2.0, dark_green.darken(10%), "Structure 6"),
    (17.0, 1.6, dark_green.darken(15%), "Structure 7"),
    (18.6, 1.5, dark_green.darken(20%), "Structure 8"),
  )

  for (idx, (x_pos, width, color, label)) in
unbatched_structures.enumerate() {
    draw_structure_rect(x_pos, unbatched_cpu_y, width, color, label,
"unbatched-" + str(idx + 1) + "-cpu")
  }

  content((21.8, unbatched_cpu_y - 1.9), text(size: 10pt, weight: "bold")
[... continues], anchor: "center")

  // CPU operation blocks in a loop
  for x in range(22) {
    let x_offset = 5.25 + x * 0.7
    if x_offset < 20 {
      draw_sim_block(
        x_offset,
        unbatched_cpu_y - 0.4,
        0.4,
        cpu_light_gray,
        "unbatched-cpu-op-" + str(x),
        task_label: "",
        opacity: 90%,
      )
    }
  }

  // GPU blocks with structure IDs and positions
  let gpu_blocks = (
    (1, dark_blue, (5.5, 6.5)),
    (2, dark_green, (7.5, 8.5)),
```

```
    (3, dark_orange, (9.8, 10.6)),
    (4, dark_red, (11.5, 12.5)),
    (5, dark_green, (13.8, 14.6, 15.4)),
    (6, dark_green.darken(10%), (16.2, 16.8, 17.4, 18.0, 18.6)),
    (7, dark_green.darken(15%), (19.2, 19.8)),
  )

  for (struct_num, color, positions) in gpu_blocks {
    for (idx, pos) in positions.enumerate() {
      draw_sim_block(
        pos,
        unbatched_gpu_y,
        0.3 * box_width_factor,
        color,
        "unbatched-" + str(struct_num) + "-gpu-" + str(idx + 1),
        task_label: "S" + str(struct_num),
      )
    }
  }

  // 2. BinningAutoBatcher
  let binning_cpu_y = strategy_y_positions.at(1) + 1.3
  let binning_gpu_y = strategy_y_positions.at(1) - 0.85

  // CPU processing blocks - simplified to only show prep batch operations
  let cpu_blocks = (
    (5.0, "Prep batch"),
    (10.4, "Prep batch"),
  )

  for (idx, (x_pos, label)) in cpu_blocks.enumerate() {
    draw_sim_block(x_pos, binning_cpu_y, 1.3, cpu_light_gray, "binning-op-
" + str(idx), task_label: label)
  }


  // Batch visualization setup
  let s_colors = (
    dark_blue,
    dark_green,
    dark_orange,
    dark_red,
    dark_blue.darken(10%),
  )
```

```
  let s_labels = ("S1", "S2", "S3", "S4", "S5")

  // Bin parameters and active patterns
  let bins = (
    (
      // Bin 1: 6 steps with completion patterns
      0,
      false,
      (
        (1, 1, 1, 1, 1), // Step 0: all active
        (1, 1, 1, 1, 1), // Step 1: all active
        (0, 1, 1, 1, 1), // Step 2: structure 1 completes earlier
        (0, 0, 1, 1, 1), // Step 3: structures 1,2 complete
        (0, 0, 0, 1, 0), // Step 4: structures 1,2,3,5 complete
        (0, 0, 0, 1, 0), // Step 5: only 4 remains
      ),
    ),
    (
      // Bin 2: 6 steps with different completion pattern
      6,
      true,
      (
        (1, 1, 1, 1, 1), // Step 0: all active
        (1, 1, 1, 1, 1), // Step 1: structure 3 completes early
        (0, 1, 0, 1, 1), // Step 2: structures 1,3 complete
        (0, 1, 0, 0, 1), // Step 3: only 4,5 remain
        (0, 1, 0, 0, 0), // Step 4: only 5 remains
        (0, 1, 0, 0, 0), // Step 5: all complete
      ),
    ),
  )

  // Draw both bins with common function
  for (start_step, is_second_bin, patterns) in bins {
    for step in range(patterns.len()) {
      let x_pos = 5.0 + (start_step + step) * (step_width - step_gap)
      let box_width = (step_width - step_gap) * box_width_factor
      let box_x_start = x_pos + ((step_width - step_gap) - box_width) / 2

      // Draw each structure slot
      for i in range(5) {
        let block_y = binning_gpu_y - 0.3 + i * (rect_height + gap)
        let binning_block_name = "binning-block-" + str(start_step) + "-"
+ str(step) + "-" + str(i)
```

```
        if patterns.at(step).at(i) == 1 {
          let color = s_colors.at(i)
          let label = if is_second_bin { "S" + str(i + 6) } else {
s_labels.at(i) }

          rect(
            (box_x_start, block_y - rect_height / 2),
            (box_x_start + box_width, block_y + rect_height / 2),
            fill: color,
            stroke: color.darken(20%),
            radius: border_radius,
            name: binning_block_name,
          )

          content(
            (box_x_start + box_width / 2, block_y),
            text(size: 7pt)[#label],
            anchor: "center",
            name: binning_block_name + "-label",
          )
        } else {
          // Empty placeholder
          rect(
            (box_x_start, block_y - rect_height / 2),
            (box_x_start + box_width, block_y + rect_height / 2),
            fill: light_red.transparentize(90%),
            stroke: (dash: "dotted", paint:
light_red.transparentize(30%)),
            radius: border_radius,
            name: binning_block_name + "-empty",
          )
        }
      }

      // Add completion label for the last step
      if step == patterns.len() - 1 {
        content(
          (box_x_start - 0.25, binning_gpu_y - 0.45),
          text(size: 8pt, fill: dark_gray, style: "italic")[Batch #if
is_second_bin [2] else [1] Complete],
          anchor: "center",
          name: "batch-" + str(if is_second_bin { 2 } else { 1 }) + "-
complete-label",
        )
      }
```

```
    }
  }

  for (x_pos, percentage) in ((5.6, 100), (8.0, 60), (9.8, 30), (11, 100),
(13.5, 60), (15.5, 20)) {
    content(
      (x_pos, binning_gpu_y + 2.7),
      text(size: 8pt, fill: dark_gray)[#percentage% GPU],
      anchor: "center",
    )
  }

  // Explanatory arrows and labels - removing the isolated red arrow
  content(
    (10.5, binning_gpu_y - 1.5),
    text(size: 7pt, fill: dark_red, style: "italic")[Must wait for batch
to complete, resulting in underutilized GPU],
    anchor: "center",
    name: "must-wait",
  )

  let batch_end_x = 5.0 + 6 * (step_width - step_gap)
  let timeline_y = strategy_y_positions.at(1) - 1.5

  line(
    "must-wait.north",
    (batch_end_x, timeline_y),
    stroke: (dash: "dotted", paint: dark_red),
    mark: (end: "stealth", fill: dark_red, scale: 0.4, offset: 0.05),
    name: "must-wait-arrow",
  )

  // 3. In-flightAutoBatcher
  let inflight_cpu_y = strategy_y_positions.at(2) + 1.3
  let inflight_gpu_y = strategy_y_positions.at(2) - 0.9

  // Structure colors and labels
  let all_colors = (
    dark_blue,
    dark_green,
    dark_orange,
    dark_red,
    dark_blue.darken(10%),
    dark_green.darken(10%),
    dark_orange.darken(10%),
```

```
    dark_red.darken(10%),
    dark_blue.darken(20%),
    dark_green.darken(20%),
  )
  let all_labels = ("S1", "S2", "S3", "S4", "S5", "S6", "S7", "S8", "S9",
"S10")

  // Structure placement by time step - fix structure continuity
  let structures_by_step = (
    (0, 1, 2, 3, 4), // t=1: S1-S5
    (0, 1, 2, 3, 4), // t=2: S1-S5 (S1 continues for longer)
    (0, 6, 2, 3, 4), // t=3: S1 continues, S6 replaces S2, S3-S5 continue
    (5, 6, 7, 3, 4), // t=4: S5, S6-S8, S4 continue (S1 completed, S5
swapped in)
    (5, 6, 7, 8, 4), // t=5: S5-S9, S4 (S3 completed, S8 swapped in)
    (5, 6, 7, 8, 9), // t=6: S5-S10 (S4 completed, S9 swapped in)
    (5, 6, 7, 8, 9), // t=7: still fully filled
    (5, 6, 7, -1, -1), // t=8: final step, partially filled (about 50%) -
using S6 instead of isolated S4
  )

  // Place CPU operations equidistant, one for each time step
  // Create the same number of prep boxes as structure steps (8 steps)

  // Draw CPU operations - one per time step
  for step in range(structures_by_step.len()) {
    let x_pos = 5.0 + step * (step_width - step_gap)
    let box_width = (step_width - step_gap) * box_width_factor * 0.7 //
Make boxes 10% wider (70% instead of 60%)

    // Left align by using 10% of remaining space as left margin instead
of centering
    let box_x_start = x_pos + ((step_width - step_gap) - box_width) * 0.1

    // Use the same generic gray for all prep blocks
    draw_sim_block(
      box_x_start,
      inflight_cpu_y,
      box_width,
      cpu_light_gray,
      "inflight-prep-" + str(step),
      task_label: "Prep",
    )
  }
```

```
  // Draw all structures by step
  for step in range(structures_by_step.len()) {
    let x_pos = 5.0 + step * (step_width - step_gap)
    let box_width = (step_width - step_gap) * box_width_factor
    let box_x_start = x_pos + ((step_width - step_gap) - box_width) / 2

    // Create a batch name for this step
    let batch_name = "inflight-batch-" + str(step)

    // Draw each structure position
    for pos in range(5) {
      let struct_idx = structures_by_step.at(step).at(pos)
      let block_y = inflight_gpu_y - 0.3 + pos * (rect_height + gap)
      let block_name = "inflight-block-" + str(step) + "-" + str(pos)

      if struct_idx != -1 {
        let color = all_colors.at(struct_idx)
        let label = all_labels.at(struct_idx)

        // Structure box
        rect(
          (box_x_start, block_y - rect_height / 2),
          (box_x_start + box_width, block_y + rect_height / 2),
          fill: color,
          stroke: color.darken(20%),
          radius: border_radius,
          name: block_name,
        )

        content(
          (box_x_start + box_width / 2, block_y),
          text(size: 7pt)[#label],
          name: block_name + "-label",
          anchor: "center",
        )

        // Swap-in indicator
        if step > 0 {
          let prev_structures = structures_by_step.at(step - 1)
          let prev_block_name = "inflight-block-" + str(step - 1) + "-" +
str(pos)

          if pos < prev_structures.len() and prev_structures.at(pos) !=
struct_idx {
            // Simple diagonal line for swap-in indicator
```

```
            line(
              (rel: (0, 0), to: block_name + ".south-west"), // End at the
block edge
              (rel: (-0.25, -0.3), to: block_name + ".south-west"), //
Start from outside
              stroke: (dash: "dotted", paint: dark_green),
              mark: (start: "stealth", fill: dark_green, scale: 0.3),
              name: block_name + "-swap-indicator",
            )
          }
        }
      } else {
        // Empty placeholders with dotted lines to indicate unused GPU
cycles
        rect(
          (box_x_start, block_y - rect_height / 2),
          (box_x_start + box_width, block_y + rect_height / 2),
          fill: light_red.transparentize(90%),
          stroke: (dash: "dotted", paint: light_red.transparentize(30%)),
          radius: border_radius,
          name: block_name + "-empty",
        )
      }
    }

    // Mark the last batch specially
    if step == structures_by_step.len() - 1 {
      // Store a reference to the last batch for later reference
      let final_block_name = "inflight-block-" + str(step) + "-0"

      // Instead of using absolute positioning, use relative positioning
      // based on the named elements we just created
      content(
        (rel: (box_width + 1.7, 0.2), to: final_block_name),
        text(size: 8pt, fill: dark_gray, style: "italic")[Final batch
partially filled],
        anchor: "center",
        name: "final-batch-label",
      )
    }
  }

  // First label with frame
  content(
    (plot_width / 2, -0.8),
```

```
    text(
      size: 9pt,
      weight: "bold",
      fill: dark_green,
    )[In-flight batching achieves highest GPU utilization and maximizes
predictions per unit time],
    frame: "rect",
    fill: light_green.transparentize(70%),
    stroke: dark_green,
    padding: 3pt,
    radius: border_radius,
  )

  // Summary notes
  content(
    (plot_width / 2, -3),
    box(width: 50em)[
      *Unbatched:* Each simulation runs sequentially with most
calculations on CPU and minimal GPU utilization\
      *Binning:* Fixed-size batches improve GPU utilization but can't
adapt to varying simulation completion times\
      *In-flight:* Dynamic reallocation eliminates GPU idle time by
immediately adding new structures when others complete. Color changes
indicate in-flight structure replacement.
    ],
    frame: "rect",
    fill: rgb("#F7FAFC"),
    stroke: 0.5pt,
    padding: (10pt, 10pt, 0pt),
    radius: border_radius,
  )
})
```